



SCHEM, MLB_LDO, K6
PVT, 3/18/10

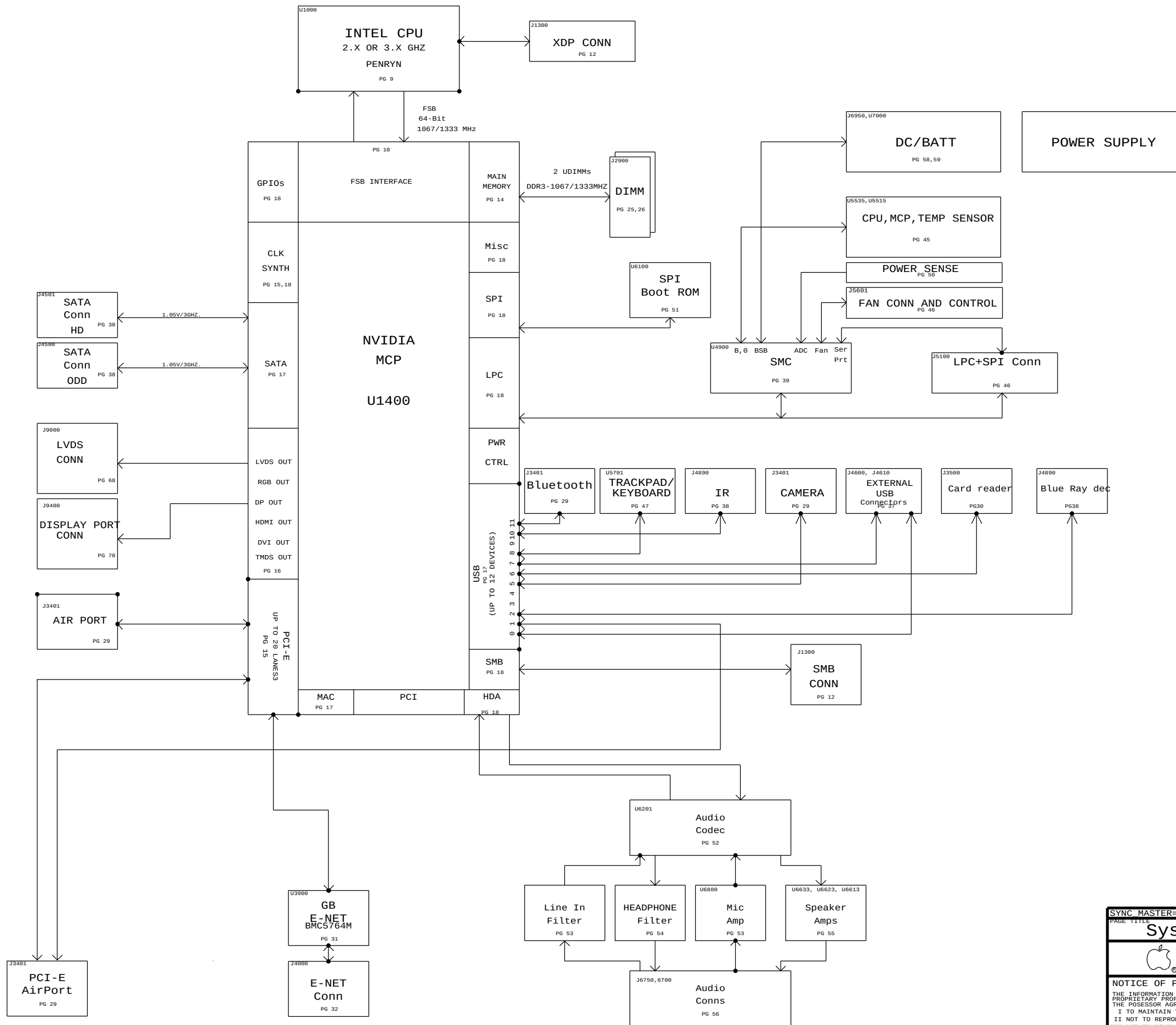
Page	Contents	Sync	Date
1	Table of Contents	K17_MLB	05/20/2009
2	System Block Diagram	K69_MLB	08/19/2009
3	Power Block Diagram	K69_MLB	08/19/2009
4	BOM Configuration	K24_MLB	07/20/2009
5	Revision History	K24_MLB	07/20/2009
6	FUNC TEST	K24_MLB	07/20/2009
7	Power Aliases	K24_MLB	07/22/2009
8	SIGNAL ALIAS	K24_MLB	07/20/2009
9	CPU FSB	T27_MLB	08/27/2009
10	CPU Power & Ground	T27_MLB	07/20/2009
11	CPU Decoupling	T27_MLB	11/23/2009
12	eXtended Debug Port (mini-XDP)	T27_MLB	07/28/2009
13	MCP CPU Interface	T27_MLB	11/05/2009
14	MCP Memory Interface	T27_MLB	08/06/2009
15	MCP PCIe Interfaces	T27_MLB	11/05/2009
16	MCP Graphics	T27_MLB	11/05/2009
17	MCP SATA, USB & Ethernet	T27_MLB	11/23/2009
18	MCP HDA, LPC & MISC	T27_MLB	11/23/2009
19	MCP Power & Ground	T27_MLB	08/06/2009
20	MCP89 Memory Rail Gating	T27_MLB	11/23/2009
21	MCP89 GFX Core Rail Gating	T27_MLB	11/23/2009
22	MCP Standard Decoupling	T27_MLB	08/15/2009
23	MCP Graphics Support	T27_MLB	08/06/2009
24	SB Misc	T27_MLB	07/28/2009
25	DDR3 SO-DIMM Connector A	T27_MLB	07/28/2009
26	DDR3 SO-DIMM Connector B	T27_MLB	07/28/2009
27	DDR3 BYTE/BIT SWAPS-K6	K18_MLB	06/19/2009
28	FSB/DDR3 Vref Margining	T27_MLB	09/29/2009
29	RIGHT CLUTCH CONNECTOR	T27_MLB	07/28/2009
30	SecureDigital Card Reader	T27_MLB	09/30/2009
31	Ethernet PHY (Caesar II/IV)	T27_MLB	08/20/2009
32	Ethernet Connector	T27_MLB	07/28/2009
33	FireWire LLC/PHY (FW643E)	T27_MLB	07/20/2009
34	FireWire Port & PHY Power	T27_MLB	12/15/2009
35	FireWire Connector	T27_MLB	07/28/2009
36	SATA Connectors	T27_MLB	08/06/2009
37	External USB Connectors	T27_MLB	08/27/2009
38	Internal USB Support	T27_MLB	08/27/2009
39	SMC	T27_MLB	09/02/2009
40	SMC Support	T27_MLB	09/02/2009
41	LPC+SPI Debug Connector	T27_MLB	08/27/2009
42	K6 SMBUS CONNECTIONS	T27_MLB	08/21/2009
43	Voltage Sensing	T27_MLB	08/27/2009
44	Current Sensing	T27_MLB	09/30/2009
45	Thermal Sensors	T27_MLB	08/27/2009

Page	Contents	Sync	Date
46	Fan	K24_MLB	07/20/2009
47	WELLSPRING 1	T27_MLB	08/15/2009
48	WELLSPRING 2	T27_MLB	08/03/2009
49	Sudden Motion Sensor (SMS)	T27_MLB	07/20/2009
50	SPI ROM	T27_MLB	10/21/2009
51	AUDIO: CODEC/REGULATOR	AUDIO	08/31/2009
52	AUDIO: LINE INPUT FILTER	AUDIO	07/17/2009
53	AUDIO: HEADPHONE FILTER	AUDIO	07/17/2009
54	AUDIO: SPEAKER AMP	AUDIO	07/17/2009
55	AUDIO: JACK	AUDIO	08/25/2009
56	AUDIO: JACK TRANSLATORS	AUDIO	08/27/2009
57	DC-In & Battery Connectors	K24_MLB	07/20/2009
58	PBus Supply & Battery Charger	T27_MLB	07/29/2009
59	5V/3.3V SUPPLY	K24_MLB	07/20/2009
60	1.5V/1.35V LVDDR3 Supply	T27_MLB	08/06/2009
61	IMVP6 CPU VCore Regulator	K24_MLB	07/20/2009
62	MCP VCore Regulator	T27_MLB	08/18/2009
63	CPU VTT(1.05V) SUPPLY	K24_MLB	07/20/2009
64	Misc Power Supplies	T27_MLB	09/30/2009
65	Power Sequencing	T27_MLB	11/24/2009
66	Power FETs	T27_MLB	08/27/2009
67	LVDS CONNECTOR	K24_MLB	07/20/2009
68	DISPLAYPORT SUPPORT	K69_MLB	08/12/2009
69	DisplayPort Connector	K24_MLB	07/20/2009
70	LCD Backlight Driver	K69_MLB	08/27/2009
71	LCD Backlight Support	T27_MLB	07/28/2009
72	CPU/FSB Constraints	T27_MLB	08/03/2009
73	Memory Constraints	T27_MLB	08/03/2009
74	MCP Constraints 1	T27_MLB	08/03/2009
75	MCP Constraints 2	T27_MLB	08/27/2009
76	Ethernet Constraints	T27_MLB	11/23/2009
77	FireWire Constraints	T27_MLB	07/20/2009
78	SMC Constraints	T27_MLB	07/28/2009
79	K6/K69 Specific Constraints	T27_MLB	09/08/2009
80	K6/K69 PCB Rule Definitions	T27_MLB	08/06/2009

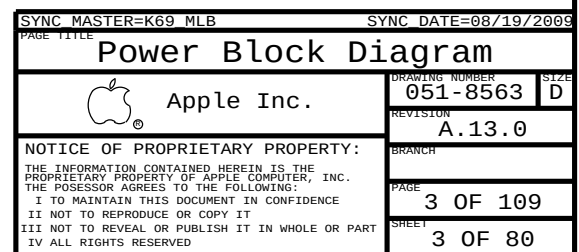
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051-8563	1	SCHEM, MLB_LDO, K6	SCH	CRITICAL	
820-2879	1	PCBF, MLB_LDO, K6	PCB	CRITICAL	

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SCHEM, MLB_LDO, K6	
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	1 OF 109
	SHEET
	1 OF 80



SYNC MASTER=K69 MLB		SYNC DATE=08/19/2009	
PAGE TITLE			
System Block Diagram			
 Apple Inc.		DRAWING NUMBER	051-8563
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-1120	PCBA, MLB_LDO, BETTER, K6	K6_COMMON, CPU:2.4GHZ, MCP89M:A02, EEEE:DD24
639-1119	PCBA, MLB_LDO, BEST, K6	K6_COMMON, CPU:2.66GHZ, MCP89M:A02, EEEE:DD23
085-1634	K6 MLB_LDO DEVELOPMENT BOM	K6_DEVEL:PVT

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DD23]	CRITICAL	EEEE:DD23
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DD24]	CRITICAL	EEEE:DD24

BOM Groups

BOM GROUP	BOM OPTIONS
K6_COMMON	COMMON, ALTERNATE, K6_MISC, K6_DEBUG:PROD, KB_BL, K6_PROGPARTS, RDRV:NO, SPI:25MHZ, CPU_CAP:15
K6_MISC	DP_ESD, MIKEY, BCM5764M, GL137, ENET_ESD, VFRQ:SLPS3, LVDDR3:YES, MCPPLL_R:REG, S0PG00D_BJT, B00ST_VOL:LOW, HDA:1.5V
K6_PROGPARTS	BOOTROM:UNLOCKED, SMC:PROG, IR:PROG, WELLSPRING:PROG
K6_DEVEL:ENG	BKLT:ENG, BMON:ENG, XDP_CONN, LPCPLUS, VREFMRGN:YES, EFI_DEBUG, S0PG00D_ISL, RDRV:IN_DEVEL
K6_DEVEL:PVT	LPCPLUS, XDP_CONN
K6_DEBUG:ENG	DEVEL_BOM, SMC_DEBUG:YES, XDP
K6_DEBUG:PVT	DEVEL_BOM, BKLT:PROD, BMON:PROD, SMC_DEBUG:YES, XDP, VREFMRGN:NO
K6_DEBUG:PROD	BKLT:PROD, BMON:PROD, SMC_DEBUG:YES, XDP, VREFMRGN:NO, LPCPLUS, MCPHVDD:P2V5, LDO:FIXED, HTOL_SENSE:YES

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3769	1	PDC, SLGVT, PRQ, 2.26, 25W, 1066, R0, 3M, BGA, P7550	U1000	CRITICAL	CPU:2.26GHZ
337S3680	1	PDC, LGDZ, PRQ, 2.40, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU:2.4GHZ
337S3756	1	PDC, SLGFG, PRQ, 2.53, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU:2.53GHZ
337S3761	1	PDC, SLGLA, PRQ, 2.66, 25W, 1066, E0, 3M, BGA	U1000	CRITICAL	CPU:2.66GHZ
337S3797	1	IC, MCP89M-A01, 31X31MM, BGA1168	U1400	CRITICAL	MCP89M:A01
337S3866	1	IC, MCP89M-A01, 31X31MM, BGA1168	U1400	CRITICAL	MCP89M:A02
341S2731	1	IC, 1MBIT, SPI FLASH, K17/18	U3990	CRITICAL	BCM5764M
343S0493	1	IC, ASIC, BCM5764M, ENET CONTROLLER, 8x8, 64QFN	U3900	CRITICAL	BCM5764M
338S0753	1	IC, FW643-E2, 1394B PHY/QNCI LINK/PCI-E, 12	U4100	CRITICAL	
353S2896	1	IC, LP8545, LED BKLT CTRLR, LLP24	U9701	CRITICAL	

Programmable Parts

338S0563	1	IC, SMC, HS8/2117, 9X9MM, TLP, HF	U4900	CRITICAL	SMC:BLANK
341T0240	1	SMC EXTERNAL, K6	U4900	CRITICAL	SMC:PROG
335S0610	1	IC, FLASH, SPI, 32MBIT, 3.3V, 86MHZ, 8-SOP	U6100	CRITICAL	BOOTROM:BLANK
341T0238	1	EFI UNLOCKED, K6/K69	U6100	CRITICAL	BOOTROM:UNLOCKED
341S2589	1	IC, EFI, LOCKED, K6	U6100	CRITICAL	BOOTROM:LOCKED
338S0633	1	IC, CYPRS, CY7C63803-LQXC, 4X4MM, USB, 24-QFN	U4800	CRITICAL	IR:BLANK
341S2384	1	IC, ENCORE II, CY7C63803-LQXC	U4800	CRITICAL	IR:PROG
337S2983	1	IC, PSOC+ W/ USB, 56 PIN, MLF, CY8C24794	U5701	CRITICAL	WELLSPRING:BLANK
341S2616	1	IC, TP PSOC, K17, K18	U5701	CRITICAL	WELLSPRING:PROG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S0693	152S0778		ALL	CYNTEC AS ALTERNATE
152S0796	152S0685		ALL	CYNTEC AS ALTERNATE
157S0058	157S0055		ALL	DELTA AS ALTERNATE
104S0018	104S0023		ALL	DALE/VISHAY AS ALTERNATE
128S0093	128S0218		ALL	KEMET AS ALTERNATE
152S0874	152S0516		ALL	MAGLAYERS AS ALTERNATE
152S0847	152S0586		ALL	MAGLAYERS AS ALTERNATE
152S1025	152S1024		ALL	TOKO AS ALTERNATE
337S3769	337S3704		ALL	INTEL P7550 CPU AS ALTERNATE
152S1135	152S0586		ALL	TOKO AS ALTERNATE
516-0213	516-0201		ALL	MOLEX AS ALTERNATE
516S0790	516S0706		ALL	MOLEX AS ALTERNATE
376S0699	376S0360		ALL	SSMP15FE AS ALTERNATE

DEVELOPMENT BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-1634	1	K6 MLB_LDO DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM

K6 BOARD STACK-UP

Top	SIGNAL
2	GROUND
3	SIGNAL(High Speed)
4	SIGNAL(High Speed)
5	GROUND
6	POWER
7	POWER
8	GROUND
9	SIGNAL(High Speed)
10	SIGNAL(High Speed)
11	GROUND
BOTTOM	SIGNAL

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		PAGE	4 OF 109
		SHEET	4 OF 80

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		PAGE	
		5 OF 109	
		SHEET	
		5 OF 80	



Functional Test Points

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Fan Connectors

PP5V_S0	TRUE	PP5V_S0	(NEED 2 TP)
FAN_RT_PWM	TRUE	FAN_RT_PWM	46
FAN_RT_TACH	TRUE	FAN_RT_TACH	46

(NEED TO ADD 3 GND TP)

MIC_FUNC_TEST

BI_MIC_LO	TRUE	BI_MIC_LO	55 56
BI_MIC_HI	TRUE	BI_MIC_HI	55 56
BI_MIC_SHIELD	TRUE	BI_MIC_SHIELD	55 56

SPEAKER_FUNC_TEST

SPKRAMP_L_N_OUT	TRUE	SPKRAMP_L_N_OUT	54 55
SPKRAMP_L_P_OUT	TRUE	SPKRAMP_L_P_OUT	54 55
SPKRAMP_R_N_OUT	TRUE	SPKRAMP_R_N_OUT	54 55
SPKRAMP_R_P_OUT	TRUE	SPKRAMP_R_P_OUT	54 55
SPKRAMP_SUB_N_OUT	TRUE	SPKRAMP_SUB_N_OUT	54 55
SPKRAMP_SUB_P_OUT	TRUE	SPKRAMP_SUB_P_OUT	54 55

LVDS_FUNC_TEST

PP3V3_LCDVDD_SW_F	TRUE	PP3V3_LCDVDD_SW_F	6 67
PP3V3_S0_LCD_F	TRUE	PP3V3_S0_LCD_F	6 67
PPVOUT_SW_LCDBKLT	TRUE	PPVOUT_SW_LCDBKLT	67 70
BKL_VSYNC	TRUE	BKL_VSYNC	67 70
LVDS_DDC_CLK	TRUE	LVDS_DDC_CLK	8 67
LVDS_DDC_DATA	TRUE	LVDS_DDC_DATA	8 67
LVDS_IG_A_DATA_N<0>	TRUE	LVDS_IG_A_DATA_N<0>	8 67 74
LVDS_IG_A_DATA_P<0>	TRUE	LVDS_IG_A_DATA_P<0>	8 67 74
LVDS_IG_A_DATA_N<1>	TRUE	LVDS_IG_A_DATA_N<1>	8 67 74
LVDS_IG_A_DATA_P<1>	TRUE	LVDS_IG_A_DATA_P<1>	8 67 74
LVDS_IG_A_DATA_N<2>	TRUE	LVDS_IG_A_DATA_N<2>	8 67 74
LVDS_IG_A_DATA_P<2>	TRUE	LVDS_IG_A_DATA_P<2>	8 67 74
LVDS_CONN_A_CLK_F_N	TRUE	LVDS_CONN_A_CLK_F_N	67 79
LVDS_CONN_A_CLK_F_P	TRUE	LVDS_CONN_A_CLK_F_P	67 79
LED_RETURN_1	TRUE	LED_RETURN_1	67 70
BKL_ISEN2	TRUE	BKL_ISEN2	70
BKL_ISEN3	TRUE	BKL_ISEN3	70
LED_RETURN_4	TRUE	LED_RETURN_4	67 70
LED_RETURN_5	TRUE	LED_RETURN_5	67 70
LED_RETURN_6	TRUE	LED_RETURN_6	67 70

(NEED TO ADD 5 GND TP)

SATA_ODD_CONN

PP5V_SW_ODD	TRUE	PP5V_SW_ODD	(NEED 4 TP) 6 8
SMC_ODD_DETECT	TRUE	SMC_ODD_DETECT	36 39
SATA_ODD_D2R_UF_P	TRUE	SATA_ODD_D2R_UF_P	36 79
SATA_ODD_D2R_UF_N	TRUE	SATA_ODD_D2R_UF_N	36 79
SATA_ODD_R2D_P	TRUE	SATA_ODD_R2D_P	36 74
SATA_ODD_R2D_N	TRUE	SATA_ODD_R2D_N	36 74

(NEED TO ADD 4 GND TP)

SATA_HDD/IR/SIL

PP5V_S0_HDD_FLT	TRUE	PP5V_S0_HDD_FLT	(NEED 3 TP) 6 36
SATA_HDD_R2D_P	TRUE	SATA_HDD_R2D_P	36 74
SATA_HDD_R2D_N	TRUE	SATA_HDD_R2D_N	36 74
SATA_HDD_D2R_C_P	TRUE	SATA_HDD_D2R_C_P	36 74
SATA_HDD_D2R_C_N	TRUE	SATA_HDD_D2R_C_N	36 74
SYS_LED_ANODE_R	TRUE	SYS_LED_ANODE_R	36
IR_RX_OUT	TRUE	IR_RX_OUT	36 38
PP5V_S3_IR_R	TRUE	PP5V_S3_IR_R	36

(NEED TO ADD 5 GND TP)

BATT_POWER_CONN

SMBUS_SMC_BSA_SCL	TRUE	SMBUS_SMC_BSA_SCL	6 42 78
SMBUS_SMC_BSA_SDA	TRUE	SMBUS_SMC_BSA_SDA	6 42 78
SYS_DETECT_L	TRUE	SYS_DETECT_L	57
PPVBAT_G3H_CONN	TRUE	PPVBAT_G3H_CONN	57 58

(NEED 3 TP)
(NEED TO ADD 4 GND TP)

BIL_CONN

PP3V42_G3H	TRUE	PP3V42_G3H	6 7
SMBUS_SMC_BSA_SCL	TRUE	SMBUS_SMC_BSA_SCL	6 42 78
SMBUS_SMC_BSA_SDA	TRUE	SMBUS_SMC_BSA_SDA	6 42 78
SMC_BIL_BUTTON_L	TRUE	SMC_BIL_BUTTON_L	39 40 57
SMC_LID_R	TRUE	SMC_LID_R	57

(NEED TO ADD 4 GND TP)

RIGHT_CLUTCH_CONN

PP5V_S3_BITCAMERA_F	TRUE	PP5V_S3_BITCAMERA_F	29
PCIE_AP_D2R_P	TRUE	PCIE_AP_D2R_P	15 29 74
PCIE_AP_D2R_N	TRUE	PCIE_AP_D2R_N	15 29 74
PCIE_AP_R2D_P	TRUE	PCIE_AP_R2D_P	29 74
PCIE_AP_R2D_N	TRUE	PCIE_AP_R2D_N	29 74
PCIE_CLK100M_AP_CONN_P	TRUE	PCIE_CLK100M_AP_CONN_P	29 79
PCIE_CLK100M_AP_CONN_N	TRUE	PCIE_CLK100M_AP_CONN_N	29 79
USB_CAMERA_CONN_P	TRUE	USB_CAMERA_CONN_P	29 79
USB_CAMERA_CONN_N	TRUE	USB_CAMERA_CONN_N	29 79
PP5V_WLAN	TRUE	PP5V_WLAN	6 29
PCIE_WAKE_L	TRUE	PCIE_WAKE_L	15 24 29
SMBUS_SMC_A_S3_SCL	TRUE	SMBUS_SMC_A_S3_SCL	6 42 78
SMBUS_SMC_A_S3_SDA	TRUE	SMBUS_SMC_A_S3_SDA	6 42 78
USB_BT_CONN_P	TRUE	USB_BT_CONN_P	29 79
USB_BT_CONN_N	TRUE	USB_BT_CONN_N	29 79
AP_CLKREQ_Q_L	TRUE	AP_CLKREQ_Q_L	29
AP_RESET_CONN_L	TRUE	AP_RESET_CONN_L	29

(NEED TO ADD 6 GND TP)

IPD_FLEX_CONN

PP3V3_S3	TRUE	PP3V3_S3	6 7
PP18V5_S3	TRUE	PP18V5_S3	6 48
Z2_CS_L	TRUE	Z2_CS_L	47 48
Z2_DEBUG3	TRUE	Z2_DEBUG3	47 48
Z2_M0SI	TRUE	Z2_M0SI	47 48
Z2_MISO	TRUE	Z2_MISO	47 48
Z2_SCLK	TRUE	Z2_SCLK	47 48
Z2_BOOTST_EN	TRUE	Z2_BOOTST_EN	48
Z2_HOST_INTN	TRUE	Z2_HOST_INTN	47 48
Z2_CLKIN	TRUE	Z2_CLKIN	47 48
Z2_KEY_ACT_L	TRUE	Z2_KEY_ACT_L	47 48
Z2_RESET	TRUE	Z2_RESET	47 48
PSOC_MISO	TRUE	PSOC_MISO	47 48
PSOC_M0SI	TRUE	PSOC_M0SI	47 48
PSOC_SCLK	TRUE	PSOC_SCLK	47 48
SMBUS_SMC_A_S3_SDA	TRUE	SMBUS_SMC_A_S3_SDA	6 42 78
SMBUS_SMC_A_S3_SCL	TRUE	SMBUS_SMC_A_S3_SCL	6 42 78
PSOC_F_CS_L	TRUE	PSOC_F_CS_L	47 48
PICKB_L	TRUE	PICKB_L	47 48

(NEED TO ADD 2 GND TP)

KEYBOARD_CONN

PP3V3_S3	TRUE	PP3V3_S3	6 7
PP3V42_G3H	TRUE	PP3V42_G3H	6 7
WS_KBD1	TRUE	WS_KBD1	47
WS_KBD2	TRUE	WS_KBD2	47
WS_KBD3	TRUE	WS_KBD3	47
WS_KBD4	TRUE	WS_KBD4	47
WS_KBD5	TRUE	WS_KBD5	47
WS_KBD6	TRUE	WS_KBD6	47
WS_KBD7	TRUE	WS_KBD7	47
WS_KBD8	TRUE	WS_KBD8	47
WS_KBD9	TRUE	WS_KBD9	47
WS_KBD10	TRUE	WS_KBD10	47
WS_KBD11	TRUE	WS_KBD11	47
WS_KBD12	TRUE	WS_KBD12	47
WS_KBD13	TRUE	WS_KBD13	47
WS_KBD14	TRUE	WS_KBD14	47
WS_KBD15_CAP	TRUE	WS_KBD15_CAP	47
WS_KBD16_NUM	TRUE	WS_KBD16_NUM	47
WS_KBD17	TRUE	WS_KBD17	47
WS_KBD18	TRUE	WS_KBD18	47
WS_KBD19	TRUE	WS_KBD19	47
WS_KBD20	TRUE	WS_KBD20	47
WS_KBD21	TRUE	WS_KBD21	47
WS_KBD22	TRUE	WS_KBD22	47
WS_KBD23	TRUE	WS_KBD23	47
WS_KBD_ONOFF_L	TRUE	WS_KBD_ONOFF_L	47
WS_LEFT_SHIFT_KBD	TRUE	WS_LEFT_SHIFT_KBD	47
WS_LEFT_OPTION_KBD	TRUE	WS_LEFT_OPTION_KBD	47
WS_CONTROL_KBD	TRUE	WS_CONTROL_KBD	47

(NEED TO ADD 2 GND TP)

KBD_BACKLIGHT_CONN

KBDLED_ANODE	TRUE	KBDLED_ANODE	48
SMC_KDBLED_PRESENT_L	TRUE	SMC_KDBLED_PRESENT_L	48

(NEED TO ADD 1 GND TP)

T57_CONN

PP5V_S3	TRUE	PP5V_S3	6 7
PP3V3_S3	TRUE	PP3V3_S3	6 7
T57_PWR_EN	TRUE	T57_PWR_EN	18
T57_RESET	TRUE	T57_RESET	18
USB_T57_N	TRUE	USB_T57_N	38 75
USB_T57_P	TRUE	USB_T57_P	38 75

(NEED TO ADD 5 GND TP)

DEBUG_VOLTAGE

PPVCORE_S0_CPU	TRUE	PPVCORE_S0_CPU	7 43
PPVCORE_S0_MCP	TRUE	PPVCORE_S0_MCP	7 43
PP1V2_ENET	TRUE	PP1V2_ENET	7
PP1V05_S0	TRUE	PP1V05_S0	7 65
PP1V5_S0	TRUE	PP1V5_S0	7 65 79
PP1V8_S0	TRUE	PP1V8_S0	7
PP3V3_S0	TRUE	PP3V3_S0	7 65 79
PP5V_S0	TRUE	PP5V_S0	6 7 65
PP3V3_S3	TRUE	PP3V3_S3	6 7
PP5V_S3	TRUE	PP5V_S3	6 7
PP0V9_S5	TRUE	PP0V9_S5	7
PP3V3_S5	TRUE	PP3V3_S5	7 65 79
PP3V42_G3H	TRUE	PP3V42_G3H	6 7
PPBUS_G3H	TRUE	PPBUS_G3H	7 43
PP3V3_ENET	TRUE	PP3V3_ENET	7
PP5V_WLAN	TRUE	PP5V_WLAN	6 29
PP5V_SW_ODD	TRUE	PP5V_SW_ODD	6 8
PP5V_S0_HDD_FLT	TRUE	PP5V_S0_HDD_FLT	6 36
PP18V5_S3	TRUE	PP18V5_S3	6 48
PP3V3_S0_LCD_F	TRUE	PP3V3_S0_LCD_F	6 67
PP3V3_LCDVDD_SW_F	TRUE	PP3V3_LCDVDD_SW_F	6 67
PP4V5_AUDIO_ANALOG	TRUE	PP4V5_AUDIO_ANALOG	51
PP1V5R1V35_S3	TRUE	PP1V5R1V35_S3	7 79
SMC_PM_G2_EN	TRUE	SMC_PM_G2_EN	39 65
PM_SLP_S4_L	TRUE	PM_SLP_S4_L	18 39 40 65
PM_SLP_S3_L	TRUE	PM_SLP_S3_L	18 39 65 69

(NEED TO ADD 6 GND TP)

SPI_DEBUG_CONN

PP3V42_G3H	TRUE	PP3V42_G3H	6 7
SPI_CS0_L	TRUE	SPI_CS0_L	41 75
SPI_CLK	TRUE	SPI_CLK	41 75
SPI_M0SI	TRUE	SPI_M0SI	41 75
SPI_MISO	TRUE	SPI_MISO	18 41 75
SPIR0M_USE_MLB	TRUE	SPIR0M_USE_MLB	18 41 50

DC_POWER_CONN

PP18V5_DCIN_FUSE	TRUE	PP18V5_DCIN_FUSE	57
ADAPTER_SENSE	TRUE	ADAPTER_SENSE	57

(NEED TO ADD 4 GND TP)

FSB_SIGNALS_WITH_NOTEST

NO_TEST=TRUE	FSB_A_L<35..3>	9 13 72
NO_TEST=TRUE	FSB_ADS_L	9 13 72
NO_TEST=TRUE	FSB_ADSTB_L<1..0>	9 13 72
NO_TEST=TRUE	FSB_D_L<63..0>	9 13 72
NO_TEST=TRUE	FSB_DINV_L<3..0>	9 13 72
NO_TEST=TRUE	FSB_DSTB_L_N<3..0>	9 13 72
NO_TEST=TRUE	FSB_DSTB_L_P<3..0>	9 13 72
NO_TEST=TRUE	FSB_HIT_L	9 13 72
NO_TEST=TRUE	FSB_HITM_L	9 13 72
NO_TEST=TRUE	FSB_LOCK_L	9 13 72
NO_TEST=TRUE	FSB_REQ_L<4..0>	9 13 72

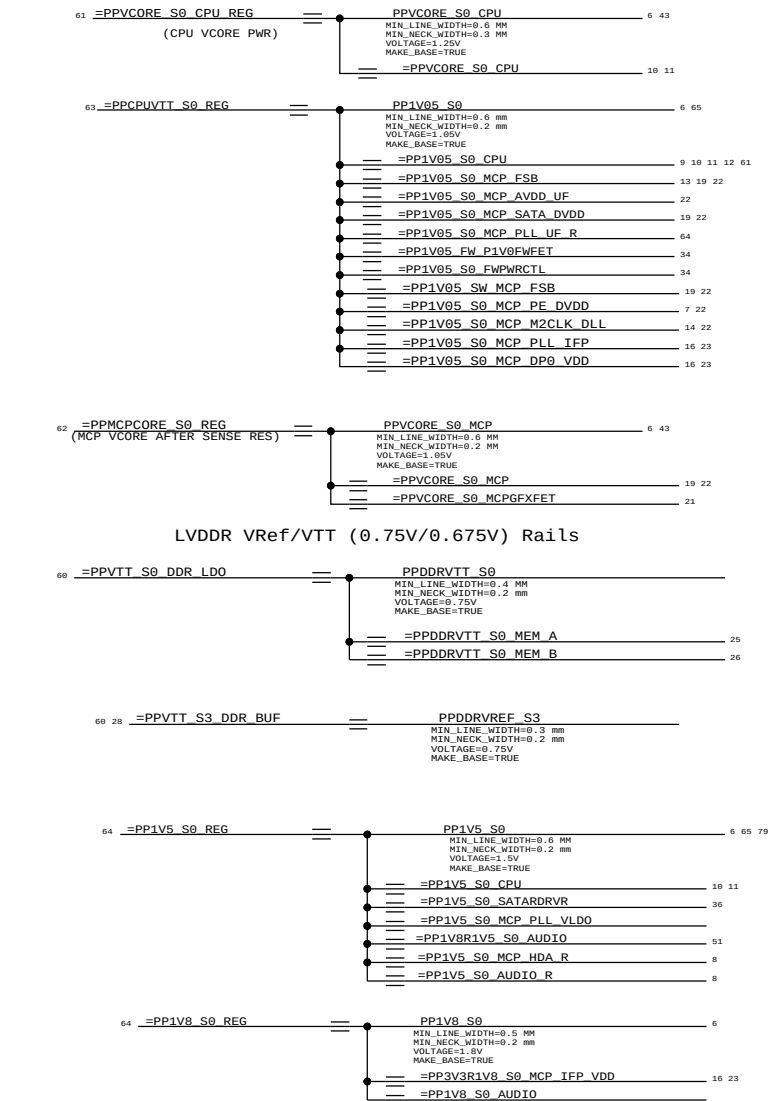
A

A

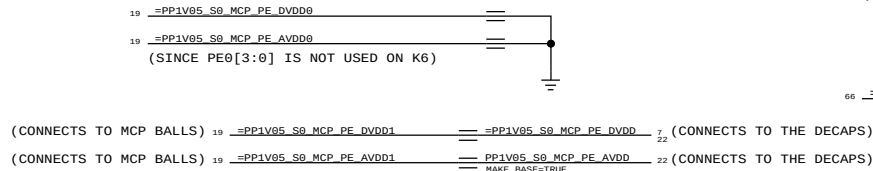
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PAGE TITLE	
FUNC TEST	
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	SIZE D
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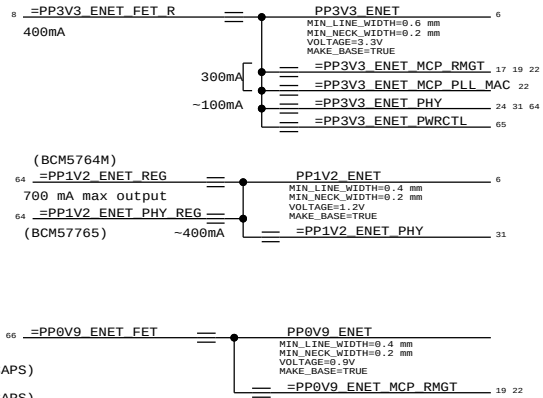
"S0,S0M" RAILS



UNUSED MCP PE0[3:0] AVDD/DVDD

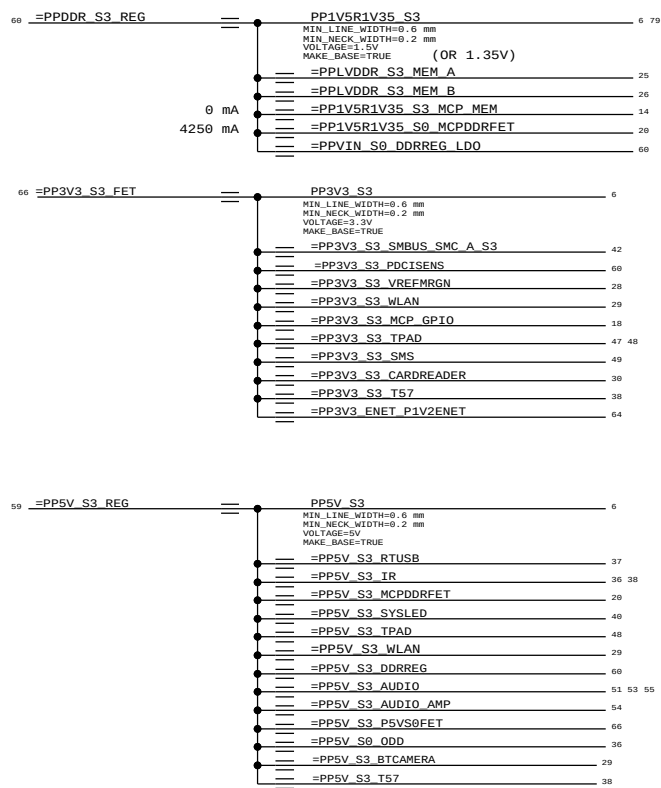


"ENET" RAILS

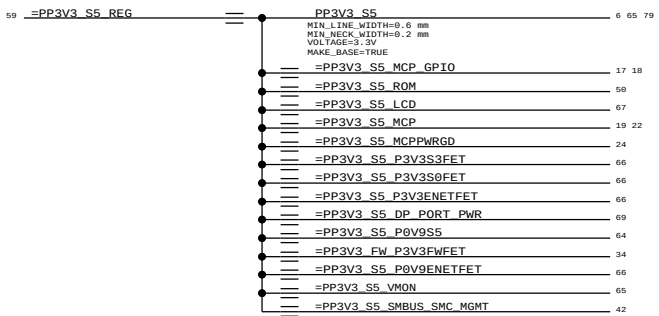


"S3" RAILS

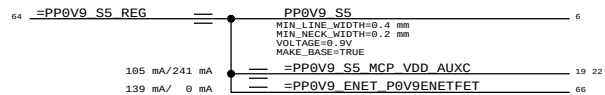
LVDDR (1.5V/1.35V) Rails



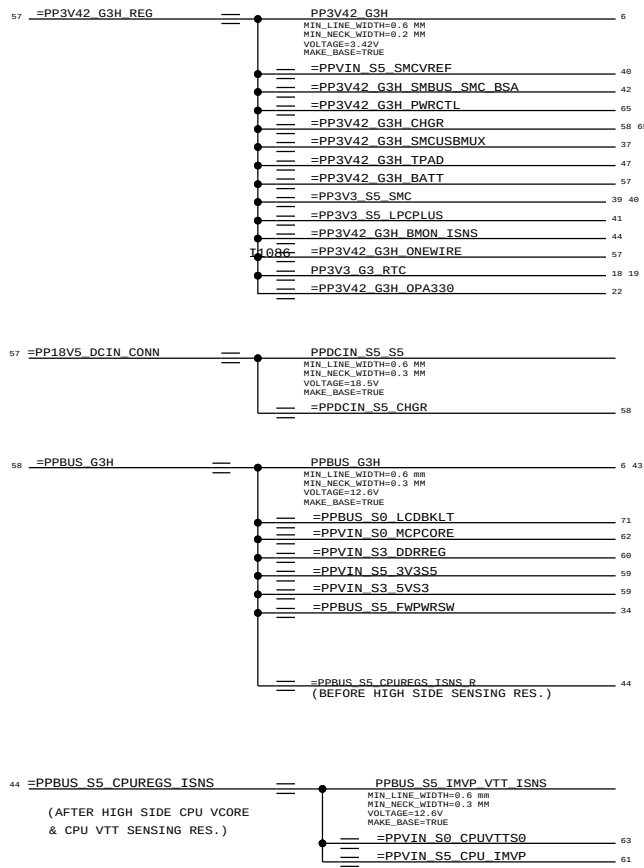
"S5" RAILS



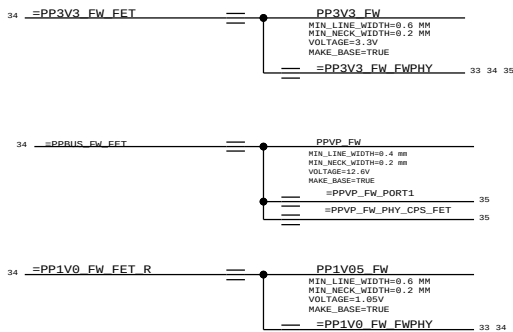
0.9V Rails



"G3H" RAILS



"FIREWIRE" RAILS

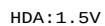


SYNC MASTER=K24 MLB		SYNC DATE=07/22/2008	
PAGE TITLE		Power Aliases	
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		REVISION	A.13.0
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LVDS ALIASES			
15	=MCP IFPA TXC P	= LVDS IG A CLK P	
			MAKE_BASE=TRUE 67 71
16	=MCP IFPA TXC N	= LVDS IG A CLK N	
			MAKE_BASE=TRUE 67 74
15	=MCP IFPA TXD P<0..2>	= LVDS IG A DATA P<0..2>	
			MAKE_BASE=TRUE 6 67 74
16	=MCP IFPA TXD N<0..2>	= LVDS IG A DATA N<0..2>	
			MAKE_BASE=TRUE 6 67 74
15	=MCP IFPA TXD P<3>	= NC LVDS IG A DATAP<3>	
			MAKE_BASE=TRUE 67 74
16	=MCP IFPA TXD N<3>	= NC LVDS IG A DATAN<3>	
			MAKE_BASE=TRUE 67 74
15	=MCP IFPB TXC P	= NC LVDS IG B CLK P	
			MAKE_BASE=TRUE 67 74
16	=MCP IFPB TXC N	= NC LVDS IG B CLK N	
			MAKE_BASE=TRUE 67 74
15	=MCP IFPB TXD P<0..3>	= NC LVDS IG B DATAP<0..3>	
			MAKE_BASE=TRUE 67 74
16	=MCP IFPB TXD N<0..3>	= NC LVDS IG B DATAN<0..3>	
			MAKE_BASE=TRUE 67 74
15	=LCD IG BKLT PWM	= LCD BKLT PWM	
			MAKE_BASE=TRUE 70
16	=LCD IG BKLT EN	= LCD BKLT EN	
			MAKE_BASE=TRUE 71
15	=MCP IFPAB DDC CLK	= LVDS DDC CLK	
			MAKE_BASE=TRUE 6 67
16	=MCP IFPAB DDC DATA	= LVDS DDC DATA	
			MAKE_BASE=TRUE 6 67

AUDIO ALIASES



CHINA

200
(166)
333
100
(480)
(RSVD)

5V ODD ALIASES



CHARGER SIGNAL

58 **CHGR_ACOK** == SMC_BC_ACOK == 39 48 57
MAKE_BASE=TRUE

MCPCOREISNS SIGNAL

62 MCPCORES0_V0 == MCPCOREISNS_N == MCPCOREISNS_N
MAKE_BASE=TRUE

62 MCPCORES0_ISP_R == MCPCOREISNS_P == MCPCOREISNS_P

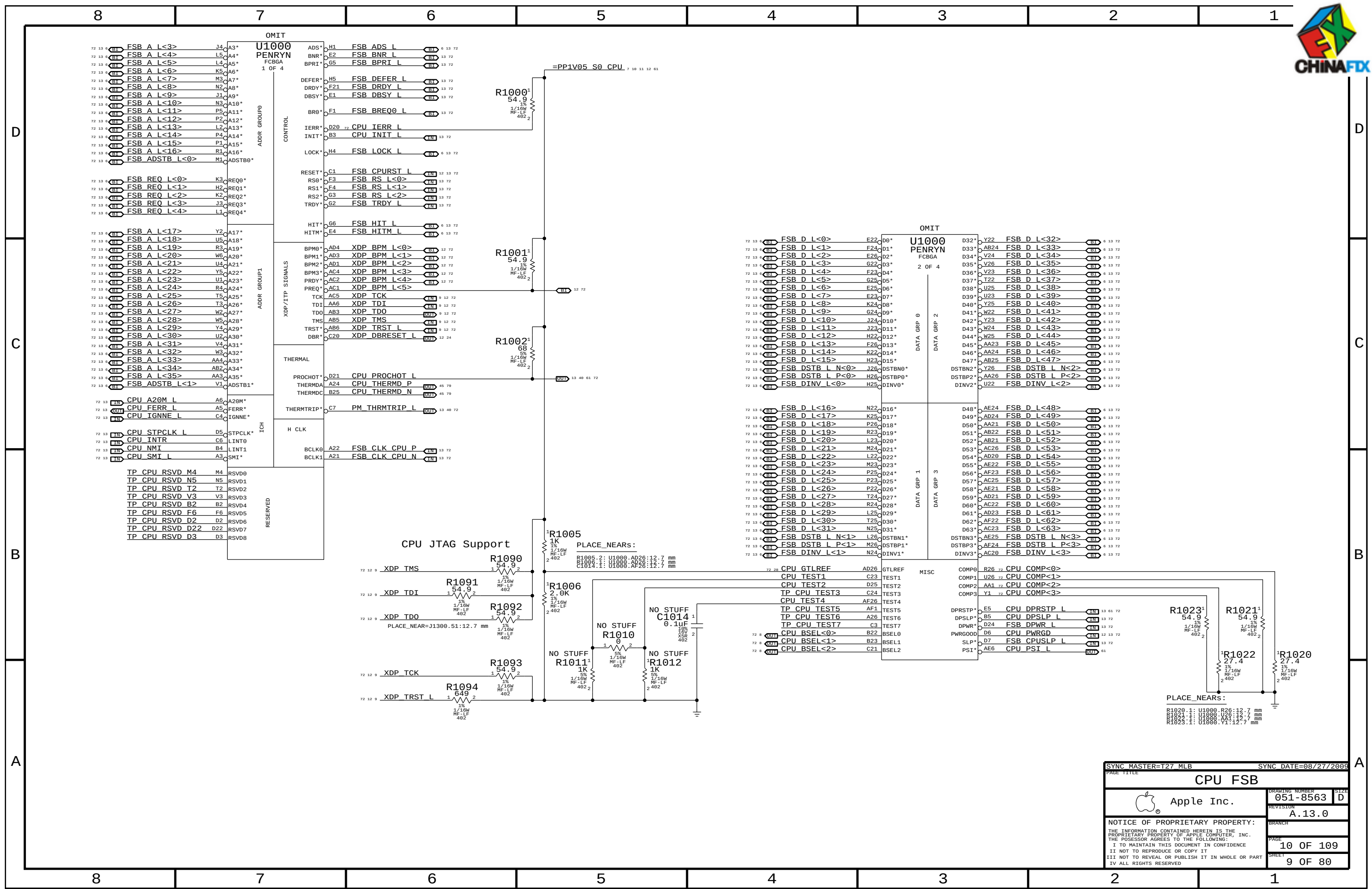
MCPCOREISNS SIGNAL

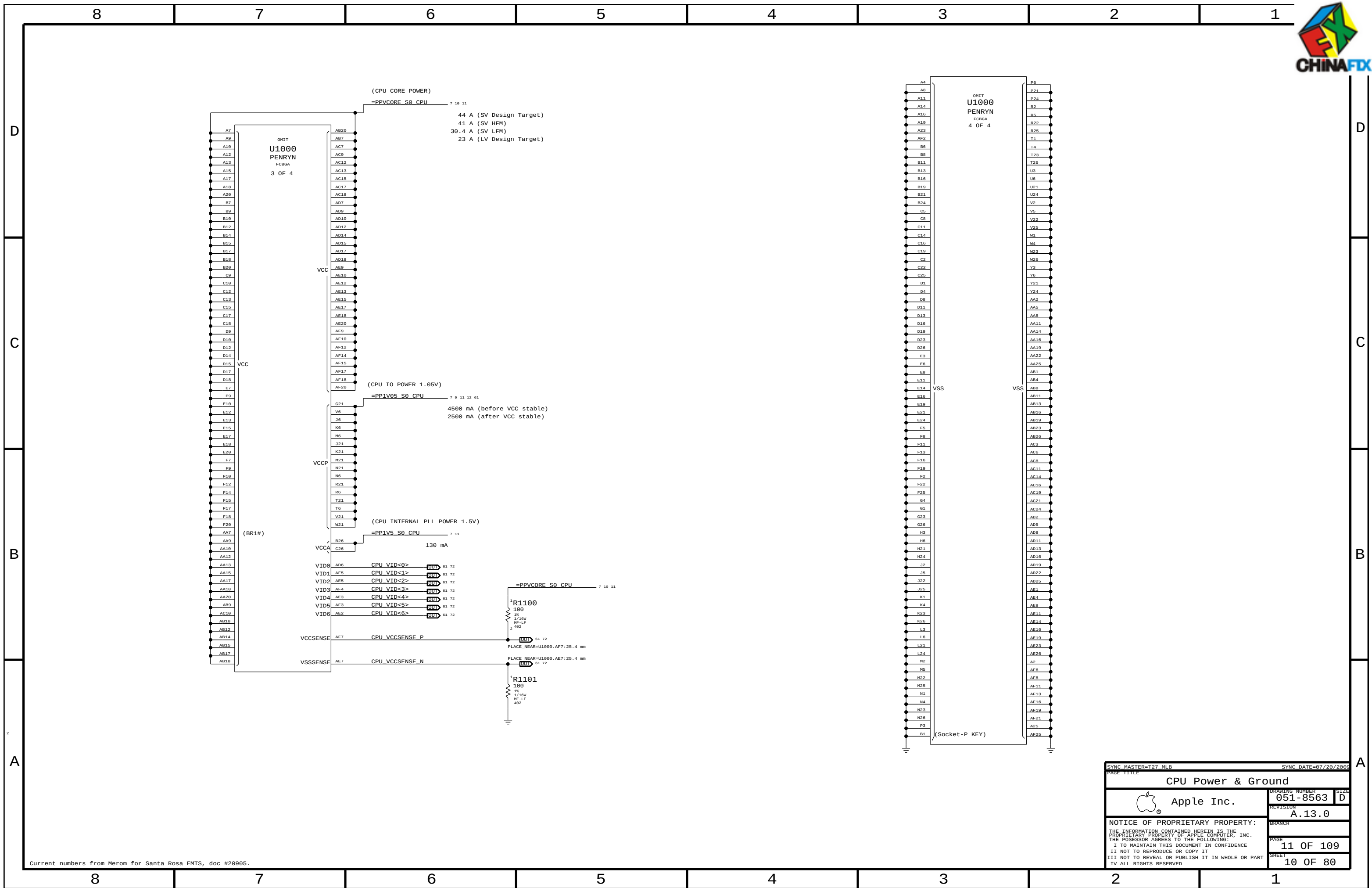
```

62 MCPCORES0 V0 == MCPCORESIS N == MCPCORESIS N 44
    MAKE_BASE=TRUE
62 MCPCORES0 ISP R == MCPCORESIS P == MCPCORESIS P 44

```

SYNC MASTER-K24 MLB	
PAGE TITLE	
SIGNAL ALIAS	
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	SHEET 8 OF 80





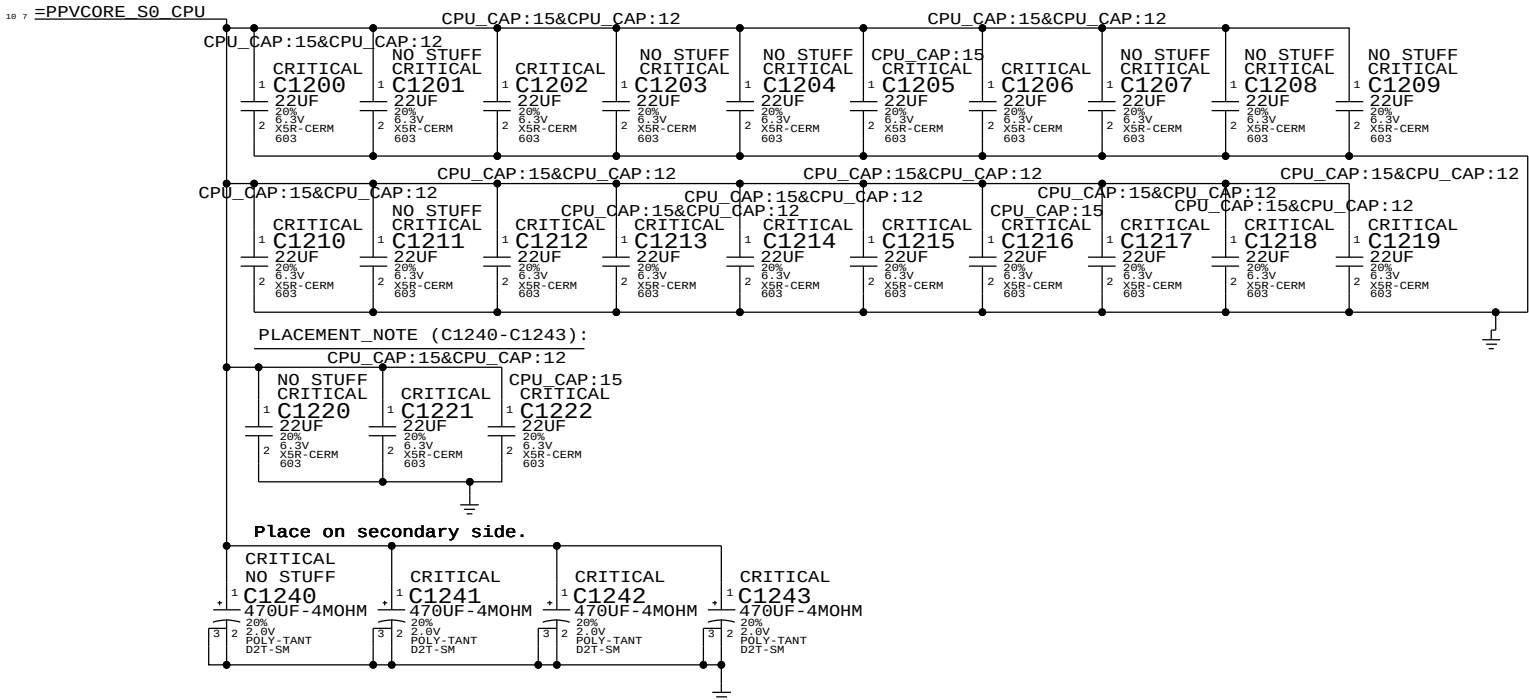
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PAGE TITLE		CPU Power & Ground	
Apple Inc.		DRAWING NUMBER	051-8563
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CPU VCore HF and Bulk Decoupling

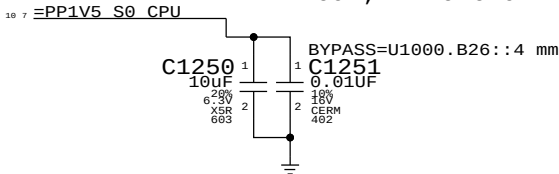
4X 330UF. 20X 22UF 0805

PLACEMENT_NOTE (C1200-C1219):
Place inside socket cavity on secondary side.



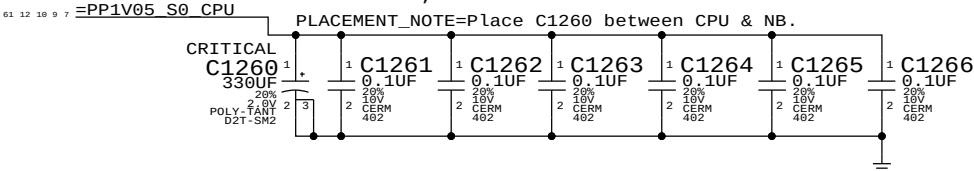
VCCA (CPU AVdd) DECOUPLING

1x 10uF, 1x 0.01uF



VCCP (CPU I/O) DECOUPLING

1x 330uF, 6x 0.1uF 0402



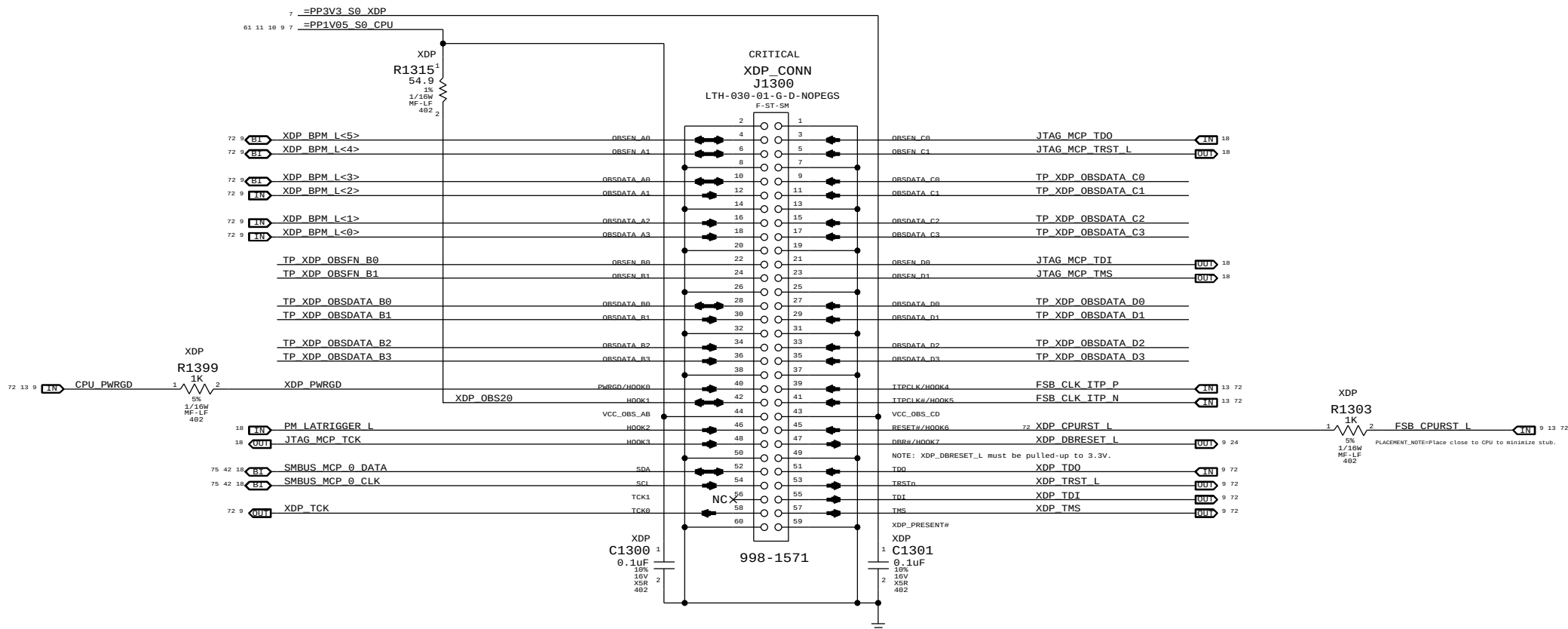
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PAGE TITLE		CPU Decoupling	
DRAWING NUMBER		051-8563	SIZE D
REVISION		A.13.0	BRANCH
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Mini-XDP Connector

NOTE: This is not the standard XDP pinout.
Use with 920-0620 adapter board to support CPU, MCP debugging.

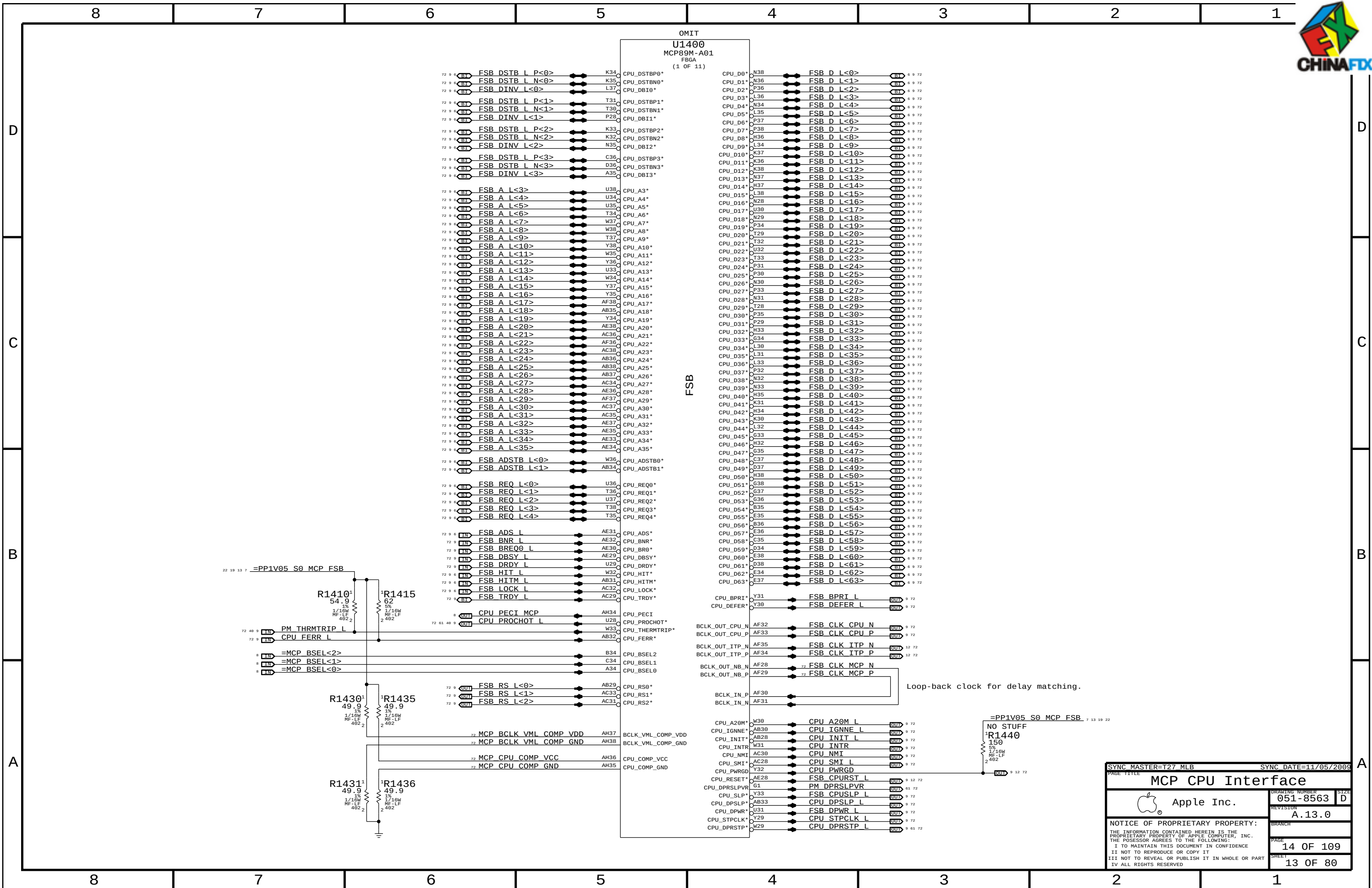
MCP89-specific pinout

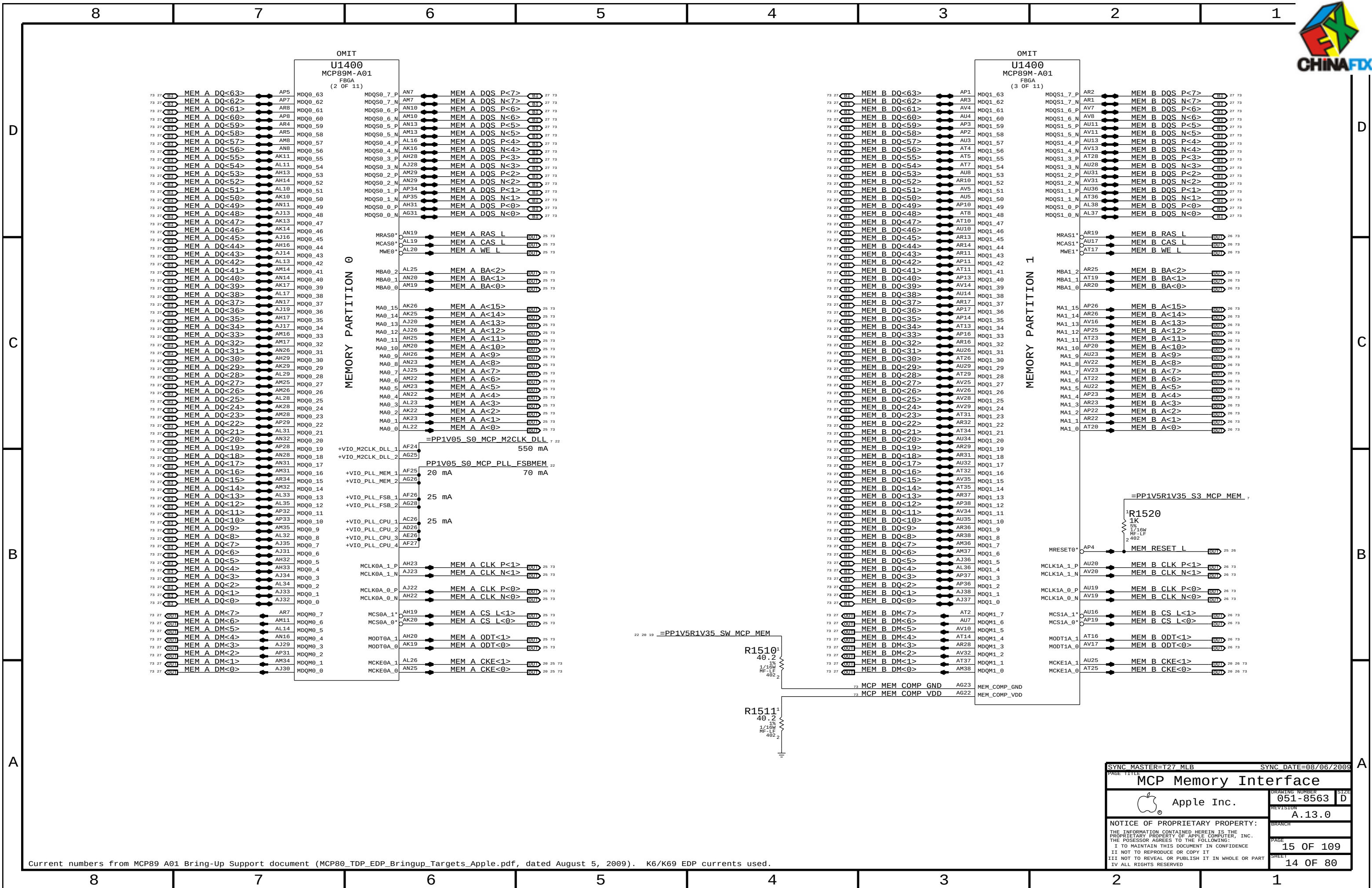


Direction of XDP module

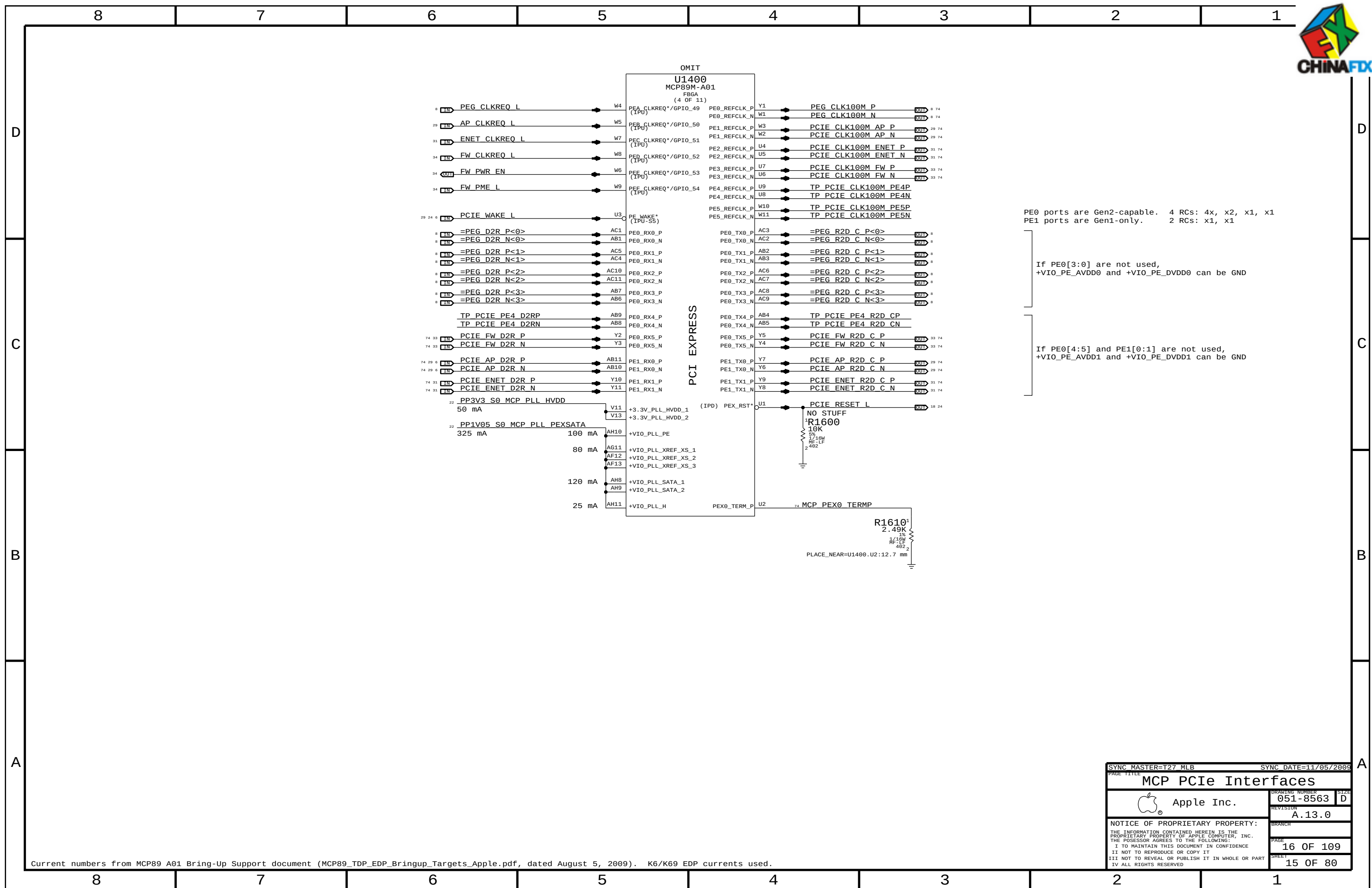
Please avoid any obstructions
on even-numbered side of J1300

SYNC MASTER=T27_MLB		SYNC DATE=07/28/2009	
PAGE TITLE		extended Debug Port (mini-XDP)	
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		SHEET	12 OF 80





SYNC MASTER=T27 MLB		SYNC DATE=08/06/2009	
PAGE TITLE		MCP Memory Interface	
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Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

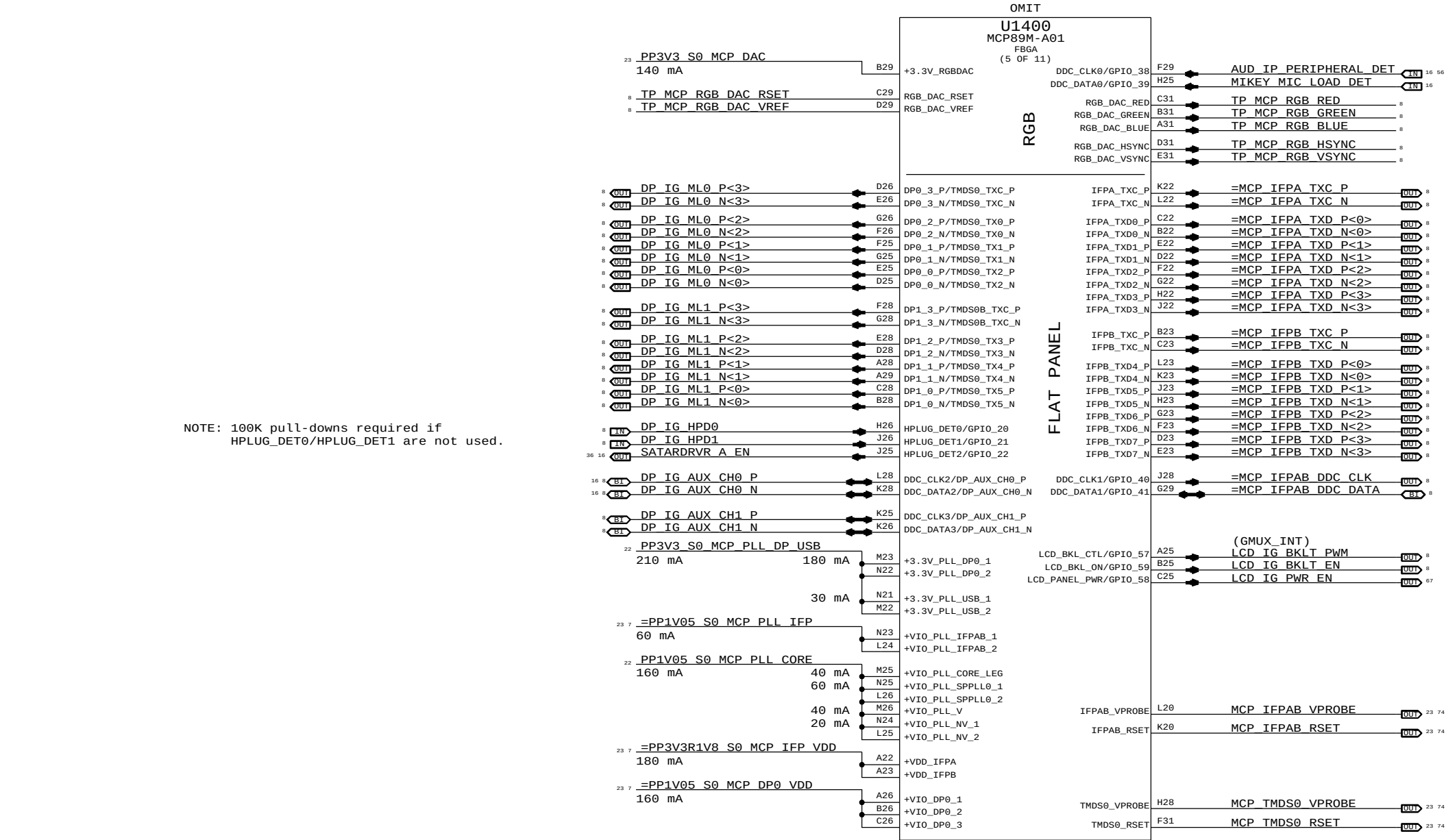
SYNC MASTER=T27 MLB		SYNC DATE=11/05/2009	
PAGE TITLE			
MCP PCIe Interfaces			
 Apple Inc.		DRAWING NUMBER	051-8563
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		SHEET	15 OF 80

D

C

B

A



NOTE: 100K pull-downs required if
HPLUG_DET0/HPLUG_DET1 are not used.

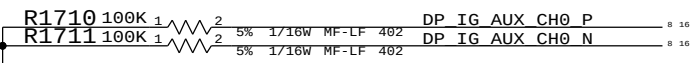
RGB DAC Disable:
Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required (or use as GPIOs).
Connect +3.3V_RGBDAC pin to GND.
NOTE: No Composite/S-Video/Component Video support on MCP89

Interface Mode		
MCP Signal	TMDS/HDMI	LVDS
=MCP_IFPA_TXC_P/N	TMDS_IG_TXC_P/N	LVDS_IG_A_CLK_P/N
=MCP_IFPA_TXD_P/N<0>	TMDS_IG_TXD_P/N<0>	LVDS_IG_A_DATA_P/N<0>
=MCP_IFPA_TXD_P/N<1>	TMDS_IG_TXD_P/N<1>	LVDS_IG_A_DATA_P/N<1>
=MCP_IFPA_TXD_P/N<2>	TMDS_IG_TXD_P/N<2>	LVDS_IG_A_DATA_P/N<2>
=MCP_IFPA_TXD_P/N<3>	TMDS_IG_TXD_P/N<3>	LVDS_IG_A_DATA_P/N<3>
=MCP_IFPB_TXC_P/N	TMDS_IG_TXD_P/N<3>	LVDS_IG_B_CLK_P/N
=MCP_IFPB_TXD_P/N<0>	TMDS_IG_TXD_P/N<4>	LVDS_IG_B_DATA_P/N<0>
=MCP_IFPB_TXD_P/N<1>	TMDS_IG_TXD_P/N<4>	LVDS_IG_B_DATA_P/N<1>
=MCP_IFPB_TXD_P/N<2>	TMDS_IG_TXD_P/N<5>	LVDS_IG_B_DATA_P/N<2>
=MCP_IFPB_TXD_P/N<3>	TMDS_IG_TXD_P/N<5>	LVDS_IG_B_DATA_P/N<3>
=MCP_IFPAB_DDC_CLK	TMDS_IG_DDC_CLK	LVDS_IG_DDC_CLK
=MCP_IFPAB_DDC_DATA	TMDS_IG_DDC_DATA	LVDS_IG_DDC_DATA

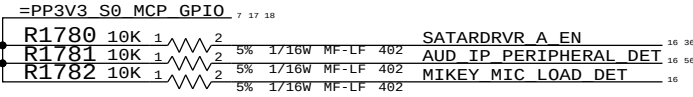
LVDS: Power +VDD_IFPx at 1.8V
TMDS: Power +VDD_IFPx at 3.3V

DDC Mode Pull-downs

NOTE: DP_AUX_CH1 also requires pull-downs if used for dual-mode DisplayPort (DP++). If unused no pulls are necessary, if used for TMDS/HDMI only then only pull-ups are necessary.



GPIO Pull-Ups



Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

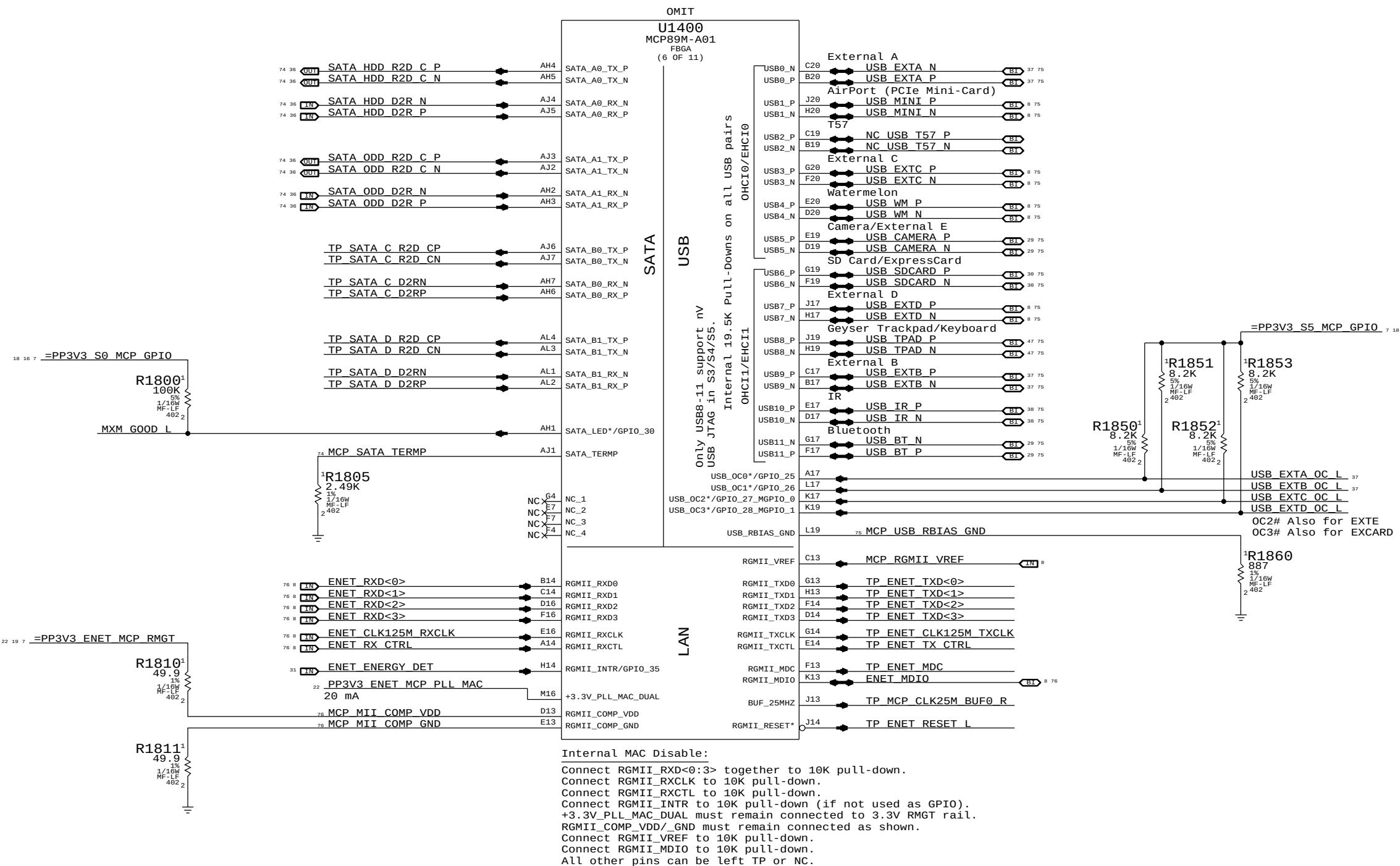
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Apple Inc.		DRAWING NUMBER	051-8563
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		SHEET	16 OF 80

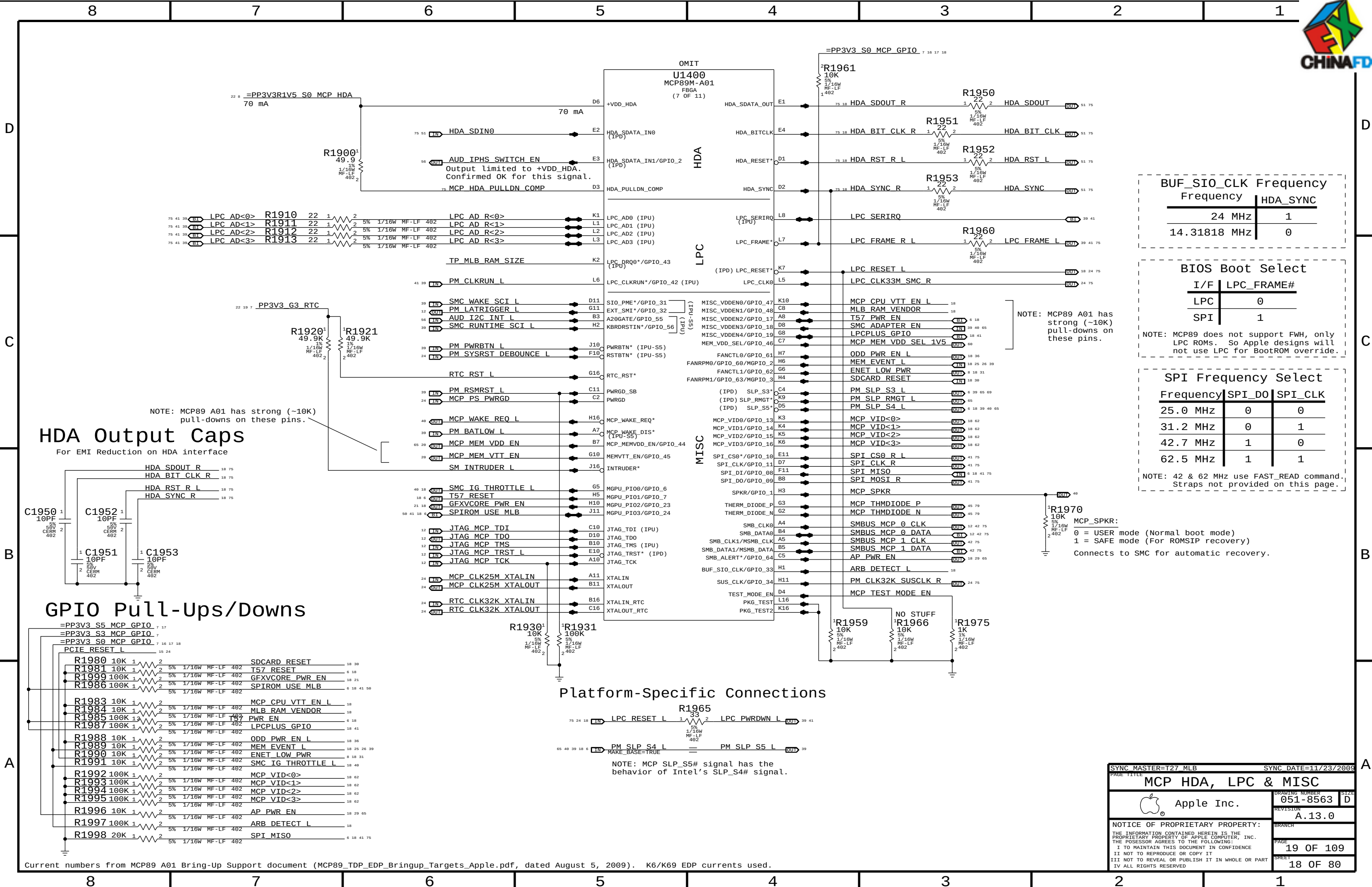
D

C

B

A





BUF_SIO_CLK Frequency

Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

BIOS Boot Select

I/F	LPC_FRAME#
LPC	0
SPI	1

SPI Frequency Select

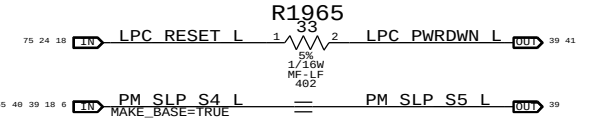
Frequency	SPI_D0	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
42.7 MHz	1	0
62.5 MHz	1	1

NOTE: 42 & 62 MHz use FAST_READ command. Straps not provided on this page.

NOTE: MCP89 A01 has strong (~10K) pull-downs on these pins.

MCP_SPKR:
0 = USER mode (Normal boot mode)
1 = SAFE mode (For ROMSIP recovery)
Connects to SMC for automatic recovery.

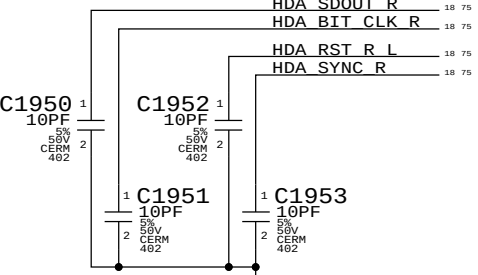
Platform-Specific Connections



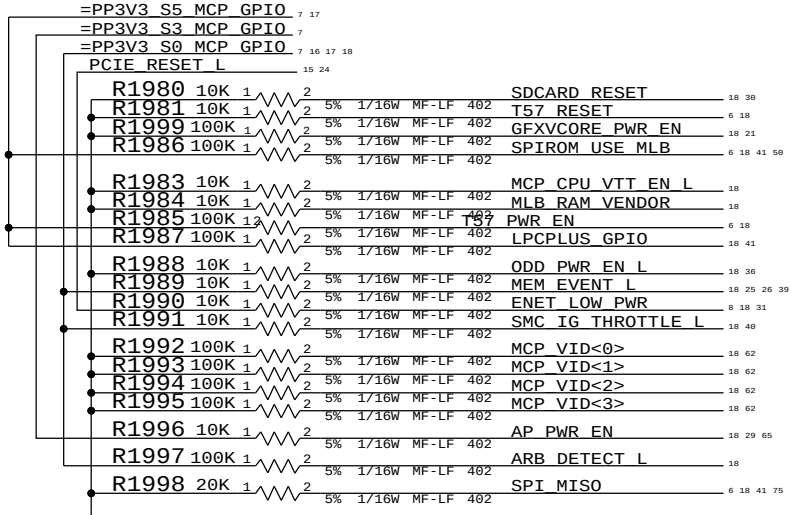
NOTE: MCP SLP_S5# signal has the behavior of Intel's SLP_S4# signal.

HDA Output Caps

For EMI Reduction on HDA interface

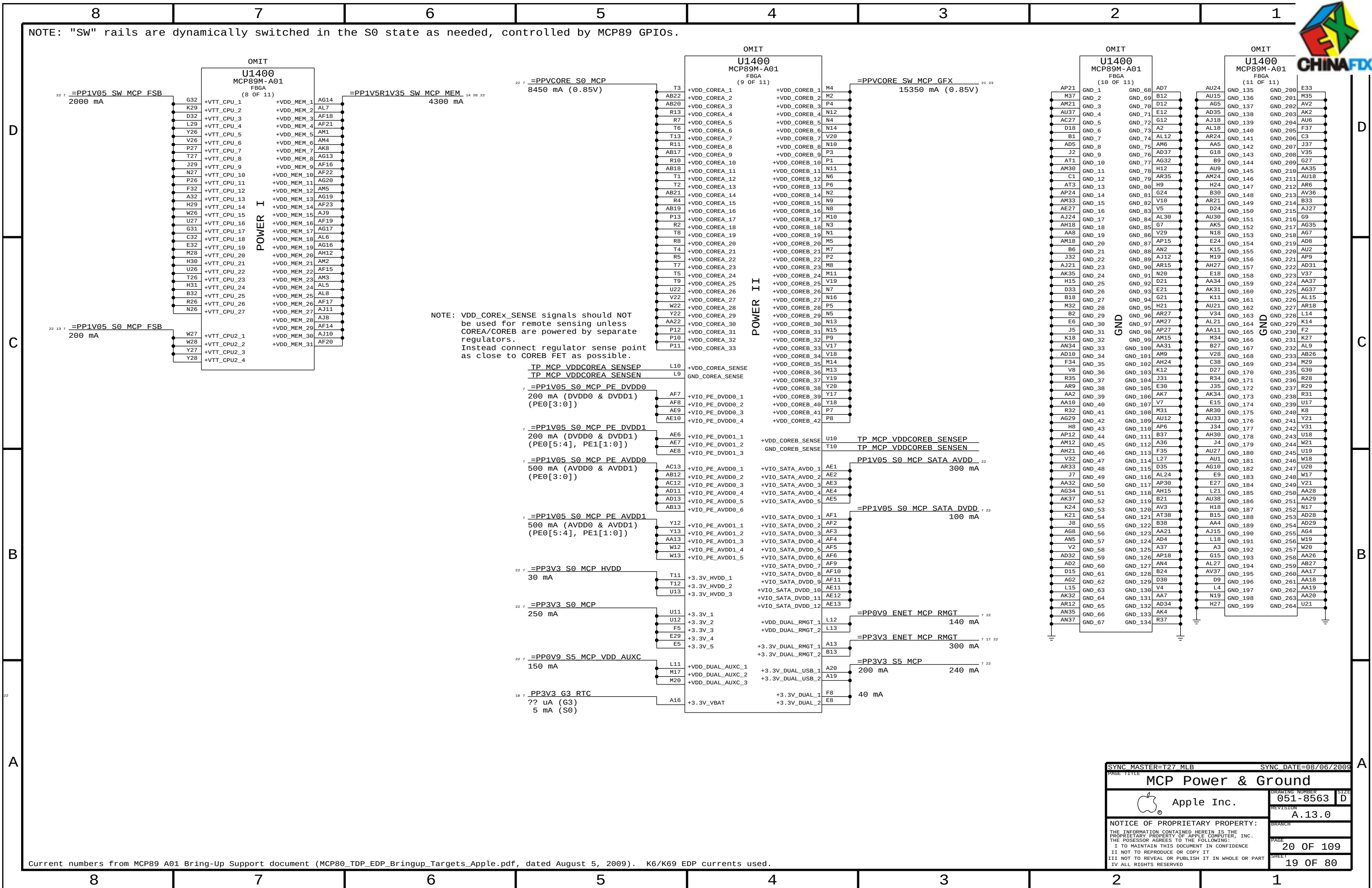


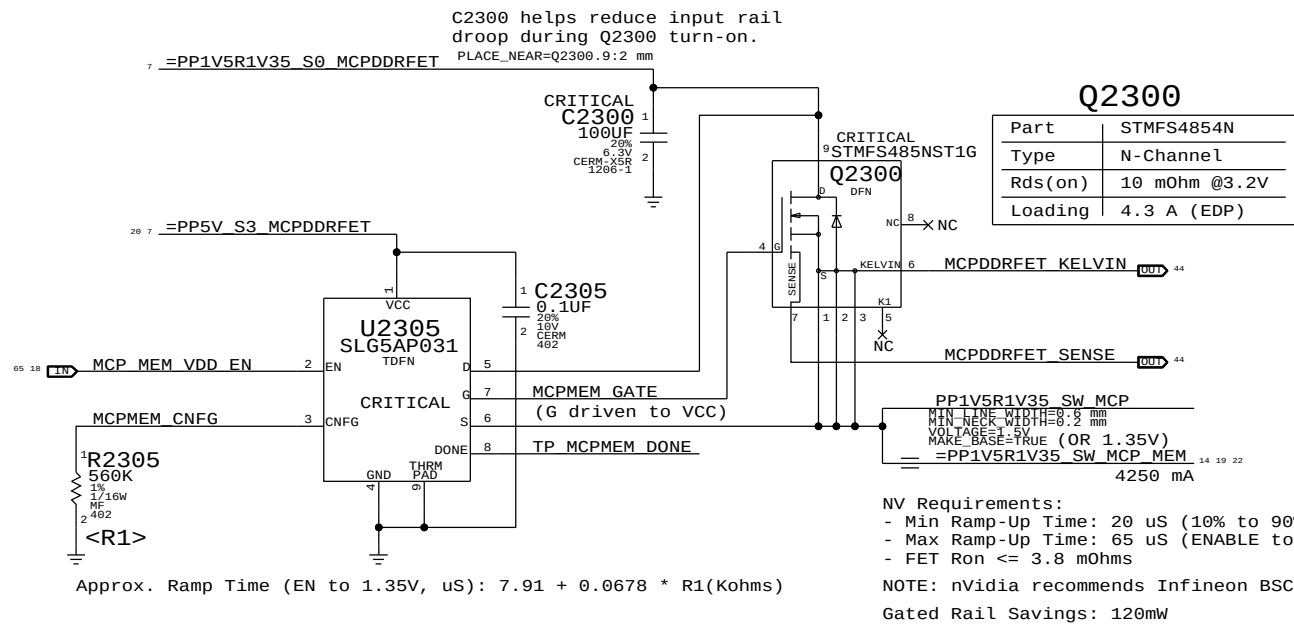
GPIO Pull-Ups/Downs



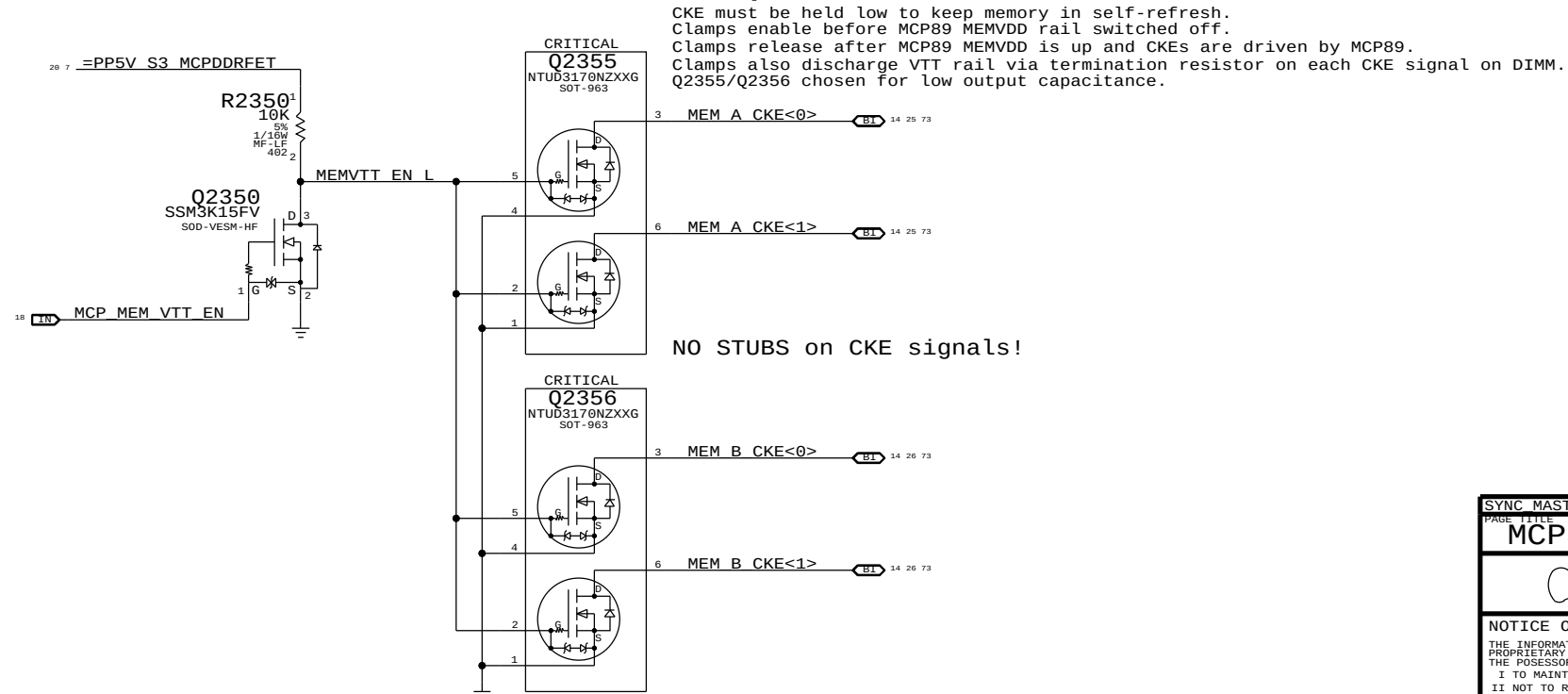
Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

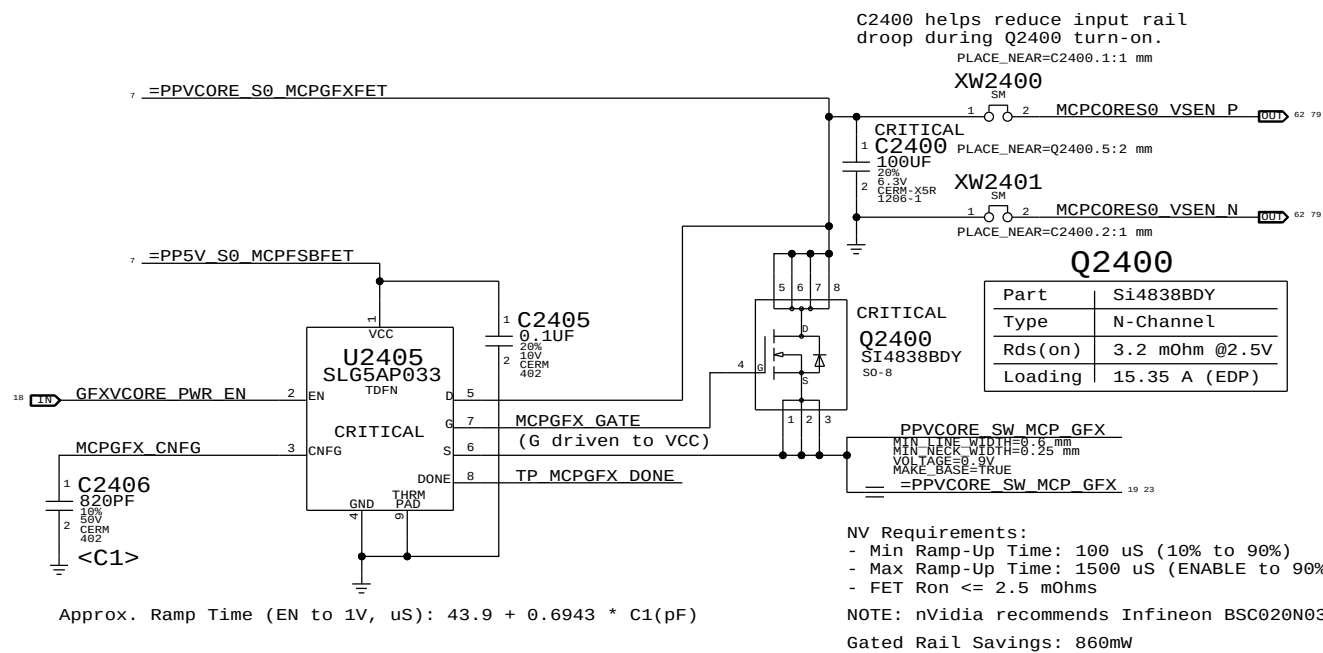
SYNC MASTER=T27 MLB		SYNC DATE=11/23/2009	
PAGE TITLE			
MCP HDA, LPC & MISC			
Apple Inc.		DRAWING NUMBER	051-8563
		REVISION	A.13.0
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PAGE		19 OF 109	
SHEET		18 OF 80	

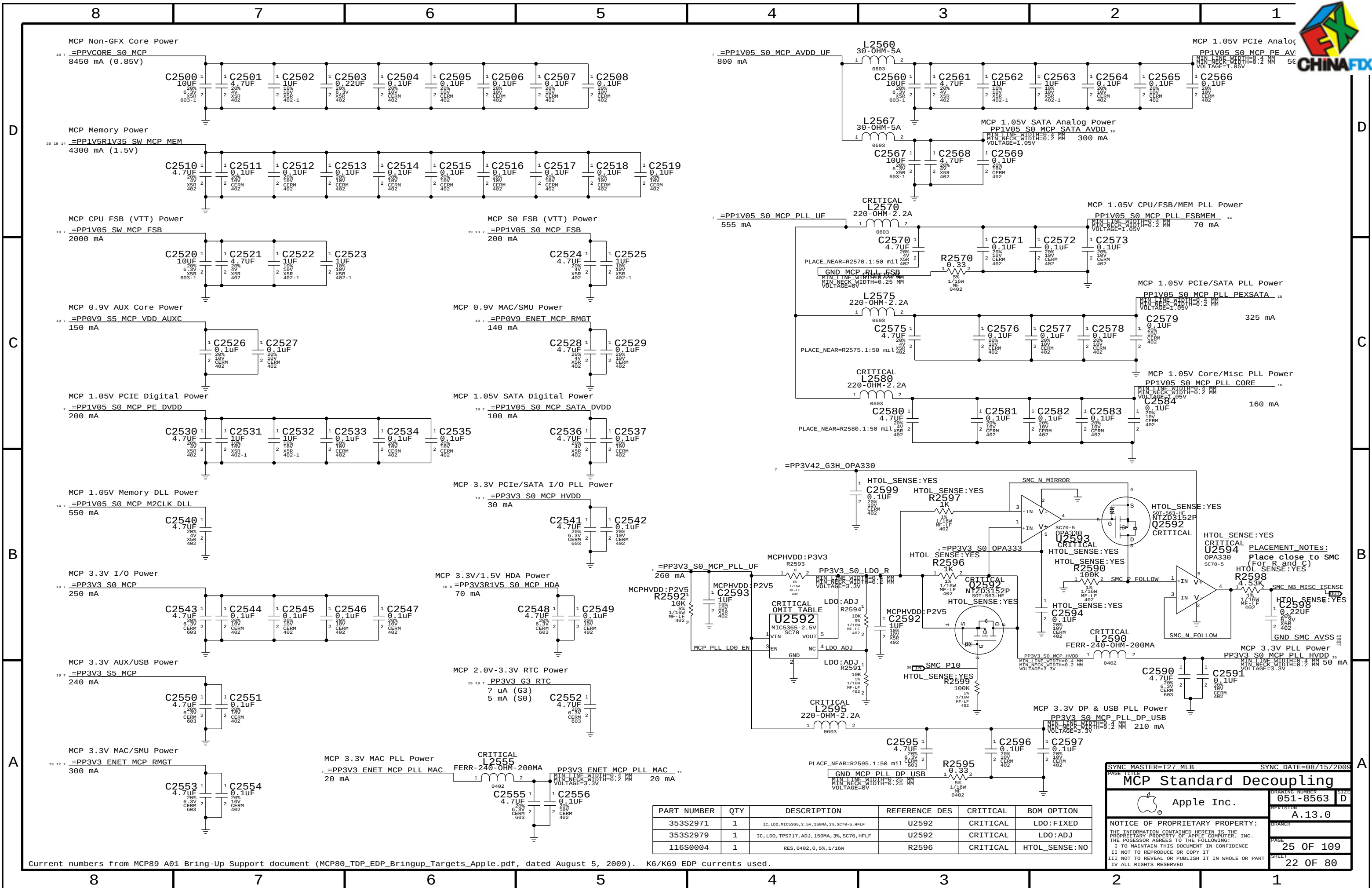


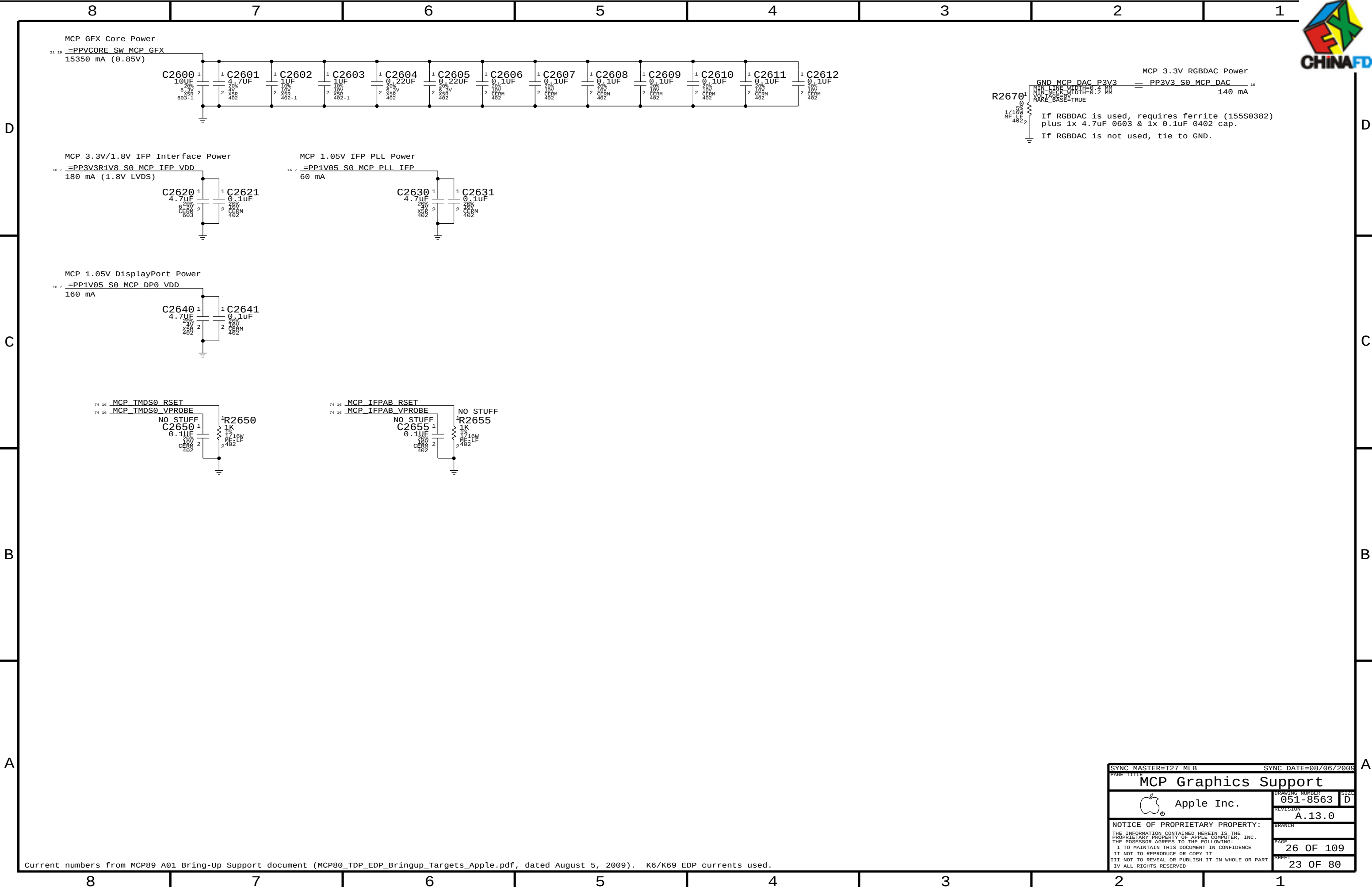


DIMM CKE Clamps







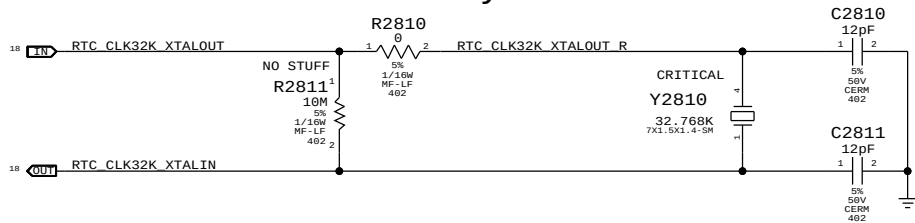


Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

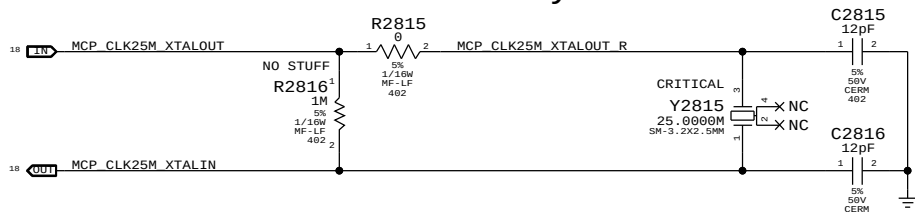
SYNC_MASTER=T27_MLB		SYNC_DATE=08/06/2009	
PAGE TITLE			
MCP Graphics Support			
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		REVISION	A.13.0
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BRANCH		PAGE	26 OF 109
SHEET		23 OF 80	



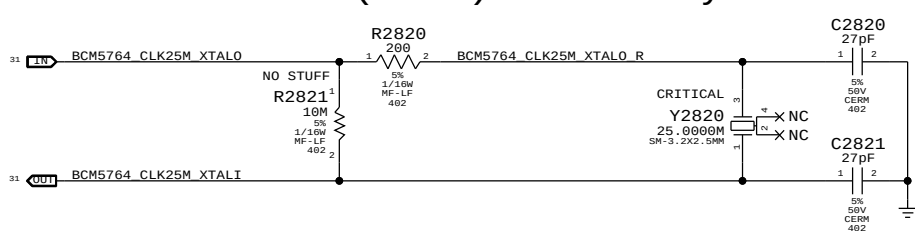
RTC Crystal



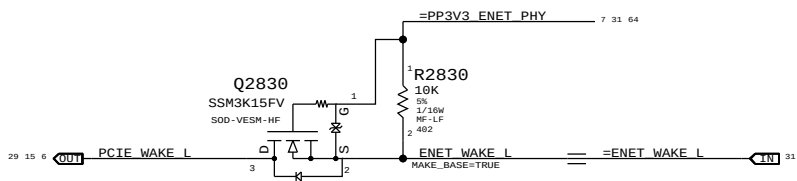
MCP 25MHz Crystal



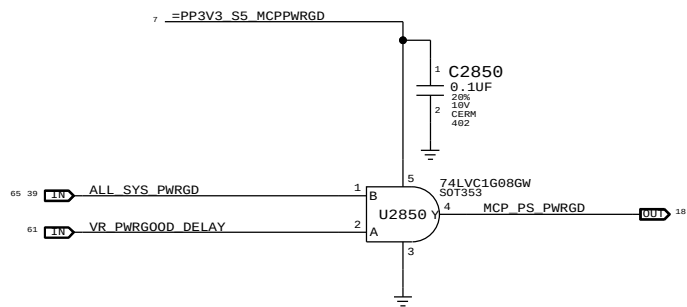
Caesar II (ENET) 25MHz Crystal



Ethernet WAKE# Isolation

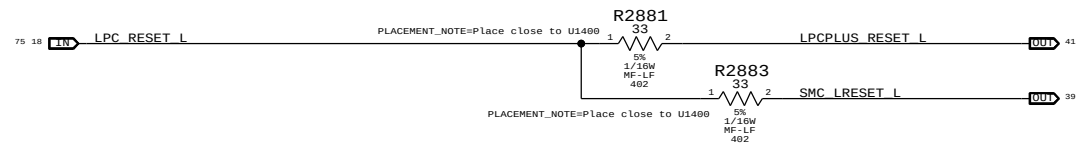


MCP S0 PWRGD & CPU_VLD

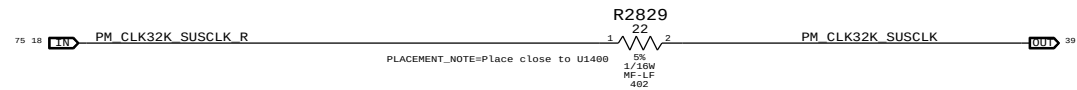
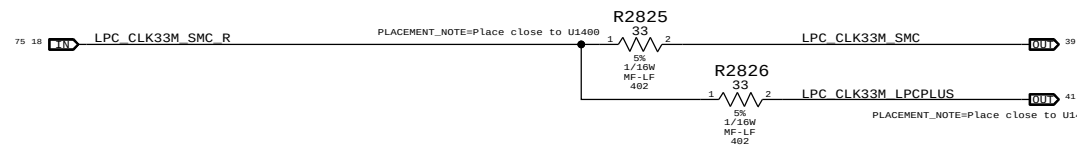
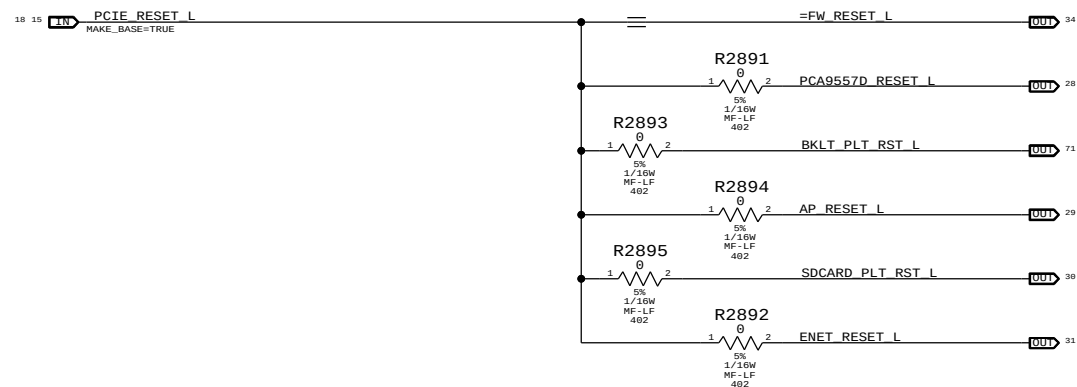


Platform Reset Connections

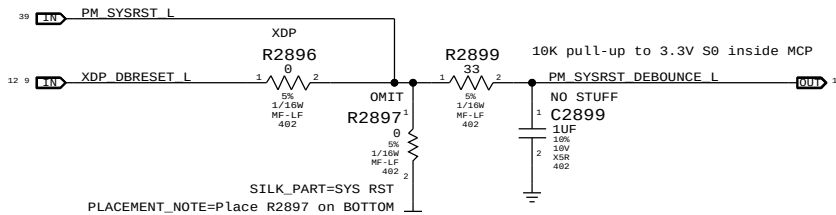
LPC Reset (Unbuffered)



PCIE Reset (Unbuffered)



System Reset Circuit



PAGE TITLE		PAGE TITLE	
SB Misc		SB Misc	
Apple Inc.		Apple Inc.	
DRAWING NUMBER		DRAWING NUMBER	
051-8563		051-8563	
REVISION		REVISION	
A.13.0		A.13.0	
BRANCH		BRANCH	
PAGE		PAGE	
28 OF 109		28 OF 109	
SHEET		SHEET	
24 OF 80		24 OF 80	



Page Notes

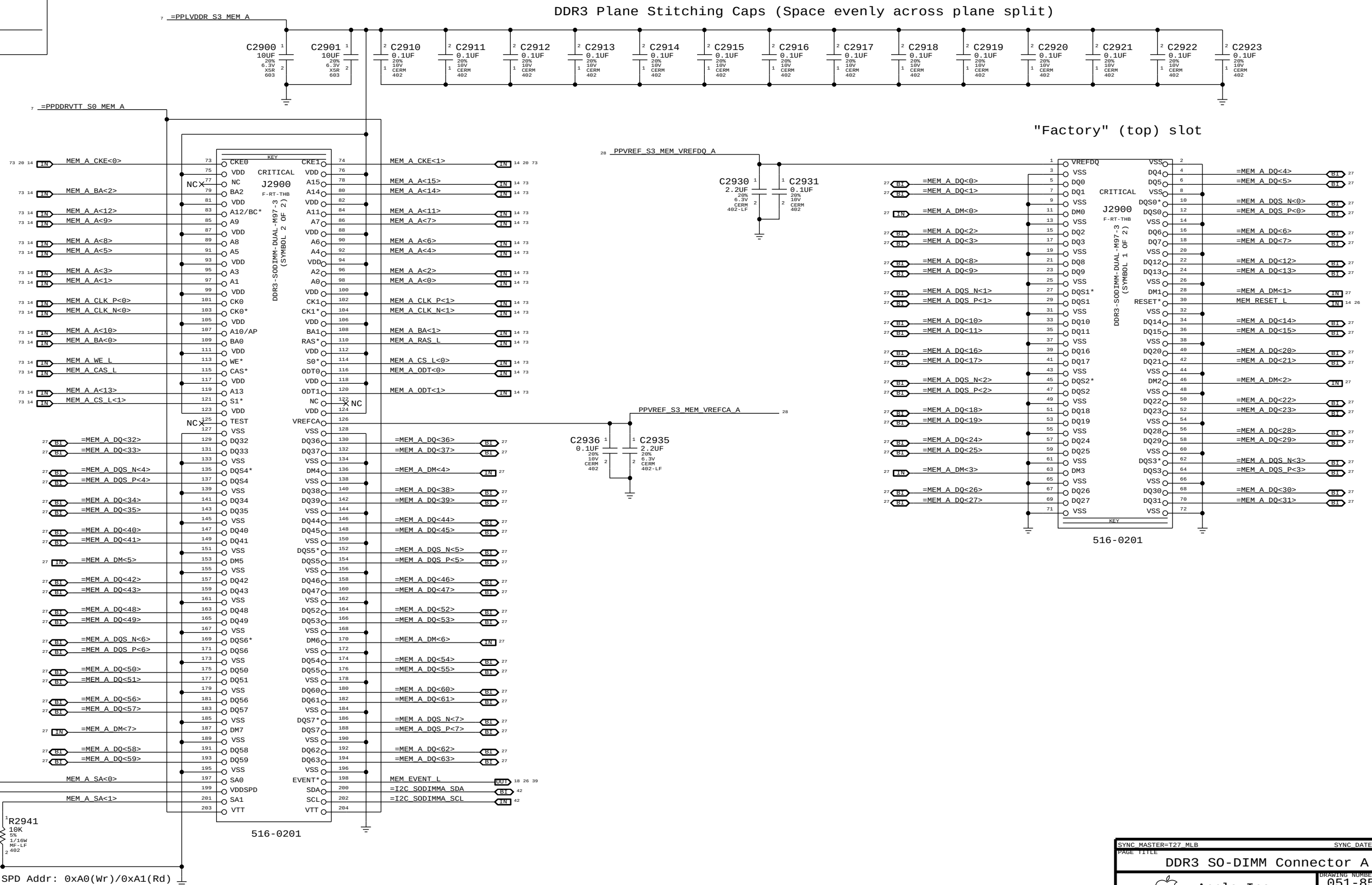
Power aliases required by this page:

- =PPLVDDR_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

```
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA
```

BOM options provided by this page:
(NONE)



SYNC MASTER-T27 MLB		SYNC DATE=07/28/2005	
PAGE TITLE			
DDR3 SO-DIMM Connector A			
		DRAWING NUMBER 051-8563	
Apple Inc.		SIZE D	
		REVISION A.13.0	
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		SHEET	
		25 OF 80	

Page Notes

Power aliases required by this page:

- =PPLVDDR_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

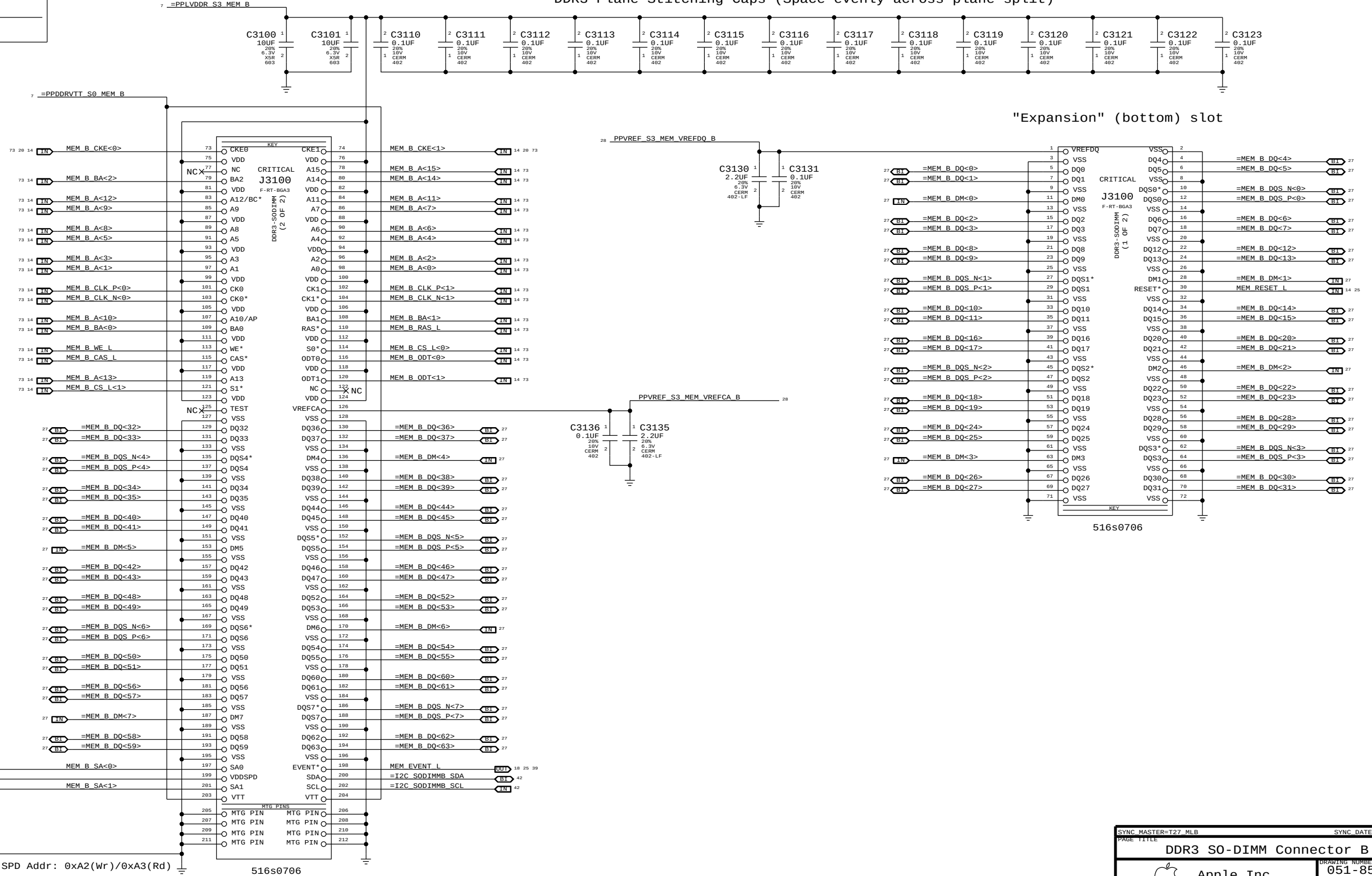
Signal aliases required by this page:

- =I2C_SODIMM_SCL
- =I2C_SODIMM_SDA

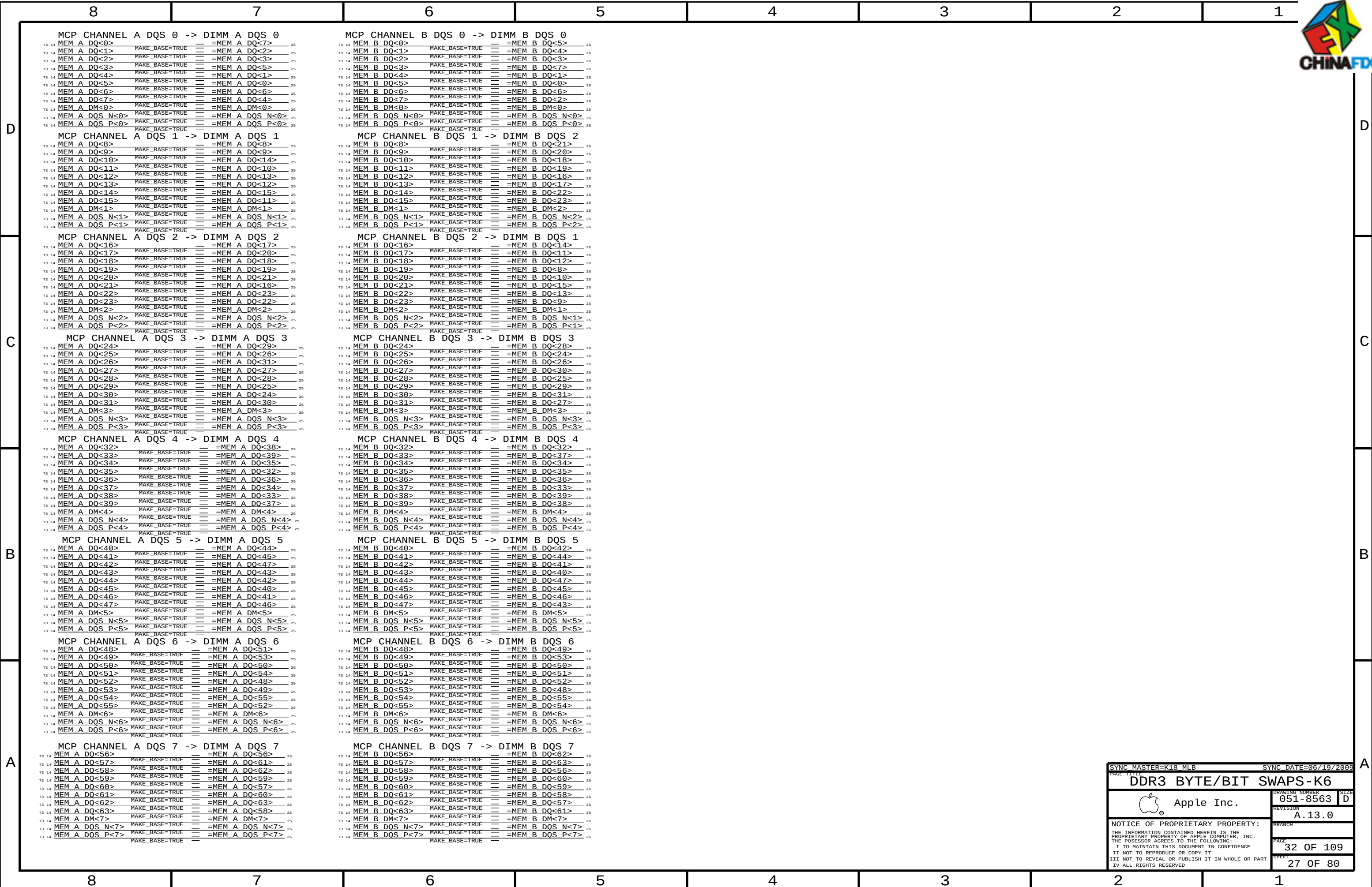
BOM options provided by this page:

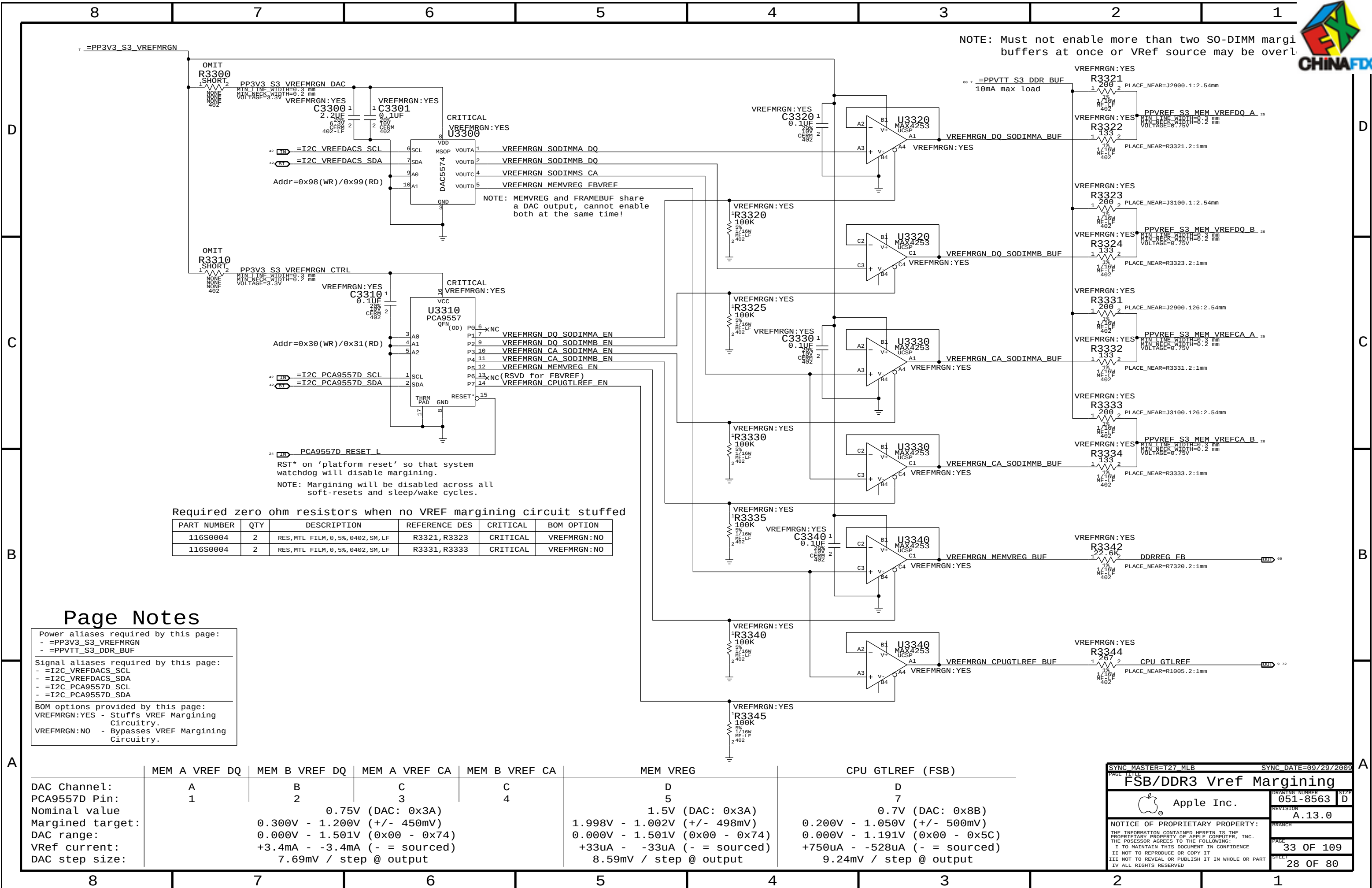
(NONE)

DDR3 Plane Stitching Caps (Space evenly across plane split)



SYNC MASTER=T27_MLB		SYNC DATE=07/28/2005	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	051-8563
		REVISION	A.13.0
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SHEET		26 OF 80	





Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

VREFMRGN:YES - Stuffs VREF Margining Circuitry.

VREFMRGN:NO - Bypasses VREF Margining Circuitry.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3321,R3323	CRITICAL	VREFMRGN:NO
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3331,R3333	CRITICAL	VREFMRGN:NO

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	CPU GTLREF (FSB)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	7
Nominal value	0.75V (DAC: 0x3A)				1.5V (DAC: 0x3A)	0.7V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)				1.998V - 1.002V (+/- 498mV)	0.200V - 1.050V (+/- 500mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)				0.000V - 1.501V (0x00 - 0x74)	0.000V - 1.191V (0x00 - 0x5C)
VRef current:	+3.4mA - -3.4mA (- = sourced)				+33uA - -33uA (- = sourced)	+750uA - -528uA (- = sourced)
DAC step size:	7.69mV / step @ output				8.59mV / step @ output	9.24mV / step @ output

SYNCH MASTER=T27_MLB		SYNCH DATE=09/29/2009	
PAGE TITLE			
FSB/DDR3 Vref Margining			
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		SHEET 28 OF 80	



5V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	26 mOhm @4.5V
LOADING	0.8 A (EDP)

CRITICAL
518S0610
J3401
20347-325E-12
F-RT-SM

PCIE AP D2R P
PCIE AP D2R N
PCIE AP R2D P
PCIE AP R2D N

AIRPORT

CRITICAL
L3401
90-OHM-100MA
DLP115
SYM-VER-1

PCIE CLK100M AP P
PCIE CLK100M AP N
PCIE WAKE L

1000 mA peak
750 mA nominal max
PP5V WLAN
MIN LINE WIDTH=1 mm
MIN NECK WIDTH=0.5 mm
VOLTAGE=5V

L3404
FERR-120-OHM-1.5A
0402-LF

CRITICAL
C3422
0.1uF
20%
16V
X5R
402

CRITICAL
C3421
0.1uF
20%
16V
X5R
402

CRITICAL
C3420
10uF
20%
16V
X5R
805

CRITICAL
Q3450
TPCP8102

23V1K-SM
C3450
0.1uF
10%
16V
X5R
402

CRITICAL
C3451
0.033uF
10%
16V
X5R
402

CRITICAL
R3451
10K
5%
1/10W
MF-LF
402

CRITICAL
R3450
33K
5%
1/10W
MF-LF
402

D

D

C

C

B

B

A

A

**ALS
CAMERA**

PP5V S3 BTCAMERA F
I2C ALS SDA
I2C ALS SCL

USB CAMERA CONN P
USB CAMERA CONN N
USB BT CONN P
USB BT CONN N

CRITICAL
L3402
90-OHM
DLP0NS
SYM-VER-1

USB CAMERA P
USB CAMERA N

BLUETOOTH

CRITICAL
L3403
90-OHM
DLP0NS
SYM-VER-1

USB BT P
USB BT N

Supervisor & CLKREQ# Isolation

CRITICAL
R3440
100K
5%
1/10W
MF-LF
402

WF: Need pull-up?

AP_RESET_CONN_L

AP_CLKREQ_Q_L

PP5V WLAN F

CRITICAL
R3453
392K
1%
1/10W
MF-LF
402

P3V3WLAN VMON

DLY = 60 ms +/- 20%

CRITICAL
R3454
97.6K
1%
1/10W
MF-LF
402

CRITICAL
U3440
SLG4AP016V
TDFN

CRITICAL
C3440
0.1uF
20%
16V
X5R
402

AP RESET L
AP PWR EN
AP CLKREQ L

SYNC MASTER=T27_MLB		SYNC DATE=07/28/2009	
PAGE TITLE		RIGHT CLUTCH CONNECTOR	
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		REVISION	A.13.0
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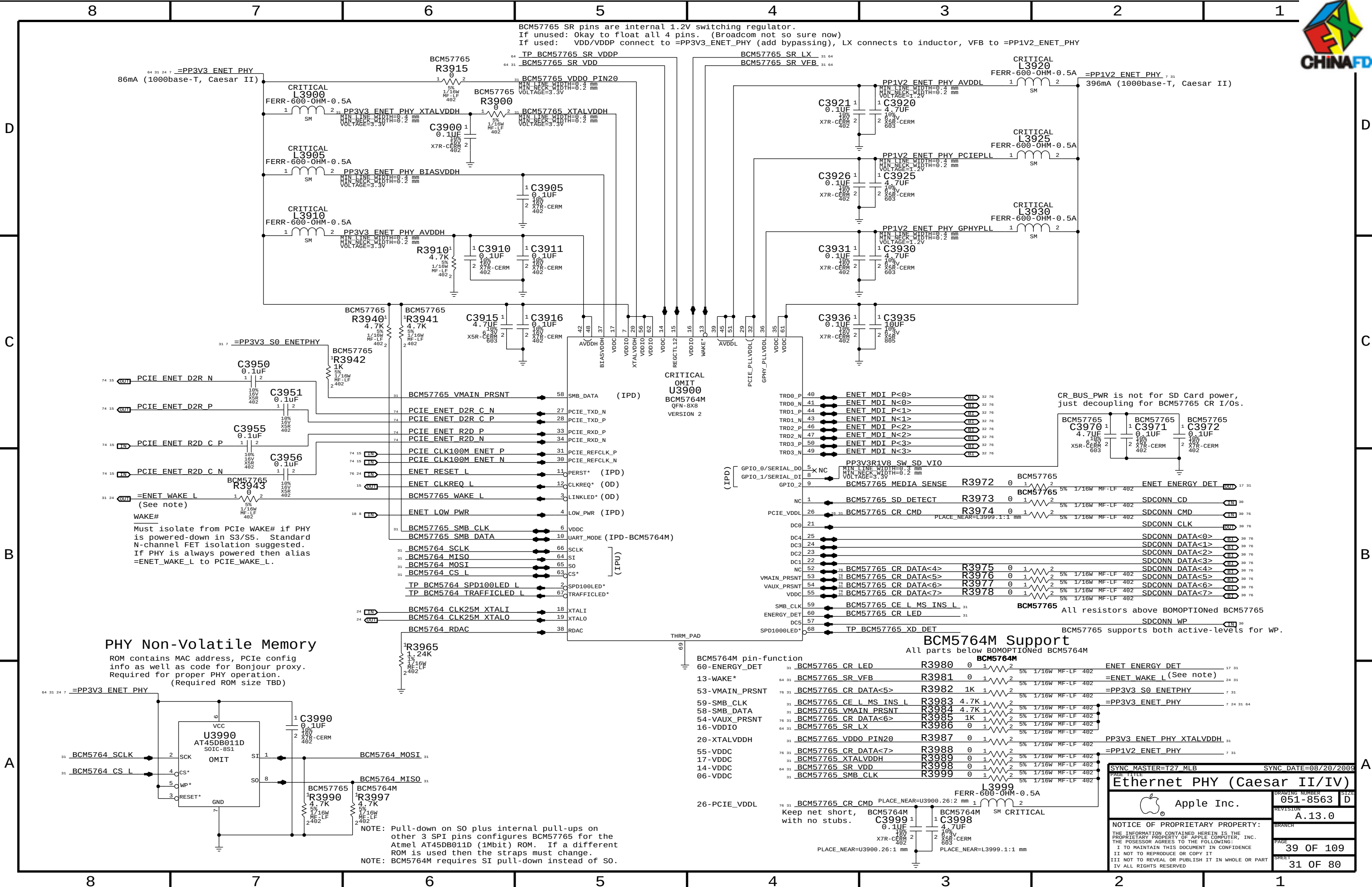


A B C D



A

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



SYNC MASTER=T27 MLB SYNC DATE=08/20/2009

Ethernet PHY (Caesar II/IV)

Apple Inc.

051-8563

REVISION A.13.0

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PAGE 39 OF 109

SHEET 31 OF 80

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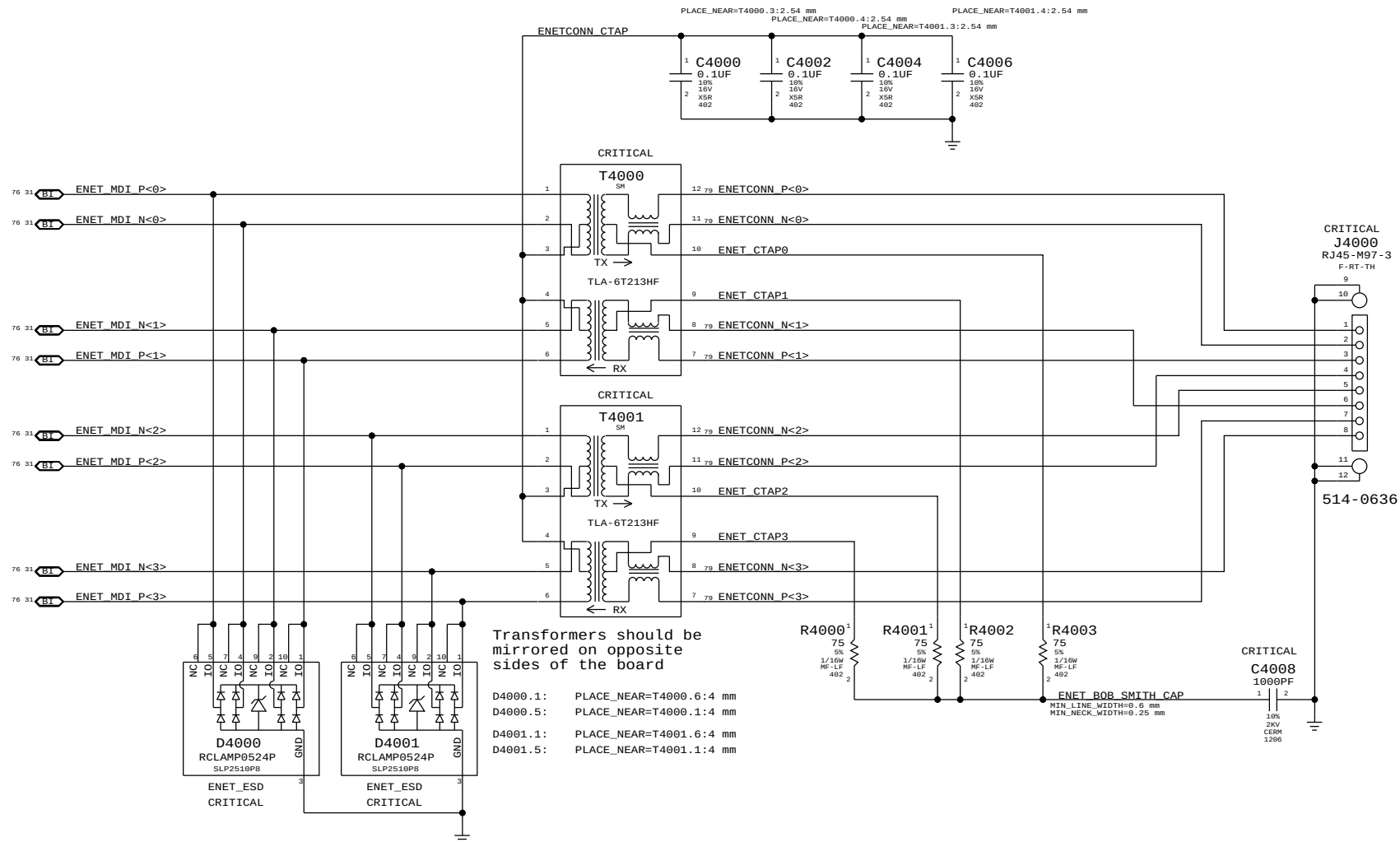


Page Notes

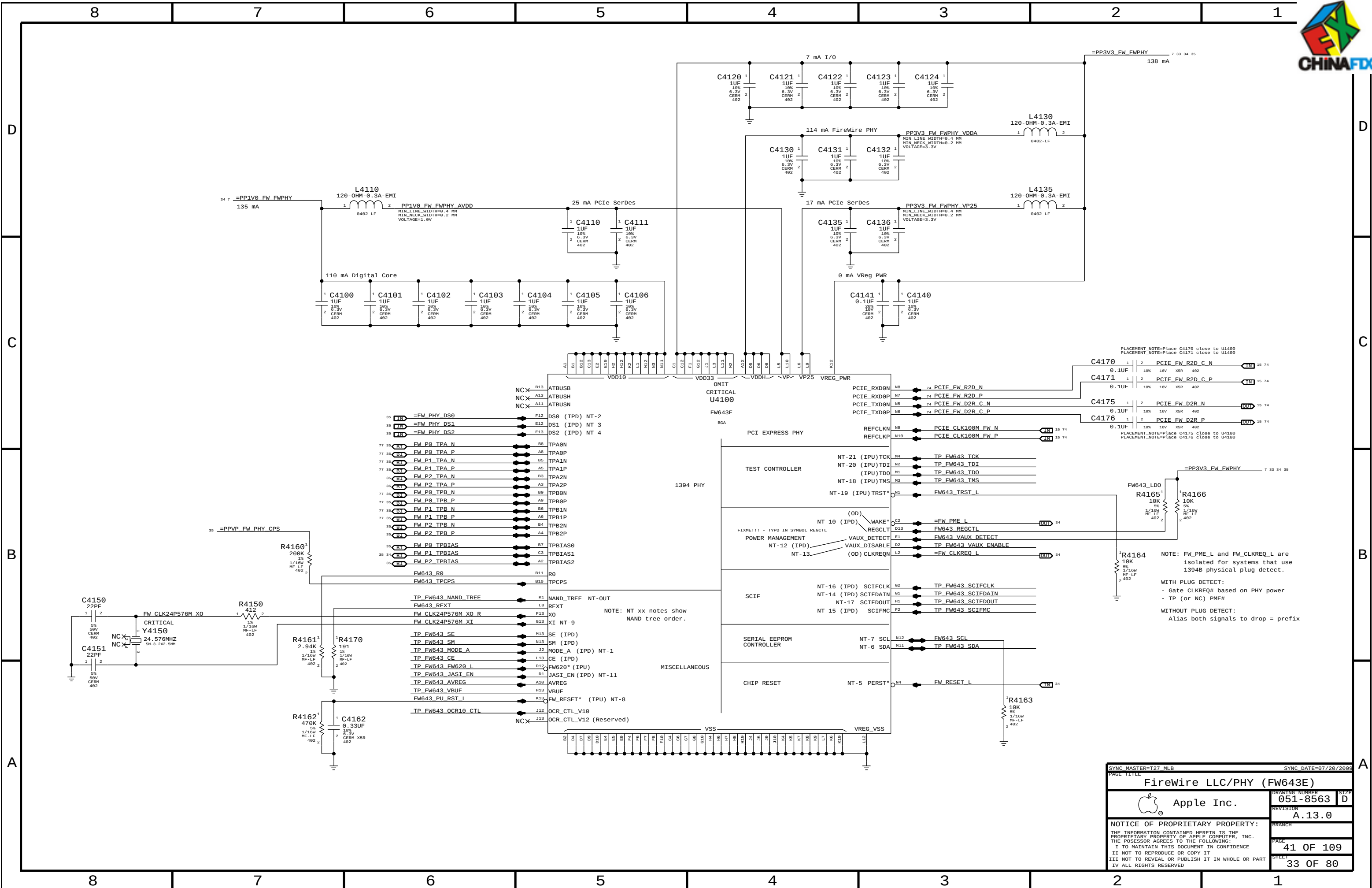
Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC MASTER=T27_MLB		SYNC DATE=07/28/2009	
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Ethernet Connector			
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		SIZE	D
		REVISION	A.13.0
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Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

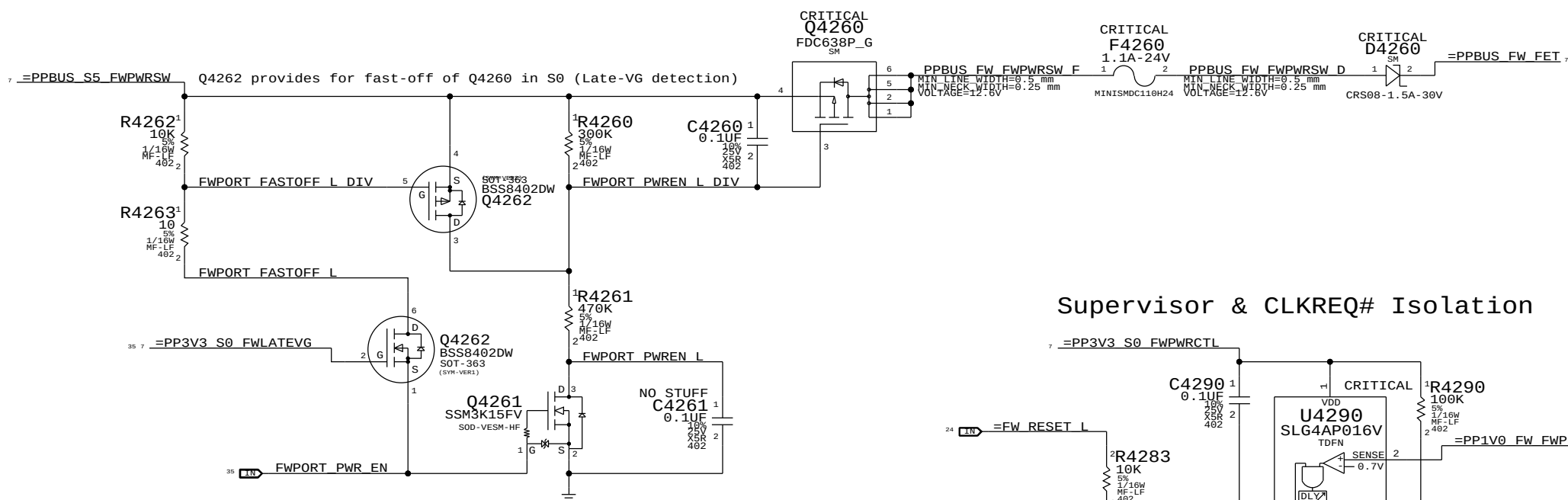
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

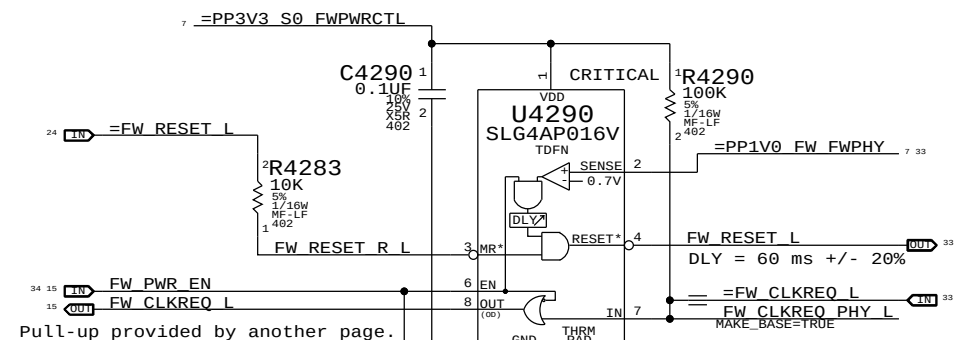
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

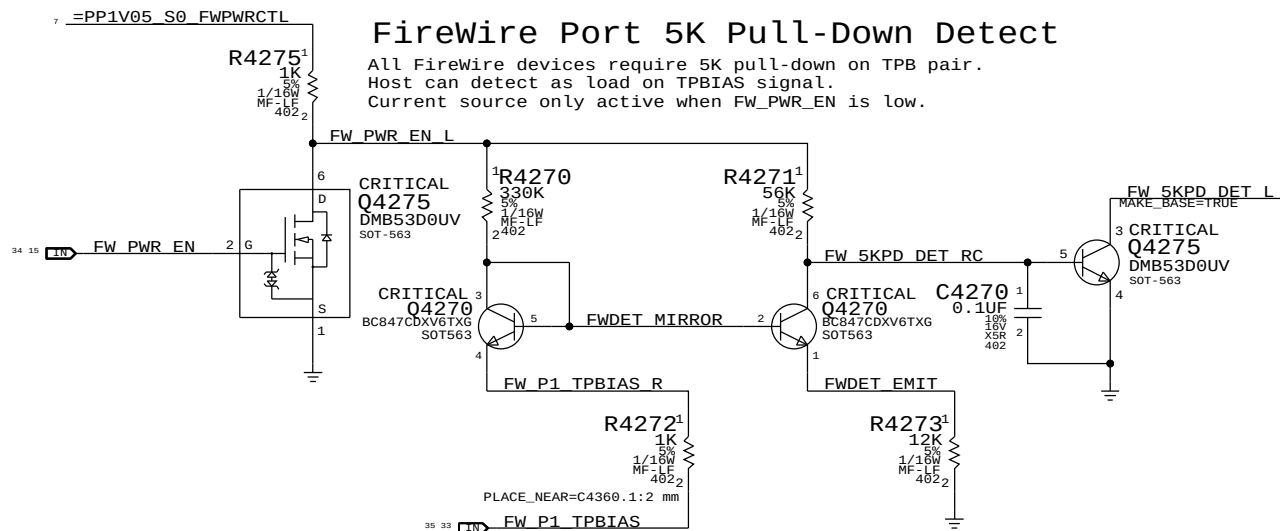


Supervisor & CLKREQ# Isolation



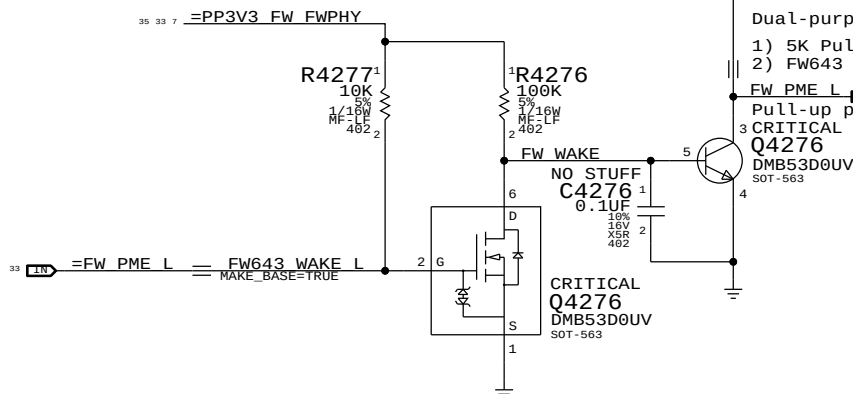
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.

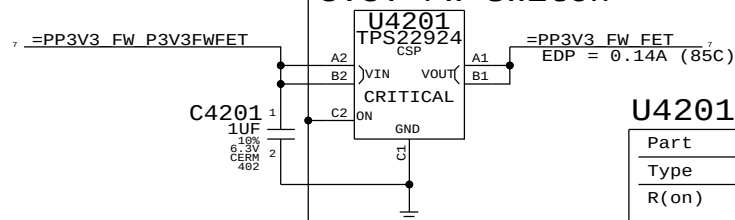


Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

3.3V FW Switch

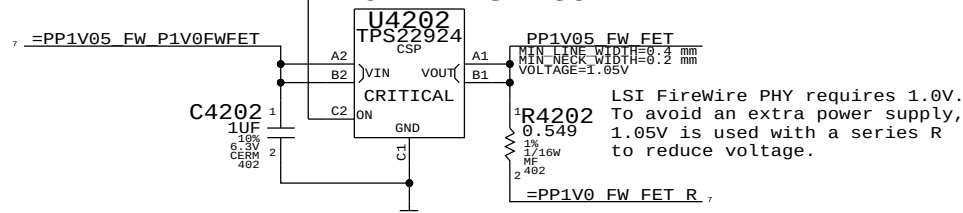


U4201 & U4202

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



TEXT NOTE FOR 3.3V RAIL CURRENT CHANGED TO EDP NUMBER.

SYNC MASTER=T27_MLB		SYNC DATE=12/15/2009	
PAGE 12/12		TITLE	
FireWire Port & PHY Power		DRAWING NUMBER	
Apple Inc.		051-8563	
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A

CANNOT SYNC THIS PAGE FROM T27, TPA AND TPB FOR U4350 IS SWAPPED

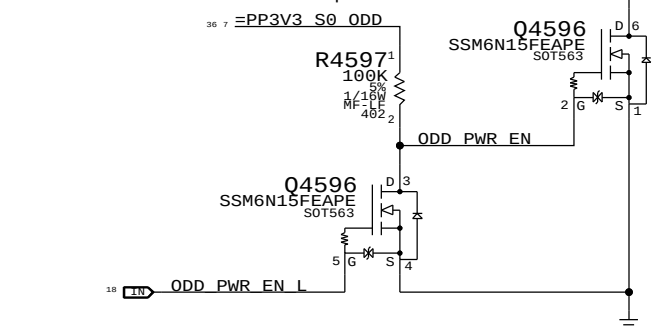
AREF needs to be isolated from all local grounds per 1394b spec

When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)

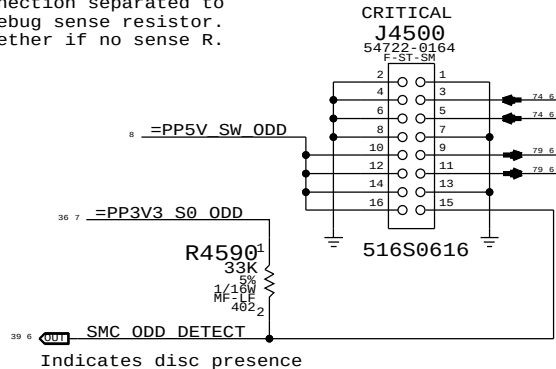
BREF should be hard-connected to logic ground for speed signaling and connection

ODD Power Control

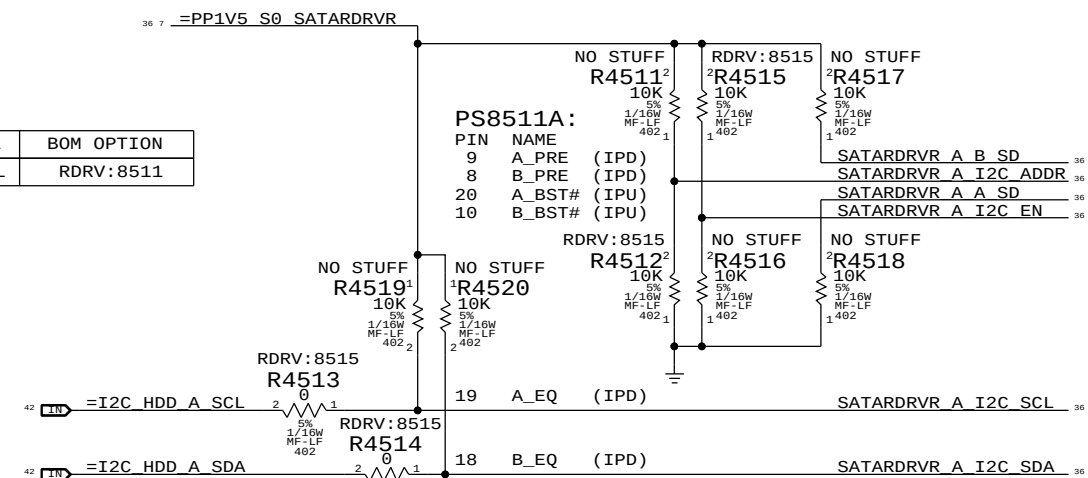
NOTE: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.



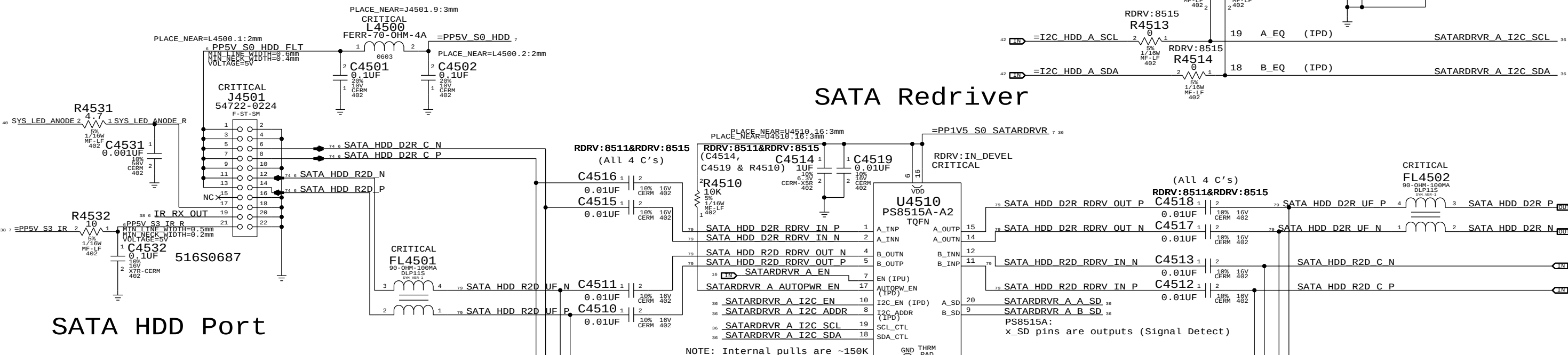
SATA ODD Port



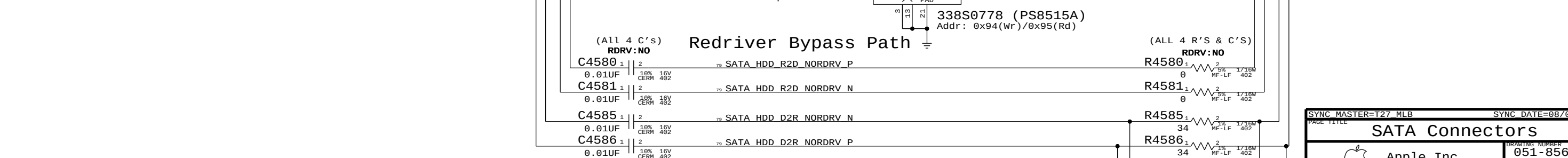
PS8511A / PS8515A Straps



SATA Redriver



SATA HDD Port



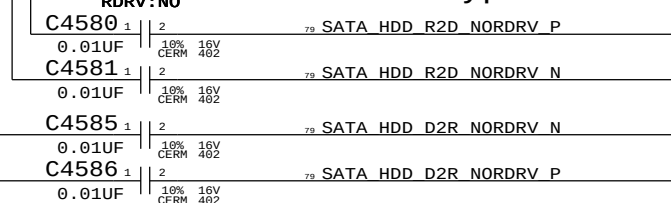
U4510 ADD NO STUFF IN PRODUCTION!!!!

J5401 PINOUTS ARE DIFFERENT FOR K6, DO NOT SYNC THIS PAGE FROM T27 DIRECTLY

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S0769	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV:8511

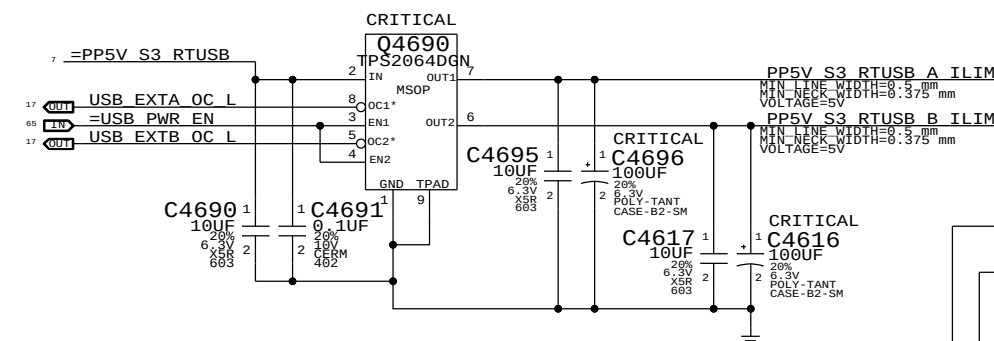
BOMOPTIONS:
 - RDRV:8511 stuffs PS8511A & associated parts (STRAPS TBD!!!)
 - RDRV:8515 stuffs PS8515A & associated parts
 - RDRV:NO stuffs bypass path (neither IC or associated parts stuffed)

Redriver Bypass Path

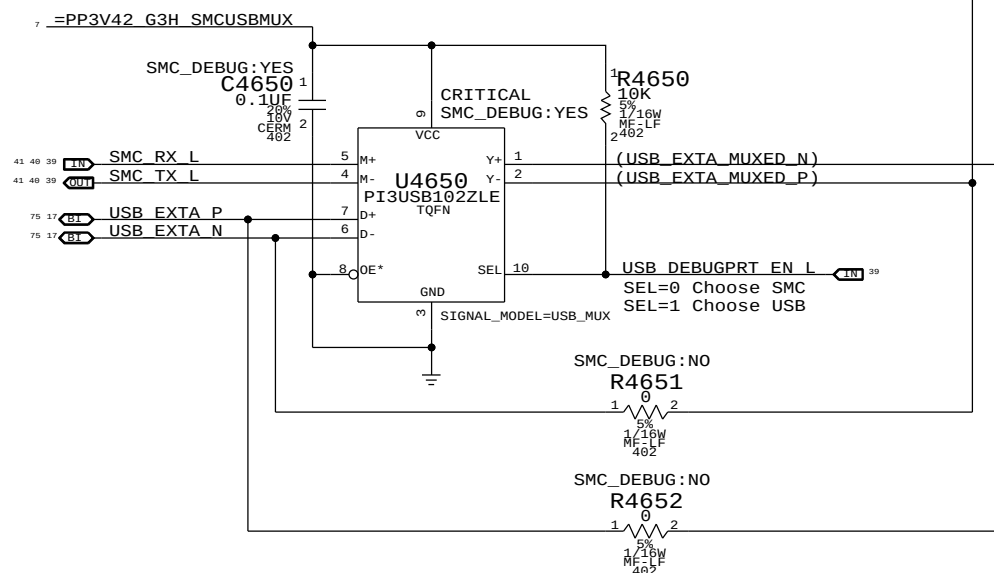


SYNC MASTER=T27 MLB	SYNC DATE=08/06/2009
SATA Connectors	
Apple Inc.	DRAWING NUMBER 051-8563
REVISION A.13.0	SIZE D
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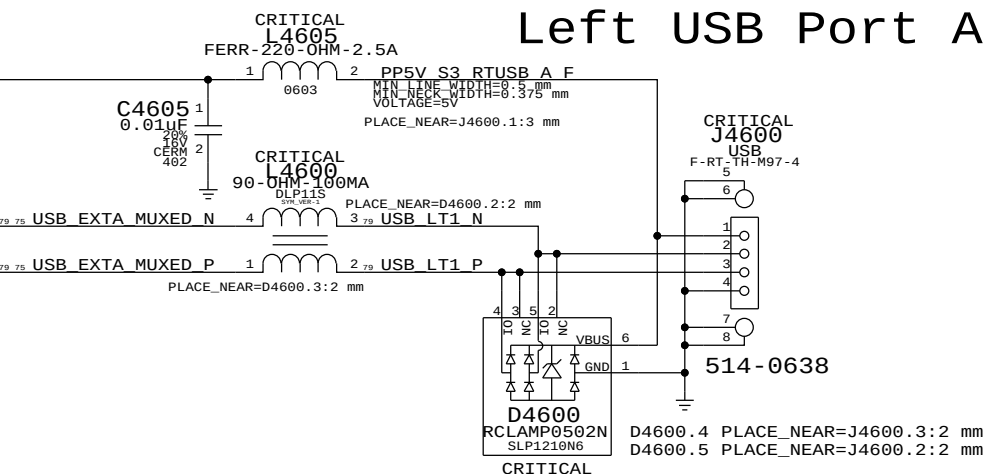
Port Power Switch



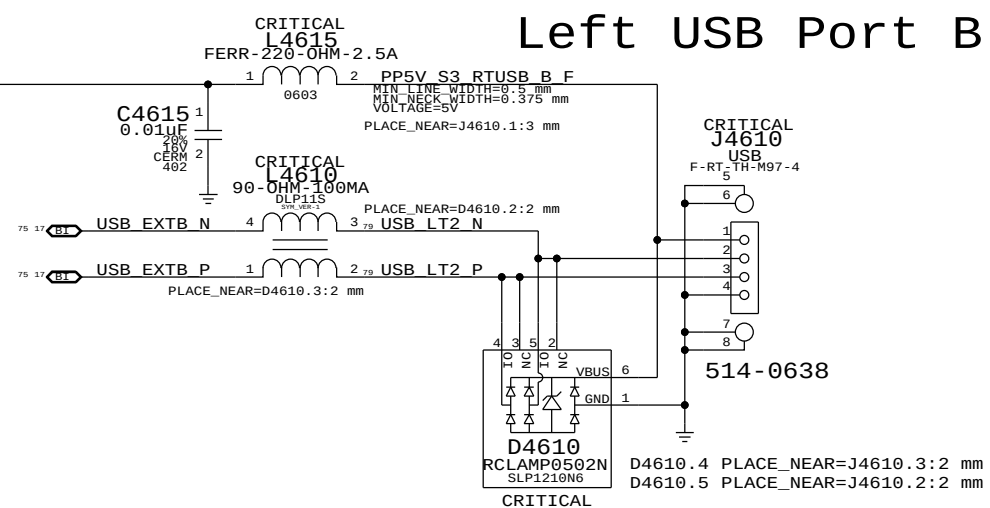
USB/SMC Debug Mux



Left USB Port A

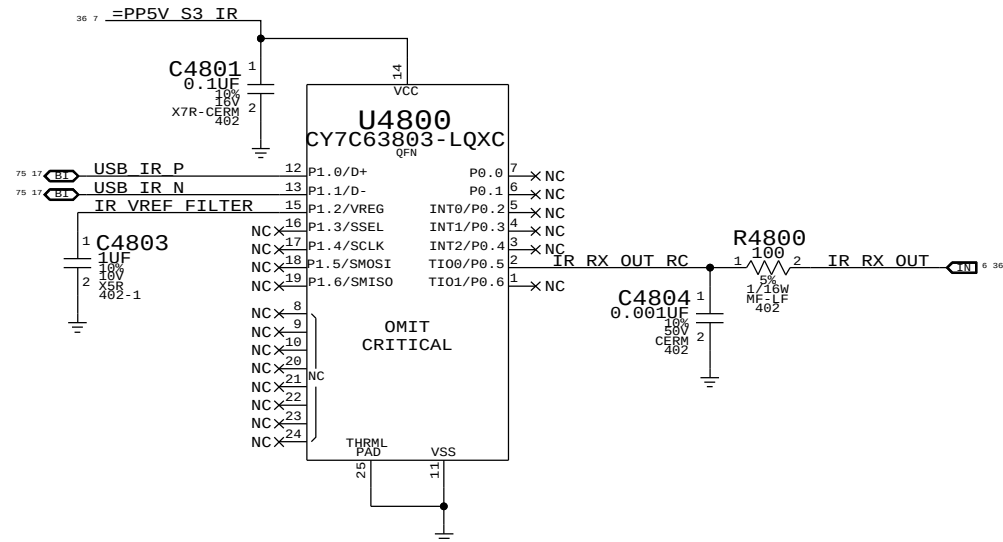


Left USB Port B

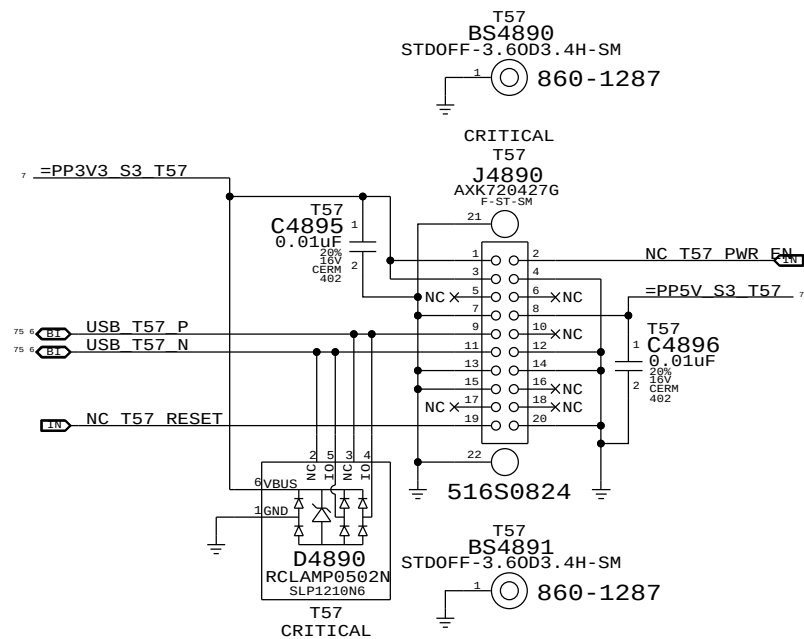




IR Support



T57 Connector



K6 NOTES : D4890 CONNECTION IS DIFFERENT,CANNOT DIRECTLY SYNC FROM T27

SYNC MASTER=T27_MLB		SYNC DATE=08/27/2009	
PAGE TITLE			
Internal USB Support			
	Apple Inc.	DRAWING NUMBER	051-8563
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		BRANCH	
		PAGE	48 OF 109
		SHEET	38 OF 80

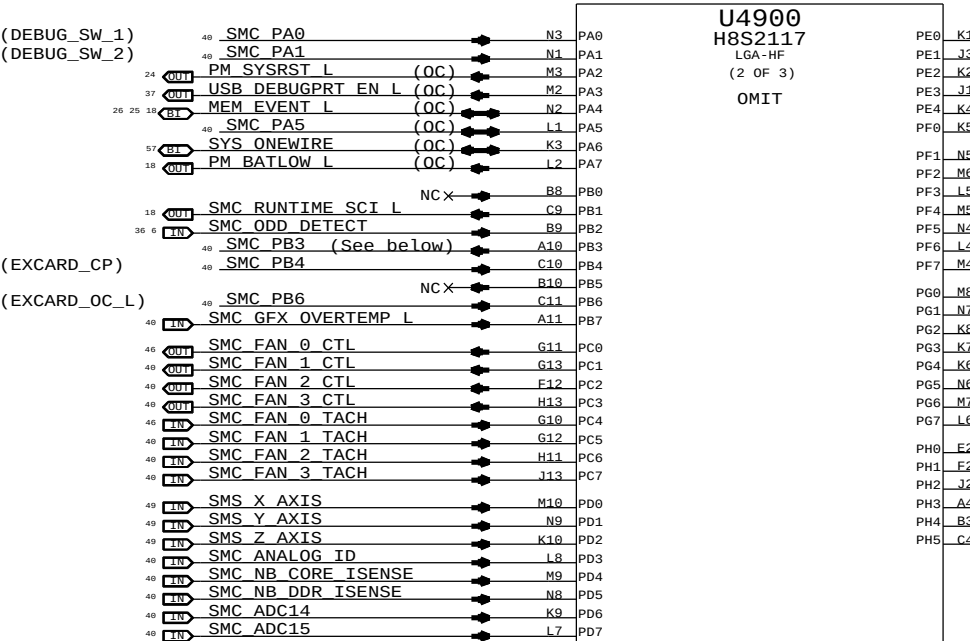
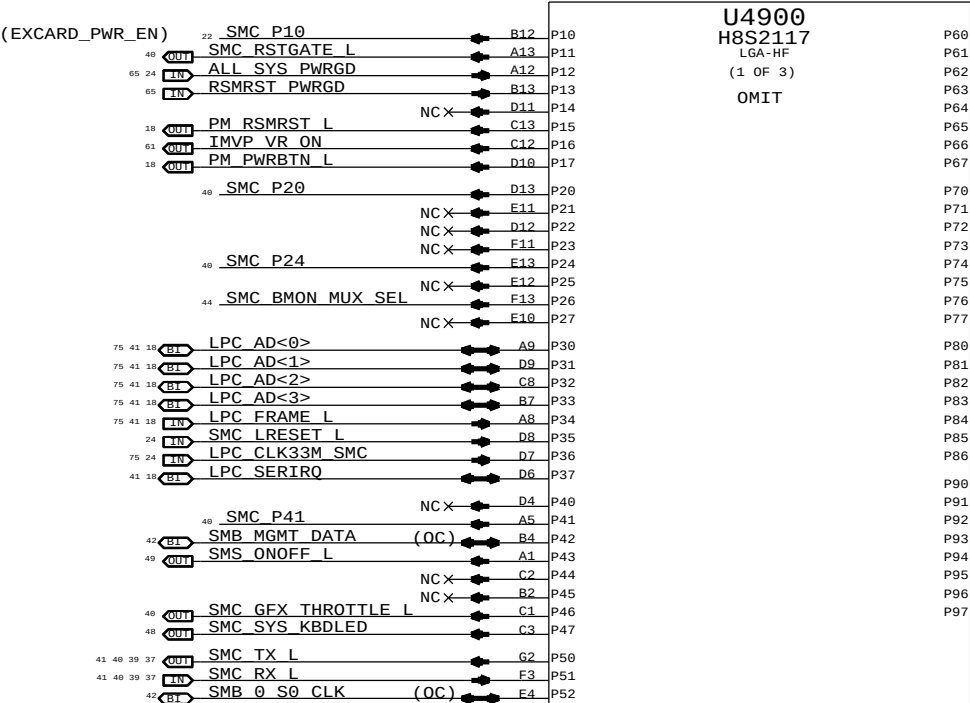
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

D

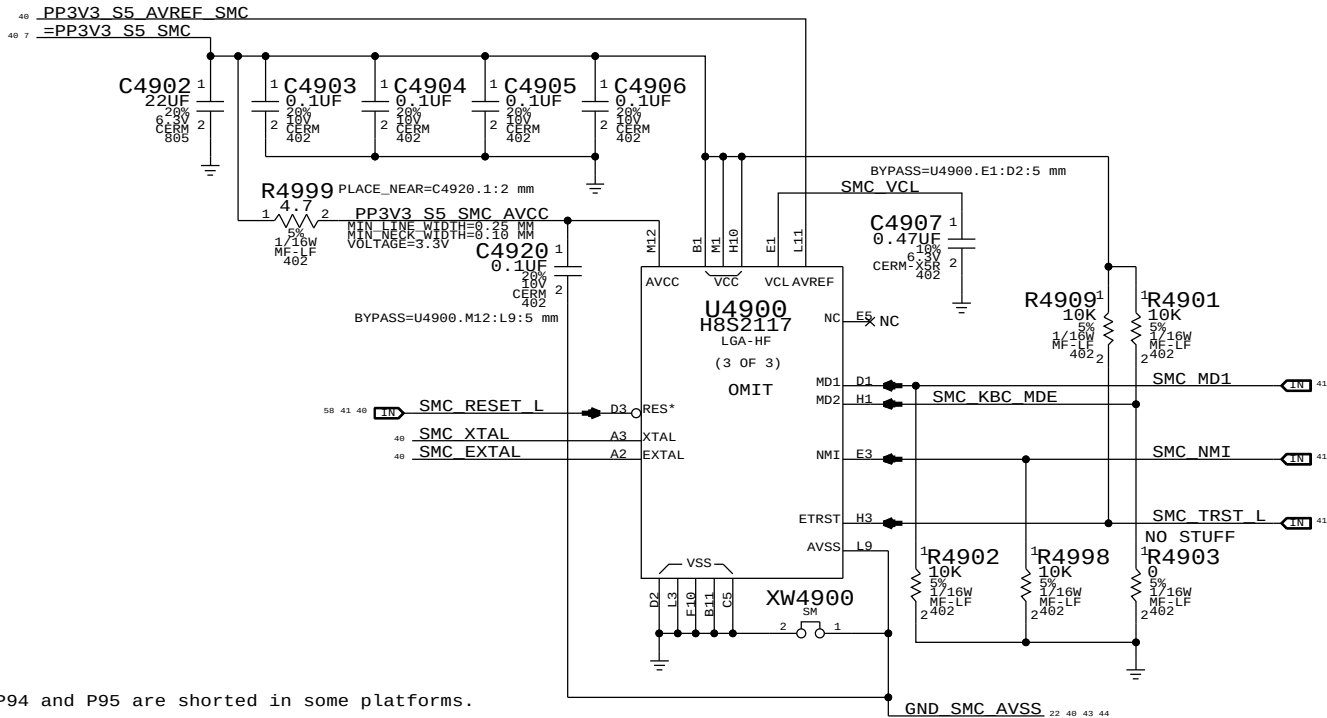
C

B

A



SMC_PB3:
SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay.



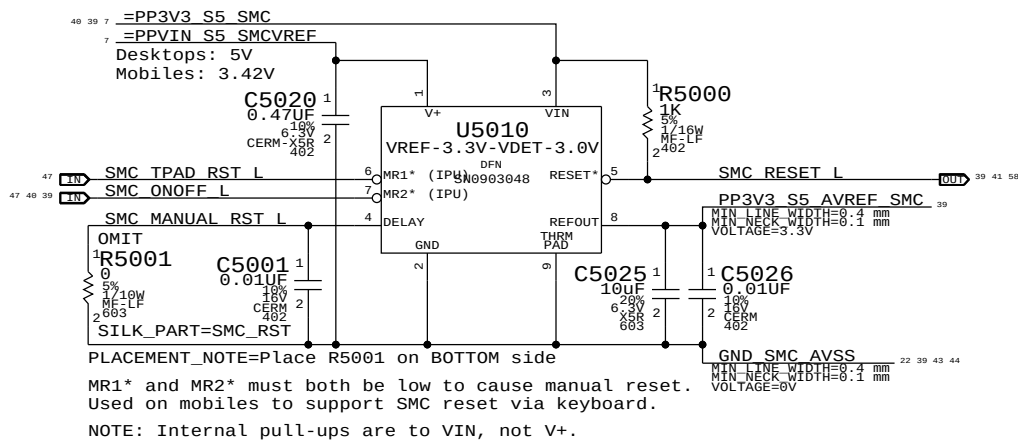
NOTE: P94 and P95 are shorted in some platforms.

NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

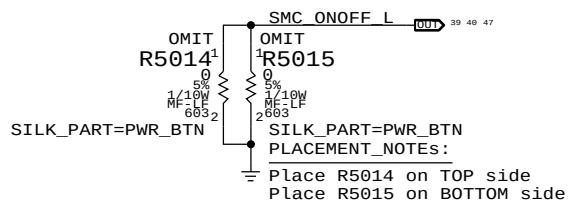
H8S2117-R:
(SMC_PECI)
(SMC_PECI_VREF)
(SMC_PECI_VSTP)

SYNC_MASTER=T27_MLB		SYNC_DATE=09/02/2009	
PAGE TITLE			
SMC			
 Apple Inc.		DRAWING NUMBER	051-8563
		REVISION	A.13.0
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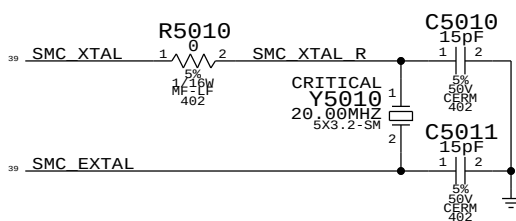
SMC Reset "Button", Supervisor & AVREF Supply



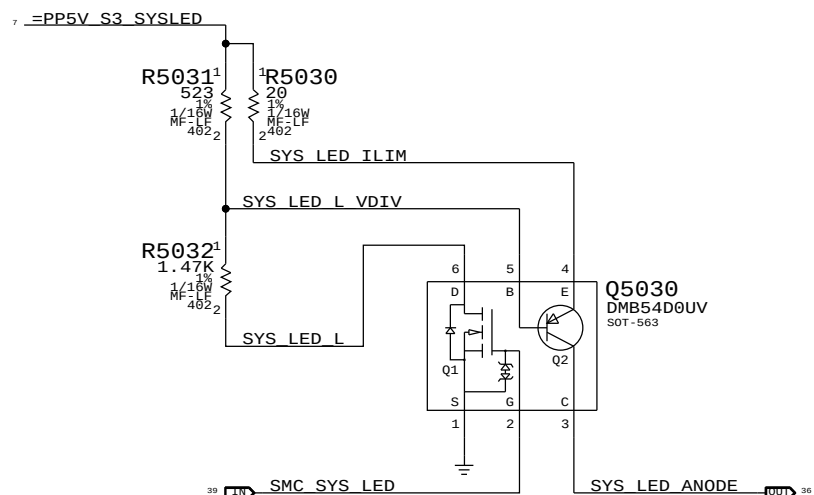
Debug Power "Buttons"



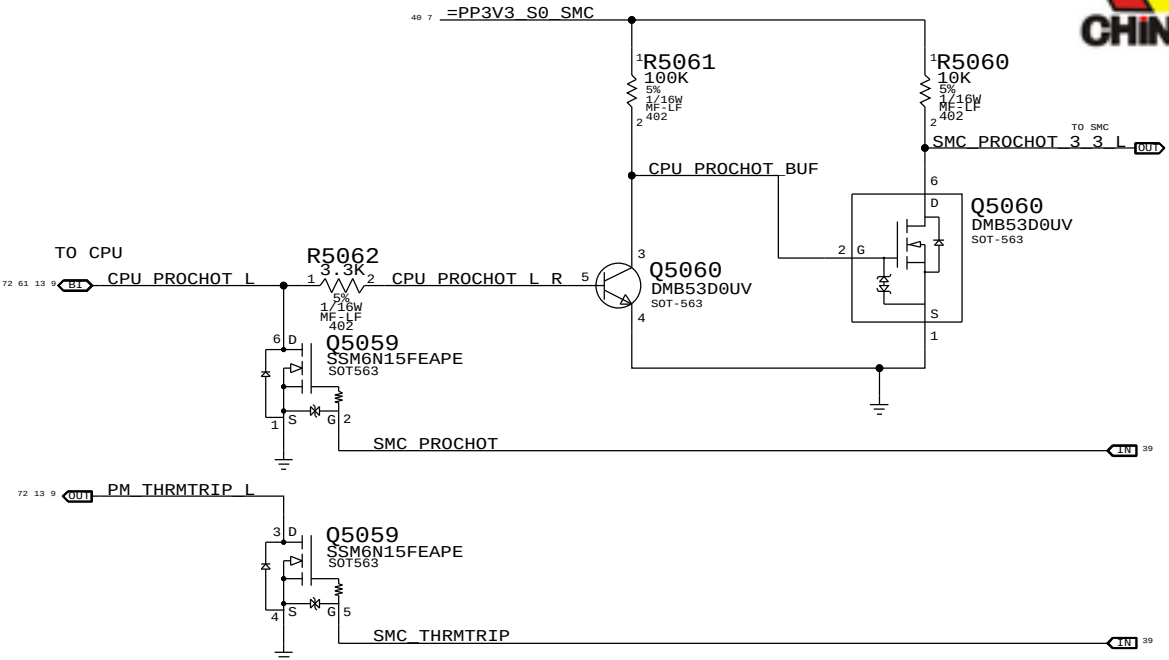
SMC Crystal Circuit



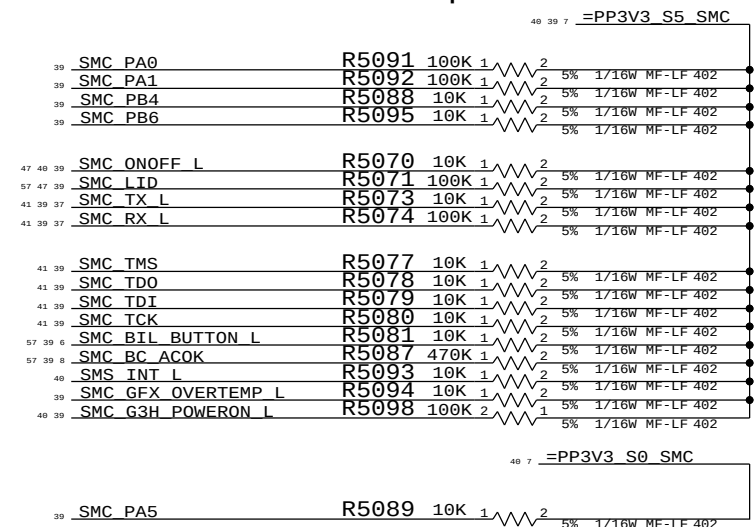
System (Sleep) LED Circuit



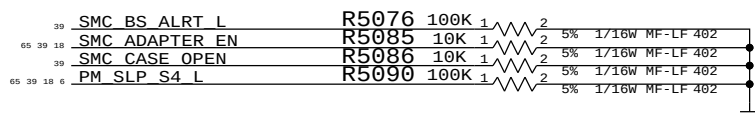
SMC FSB to 3.3V Level Shifting



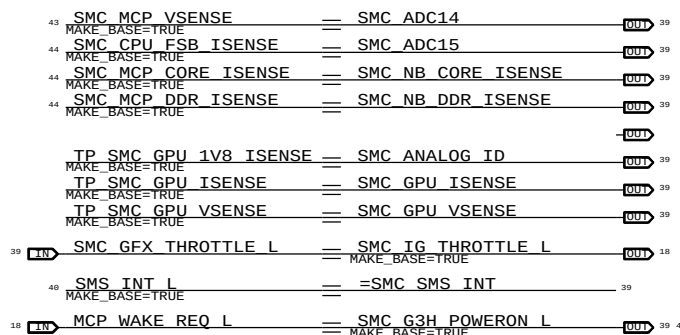
SMC Pull-ups



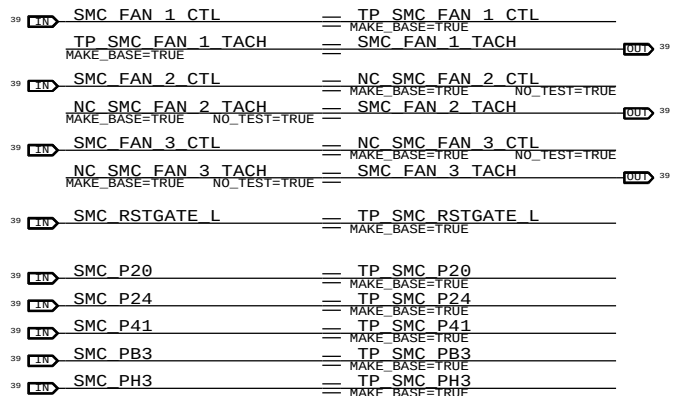
SMC Pull-downs



SMC Aliases

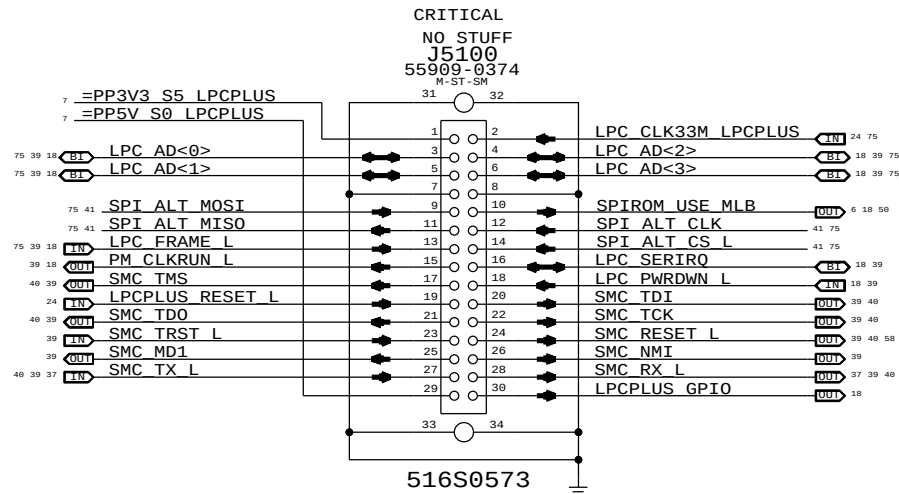


Unused Pins

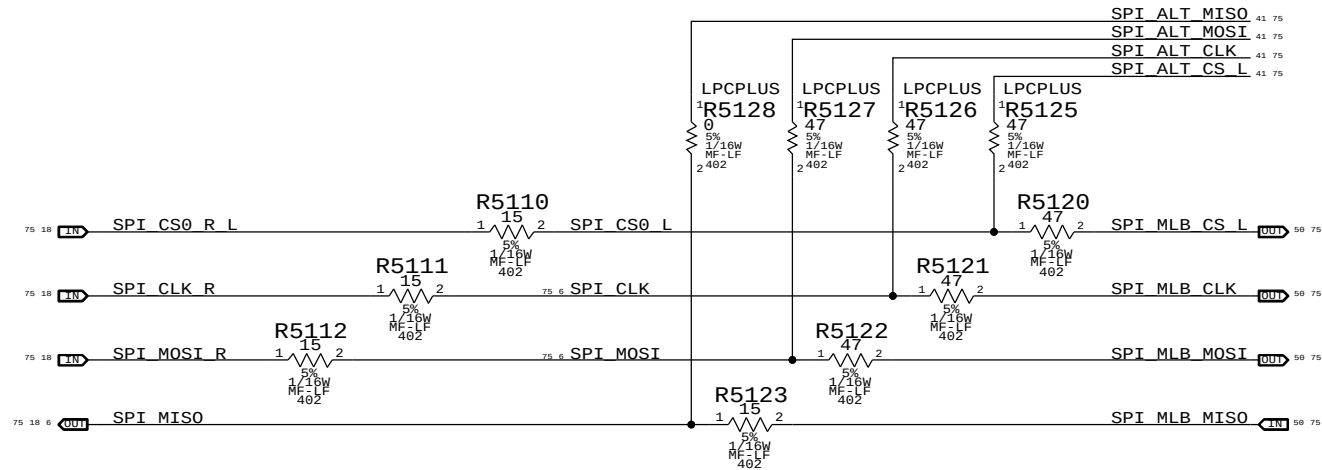




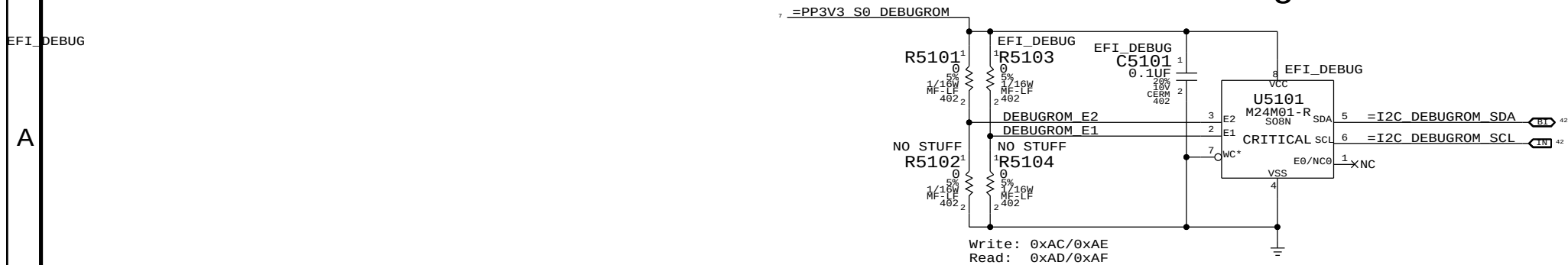
LPC+SPI Connector



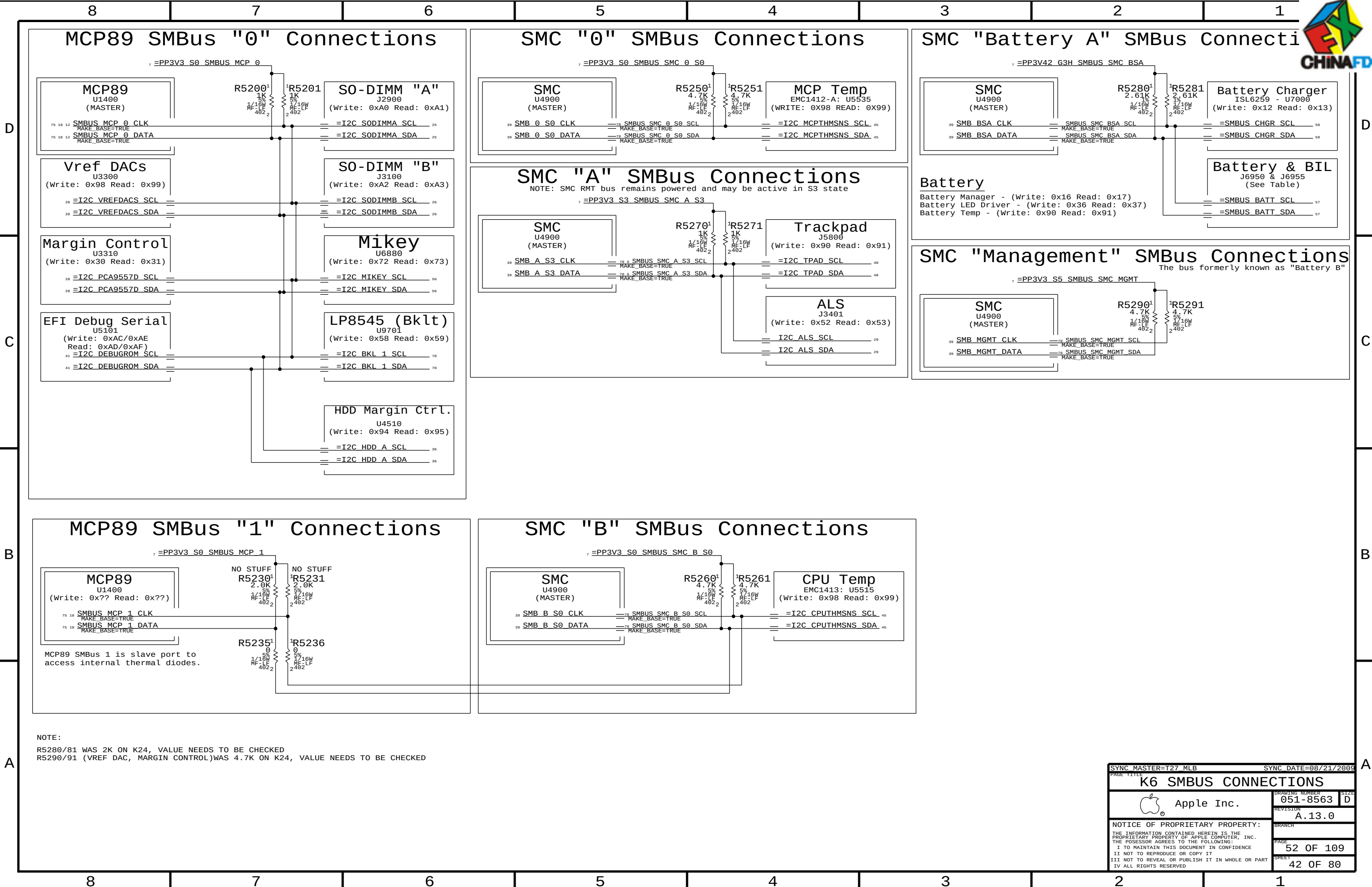
SPI Bus Series Termination



EFI Debug ROM

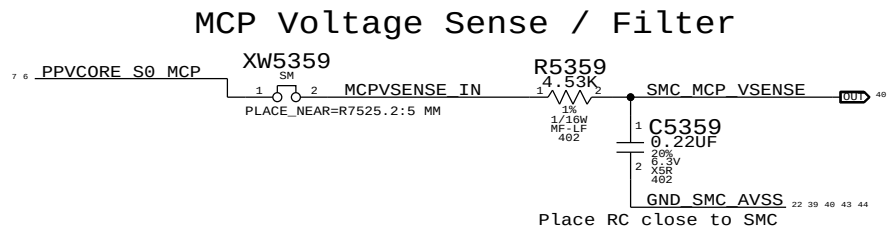
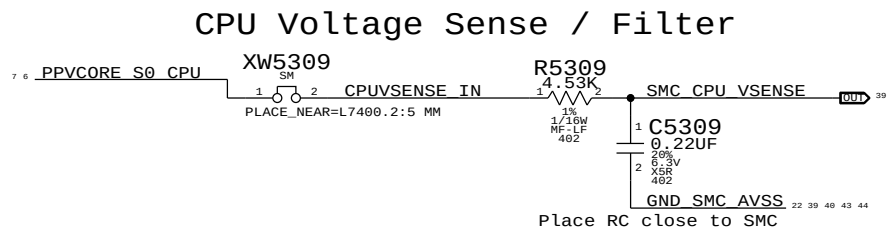


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LPC+SPI Debug Connector		LPC+SPI Debug Connector	
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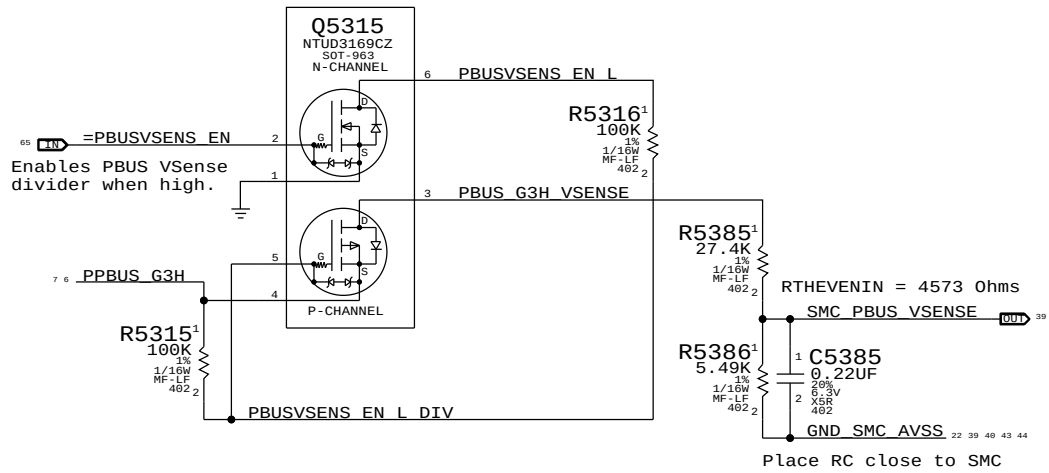


NOTE:
R5280/81 WAS 2K ON K24, VALUE NEEDS TO BE CHECKED
R5290/91 (VREF DAC, MARGIN CONTROL) WAS 4.7K ON K24, VALUE NEEDS TO BE CHECKED

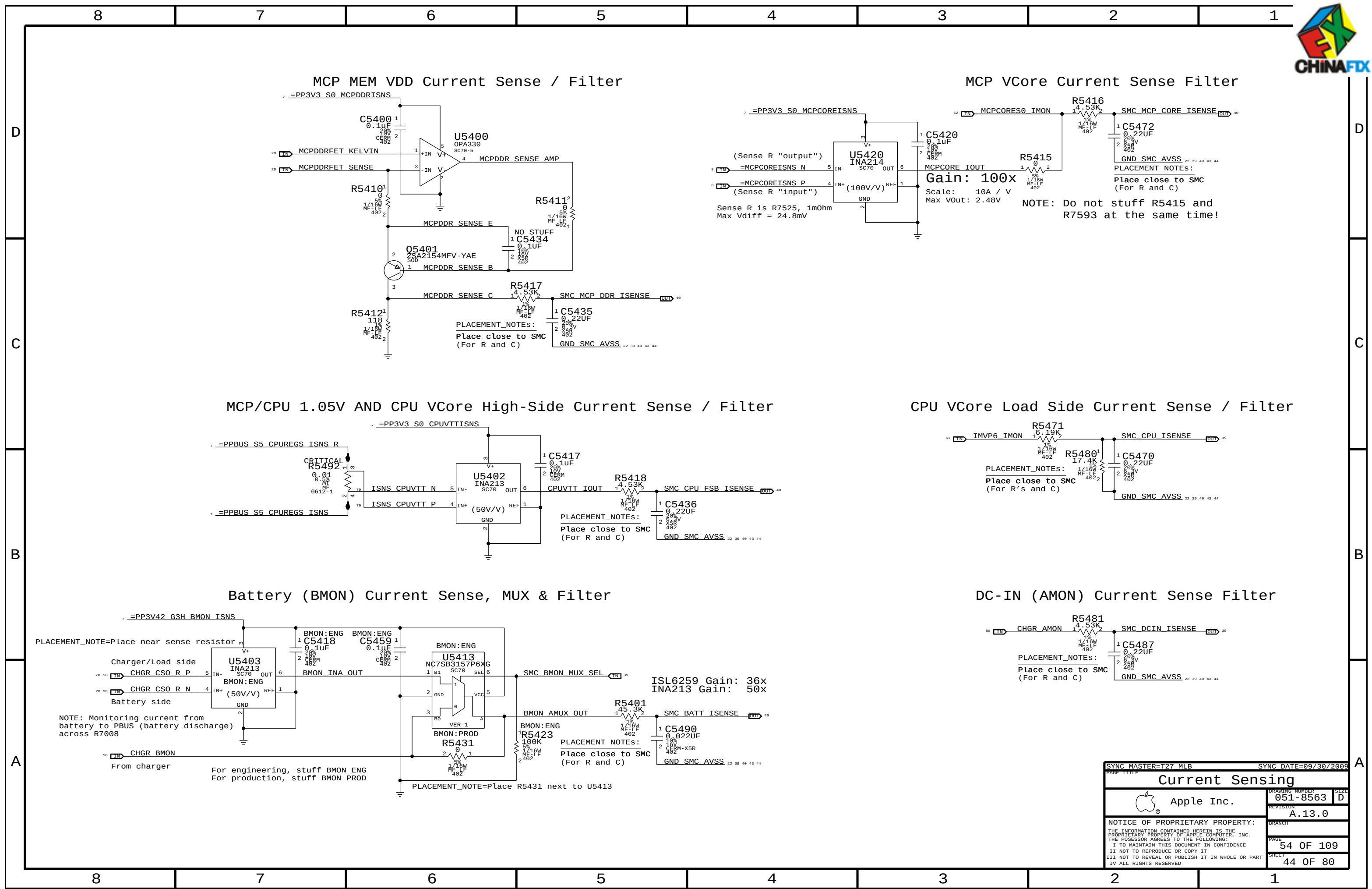
SYNC MASTER=T27 MLB		SYNC DATE=08/21/2009	
PAGE TITLE			
K6 SMBUS CONNECTIONS			
 Apple Inc.		DRAWING NUMBER	051-8563
		REVISION	A.13.0
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		PAGE	52 OF 109
		SHEET	42 OF 80



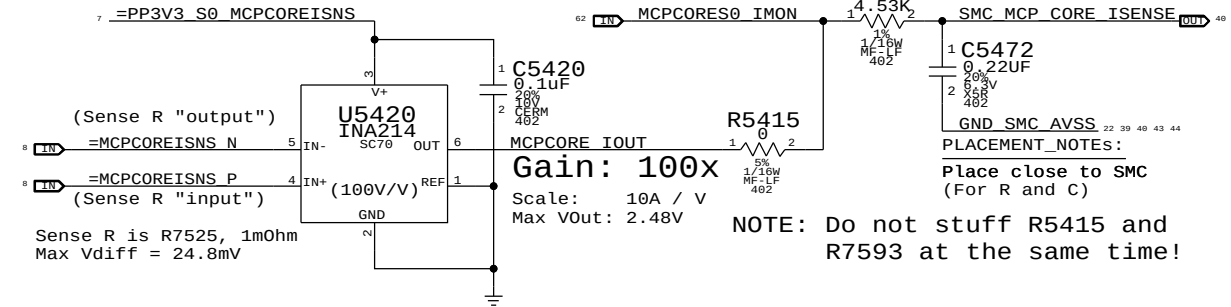
PBUS Voltage Sense Enable & Filter



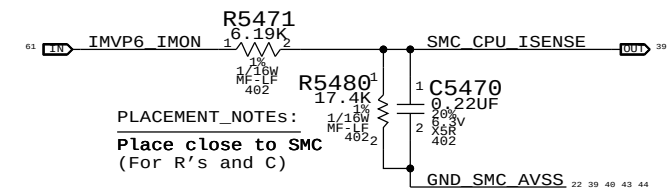
SYNC_MASTER=T27_MLB		SYNC_DATE=08/27/2009	
PAGE TITLE			
Voltage Sensing			
 Apple Inc.		DRAWING NUMBER	051-8563
		SIZE	D
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		BRANCH	
		PAGE	53 OF 109
		SHEET	43 OF 80



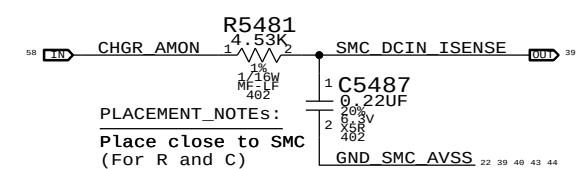
MCP VCore Current Sense Filter



CPU VCore Load Side Current Sense / Filter



DC-IN (AMON) Current Sense Filter



SYNC MASTER=T27 MLB		SYNC DATE=09/30/2009	
PAGE TITLE			
Current Sensing			
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		BRANCH 	PAGE 54 OF 109
		SHEET 44 OF 80	



D |



C



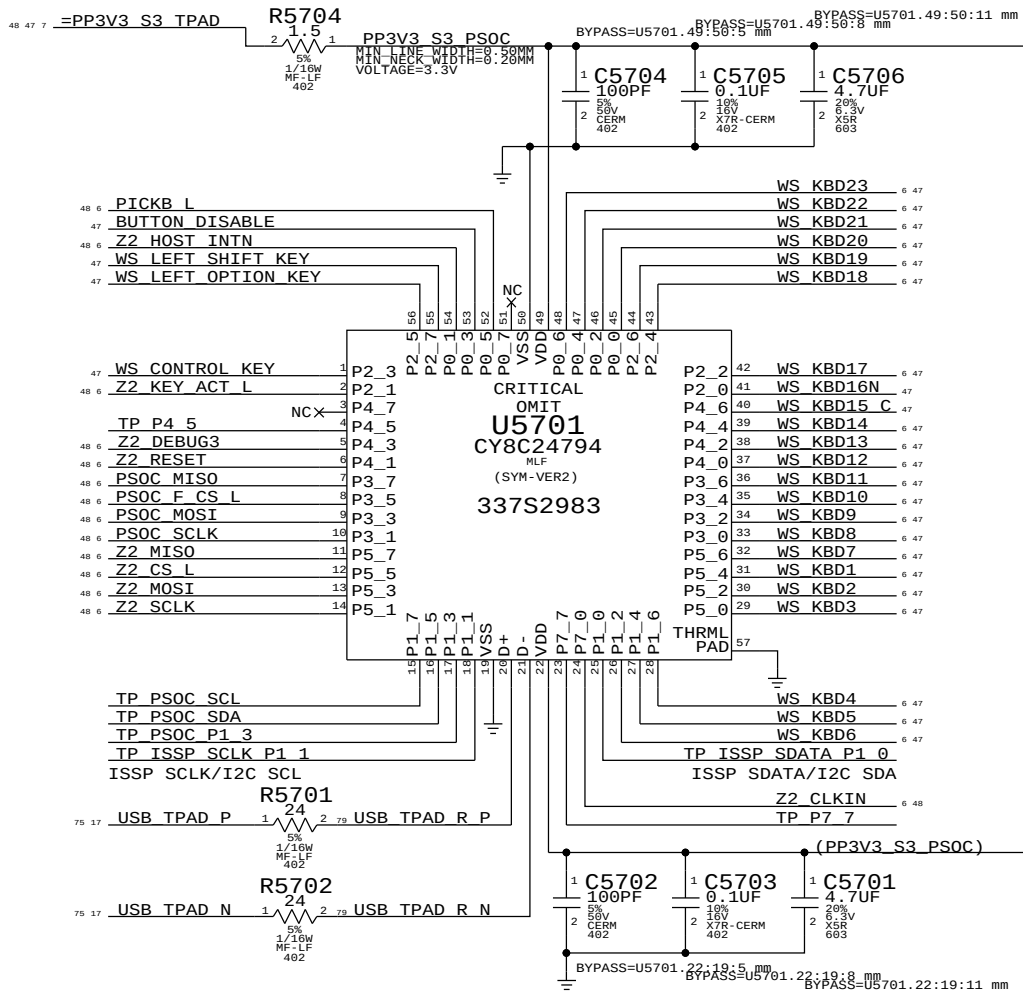
A |

Apple Inc.

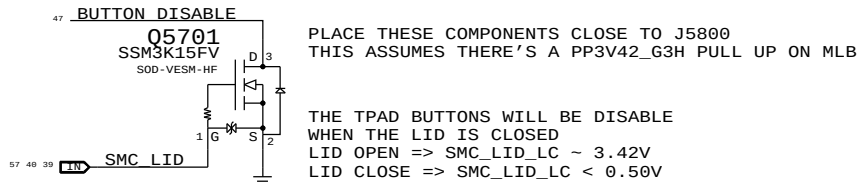


PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

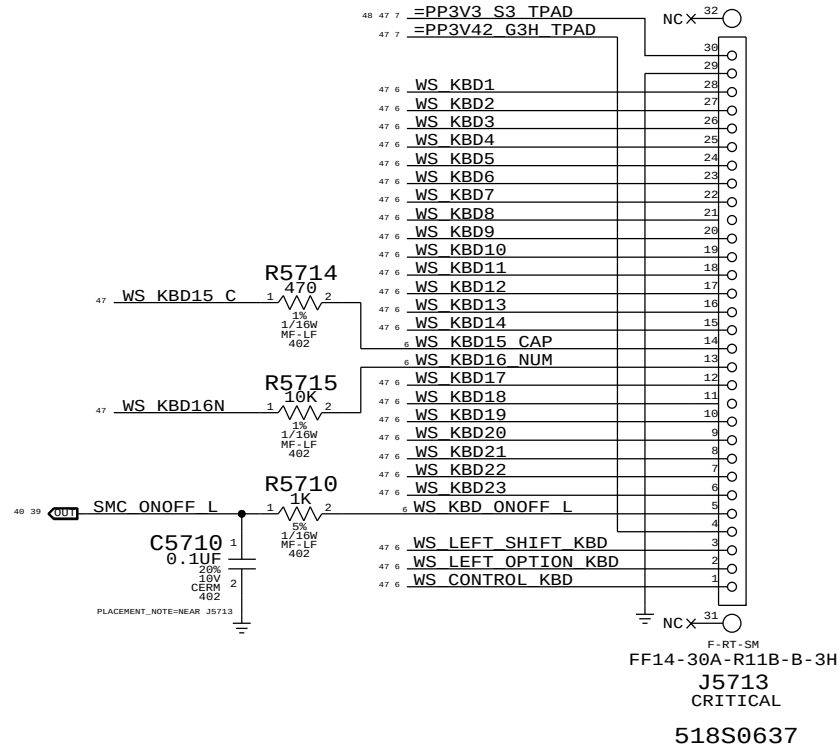


TPAD Buttons Disable



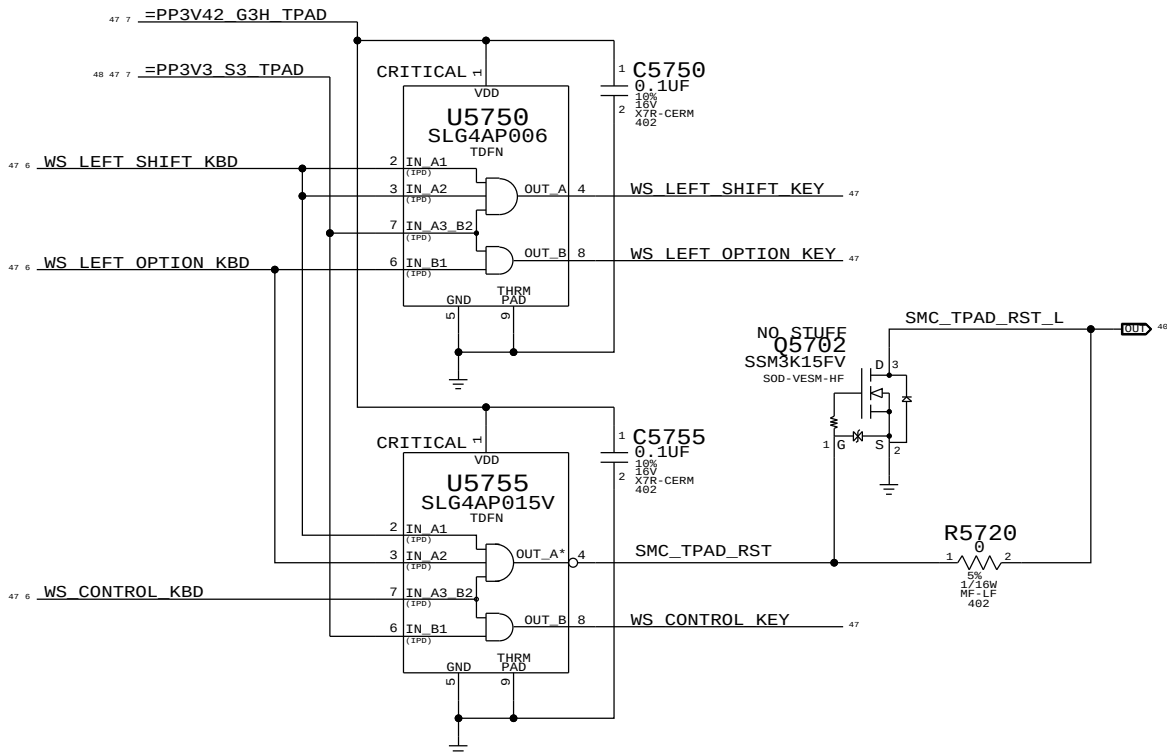
IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

Keyboard Connector



SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSOC power to isolate when PSOC is not powered.

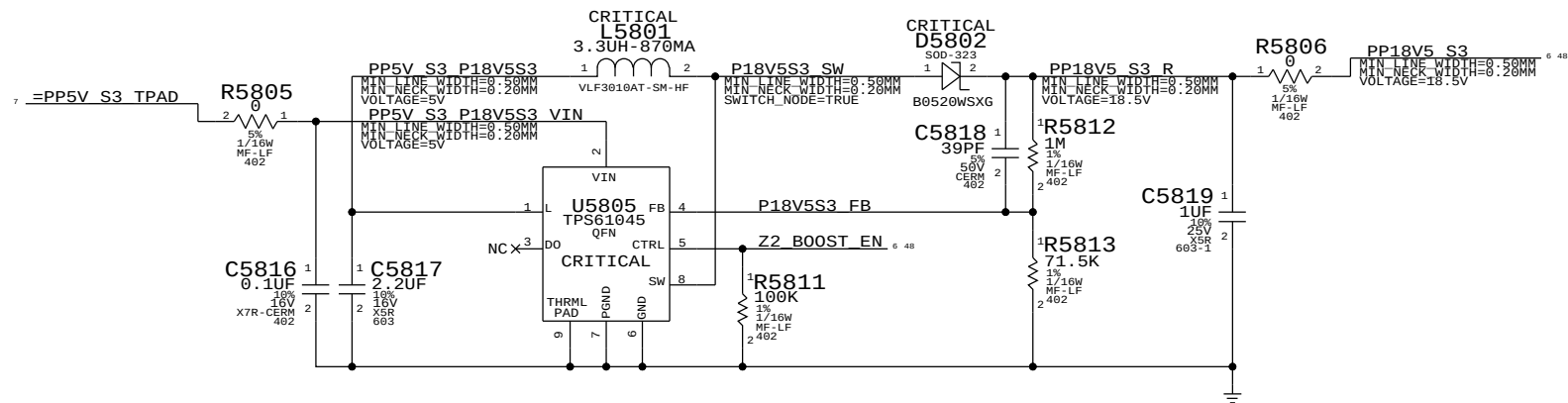


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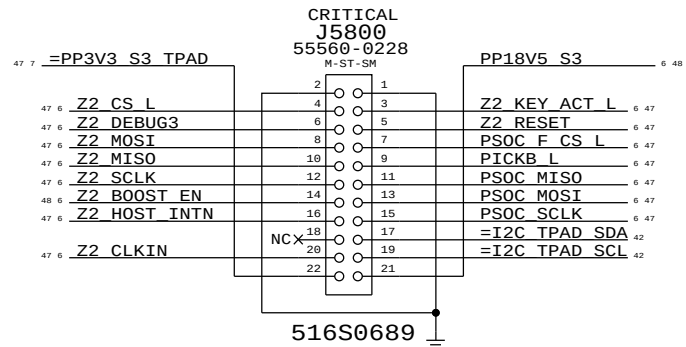
BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

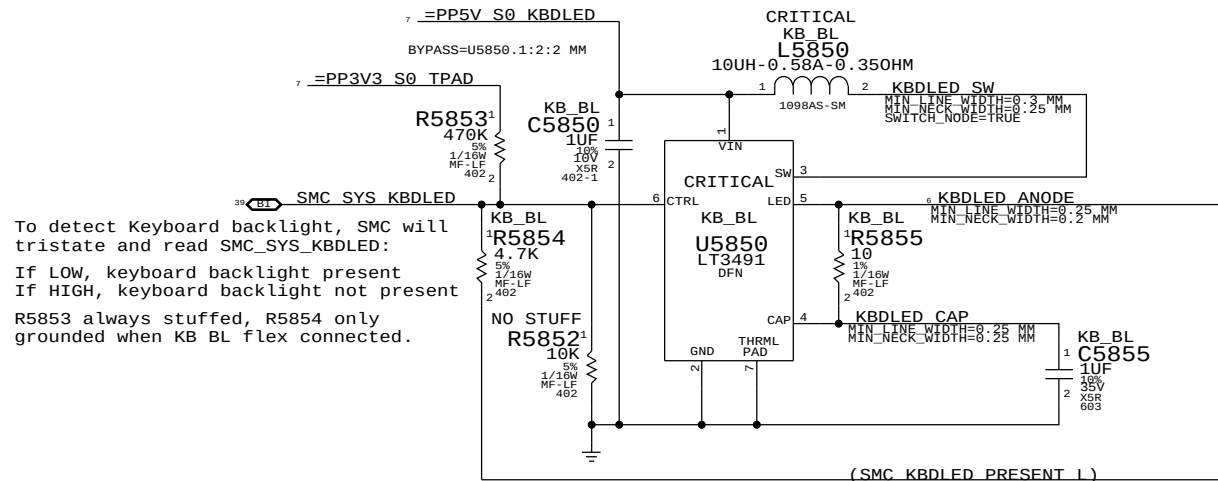
- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED



IPD Flex Connector

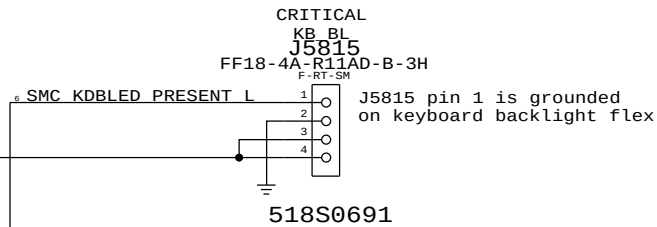


Keyboard Backlight Driver & Detection

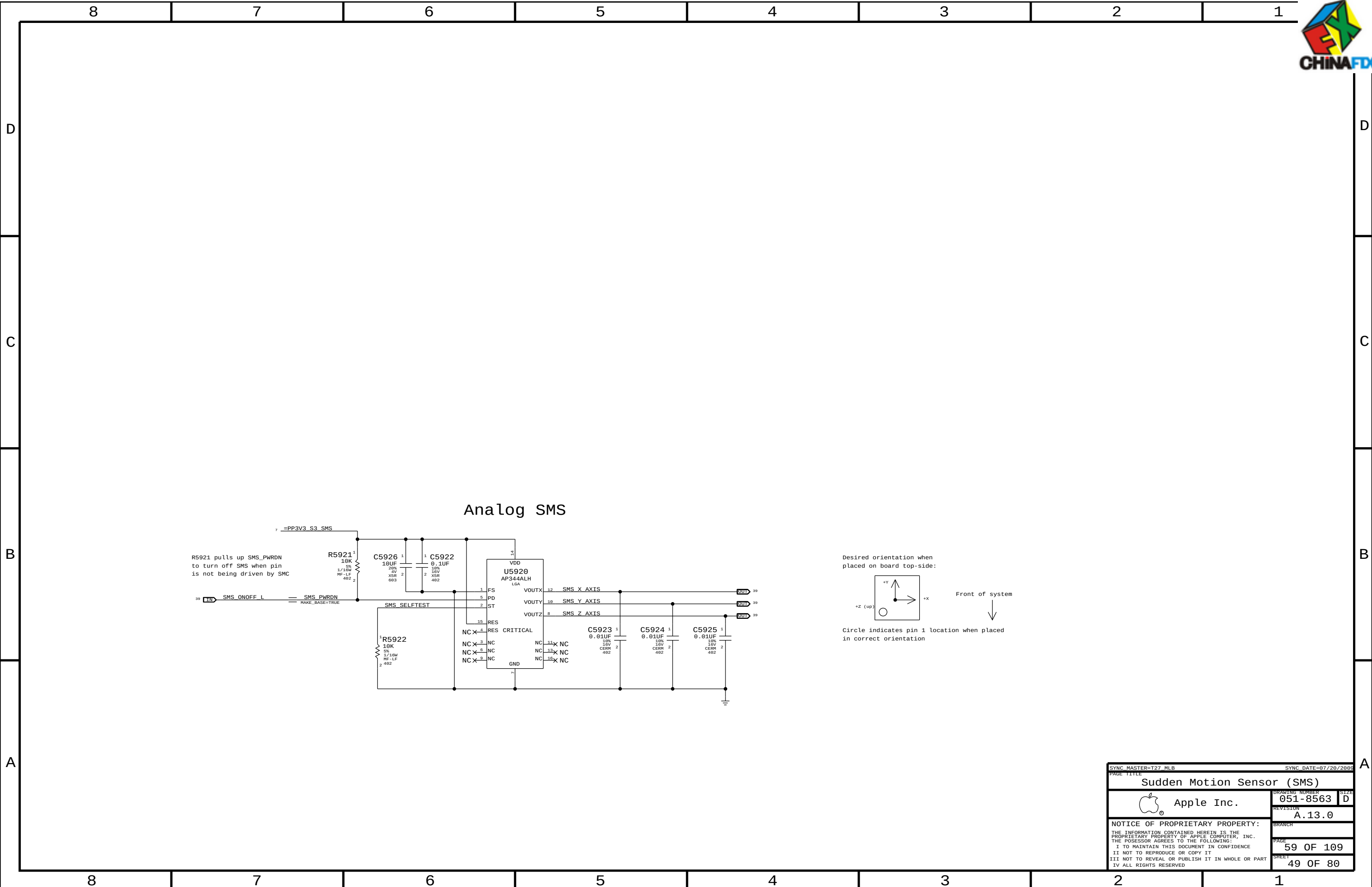


To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:
If LOW, keyboard backlight present
If HIGH, keyboard backlight not present
R5853 always stuffed, R5854 only grounded when KB BL flex connected.

Keyboard Backlight Connector



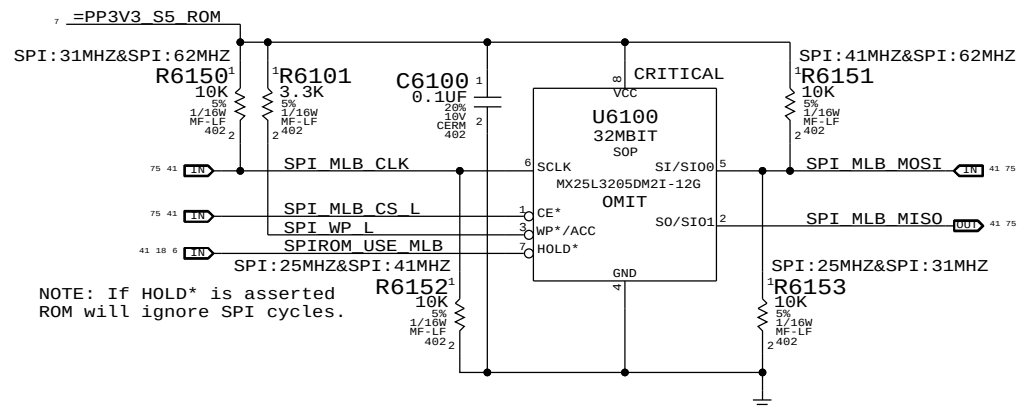
J5815 pin 1 is grounded on keyboard backlight flex

[illegible]

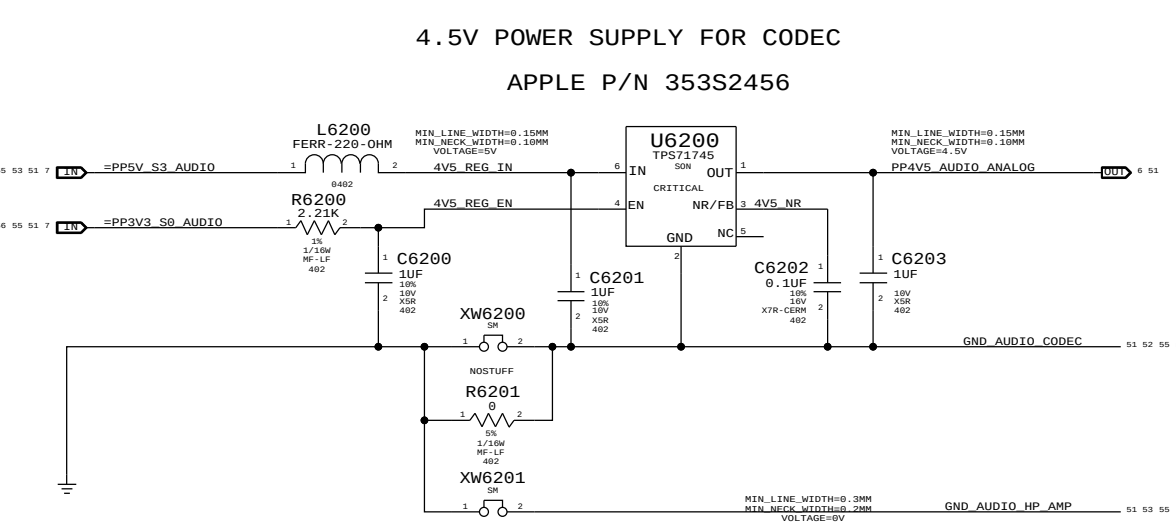
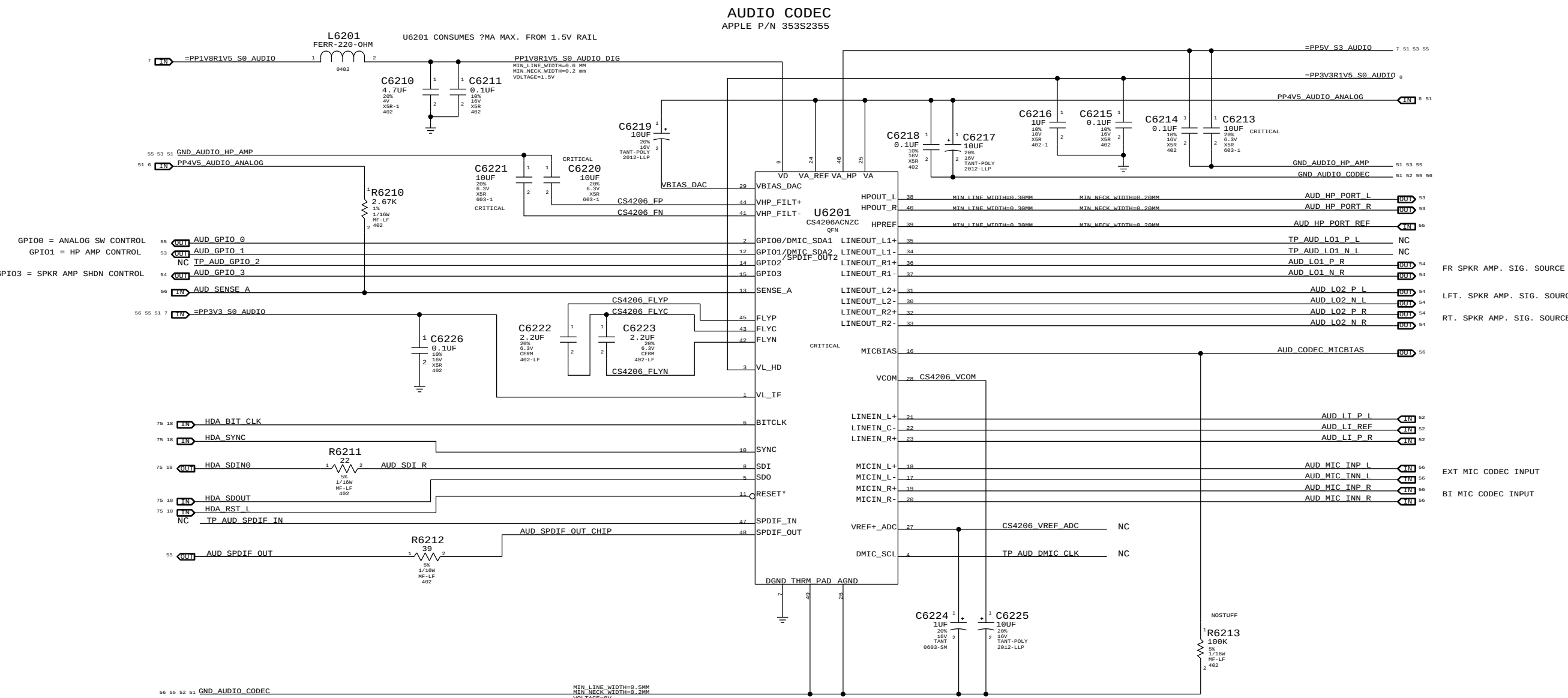
A diagram showing a coordinate system. A circle is located at the bottom left. An arrow points upwards from a point, labeled '+Y'. Another arrow points to the right from the same point, labeled '+Z (up)'.

Front of system

SYMC MASTER-T27 MLB		SYMC DATE=07/20/2005		A
PAGE TITLE				
Sudden Motion Sensor (SMS)				
 Apple Inc.	DRAWING NUMBER		SIZE	
	051-8563		D	
REVISION				
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BRANCH				
PAGE		59 OF 109		
SHEET		49 OF 80		



MCP89 SPI Frequency Select		
Frequency	SPI_MOSI	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
41.7 MHz	1	0
62.5 MHz	1	1
NOTE: 42 & 62 MHz use FAST_READ command.		



NOTES ON CODEC I/O

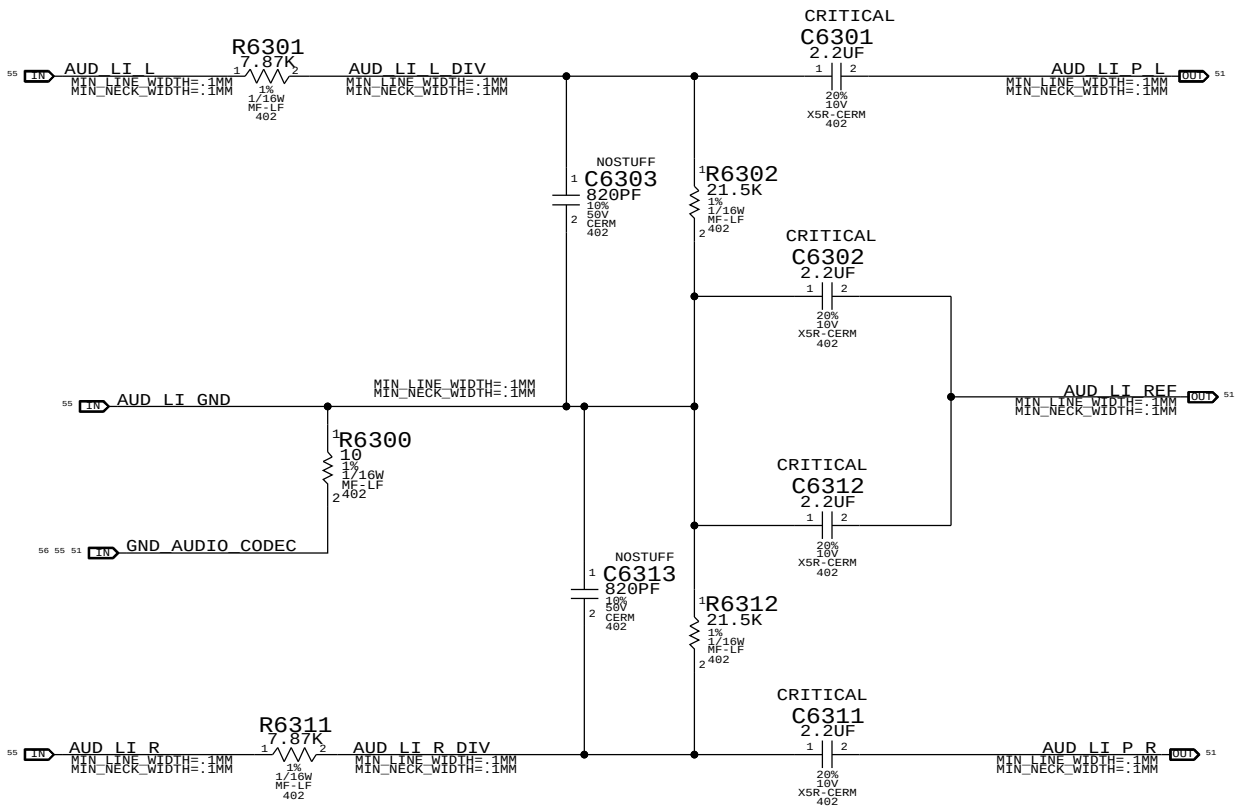
DIFF FSINPUT= 2.45VRMS
 SE FSINPUT= 1.22VRMS
 DAC1 FSOUTPUT= 1.34VRMS
 DAC2/3 FSOUTPUTDIFF= 2.67VRMS
 DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=AUDIO		SYNC DATE=08/31/2008	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
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		REVISION	A.13.0
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LINE INPUT VOLTAGE DIVIDER

CODEC RIN = 20K OHMS
NET RIN = 10.36K OHMS (INCLUDING PULL-DOWNS AT ANALOG SWITCH COM PINS)
FC_HP = 3.6 HZ
FC_LP = 43KHZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS

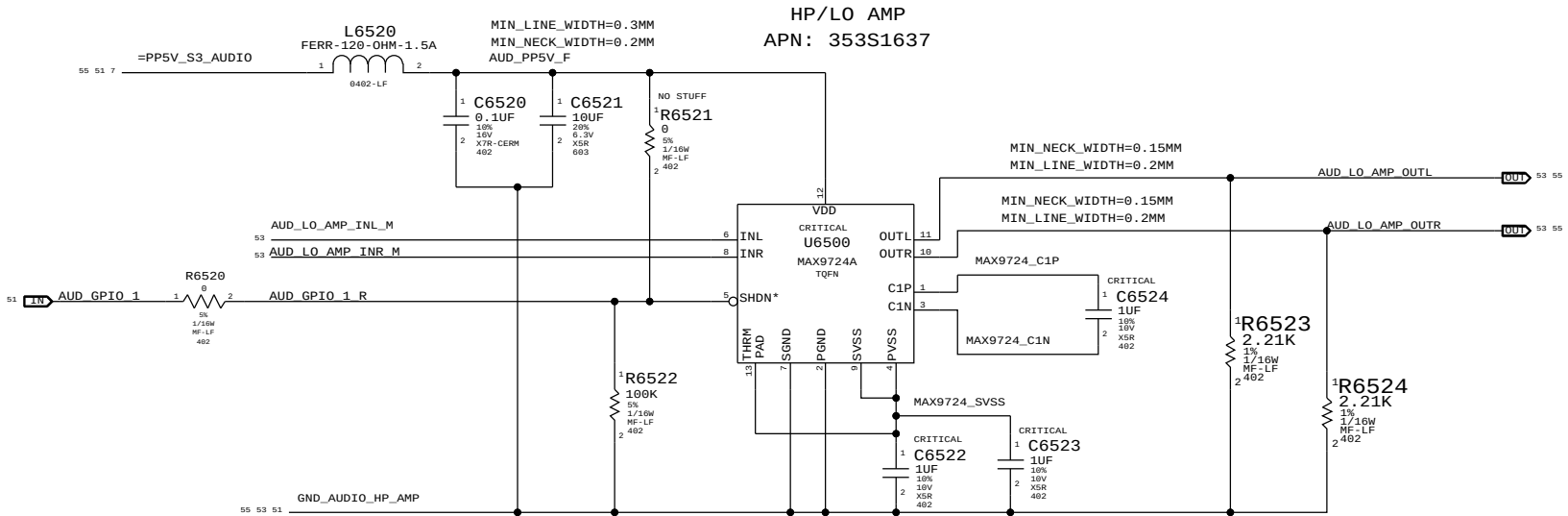
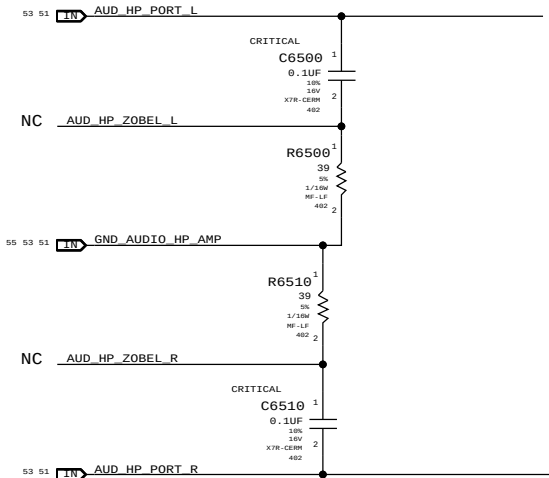


PAGE TITLE		
AUDIO: LINE INPUT FILTER		
 Apple Inc.	DRAWING NUMBER	051-8563
	REVISION	A.13.0
	BRANCH	
	PAGE	63 OF 109
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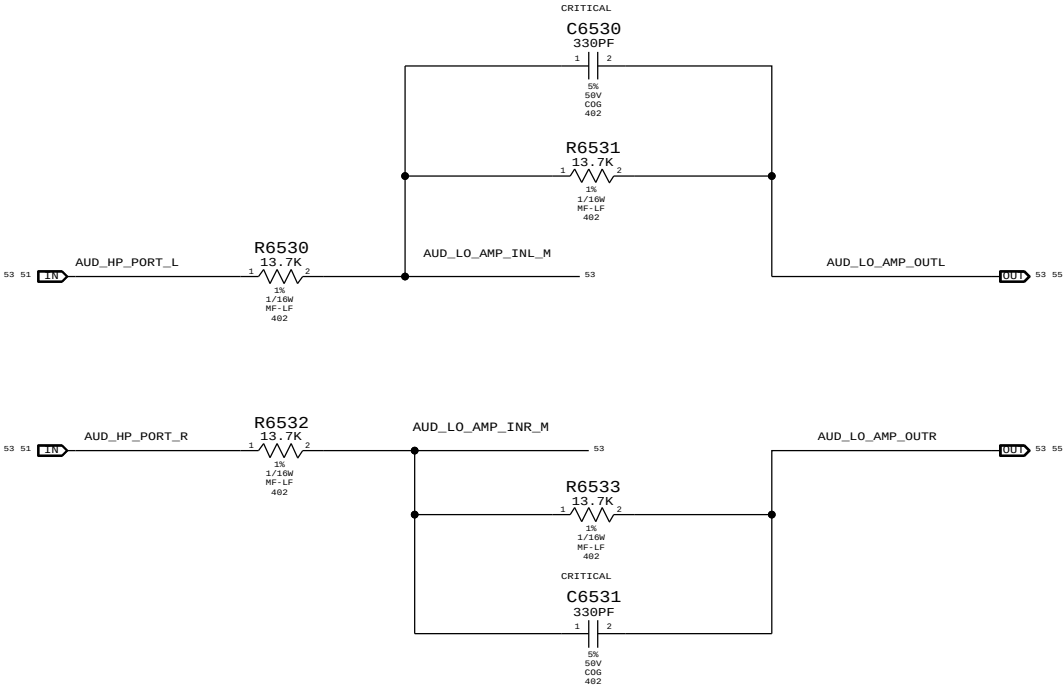
FOR PROT02, STUFF R6521 AND NO STUFF R6520 AND R6522 UNTIL
RE-TASKABLE IO SW SUPPORT AVAILABLE (FORCES IO INTO OUTPUT MODE).

ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER



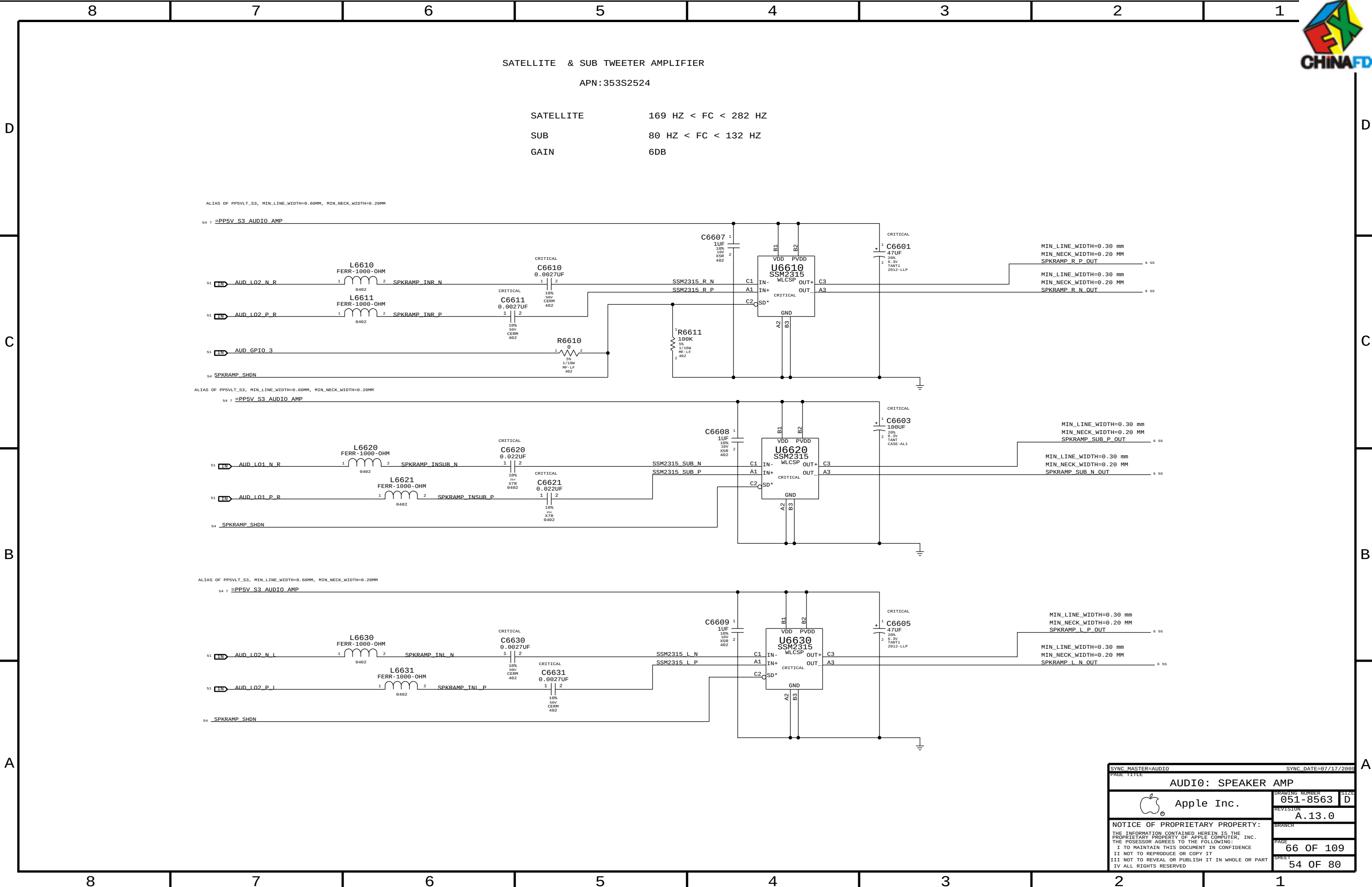
MAX9724 GAIN/FILTER COMPONENTS

AV_PB = -1V/V, FC_LPF = 35.2KHZ



SYNC MASTER=AUDIO		SYNC DATE=07/17/2009	
PAGE TITLE		AUDIO: HEADPHONE FILTER	
DRAWING NUMBER		051-8563	SIZE D
REVISION		A.13.0	BRANCH
PAGE		65 OF 109	SHEET
SHEET		53 OF 80	

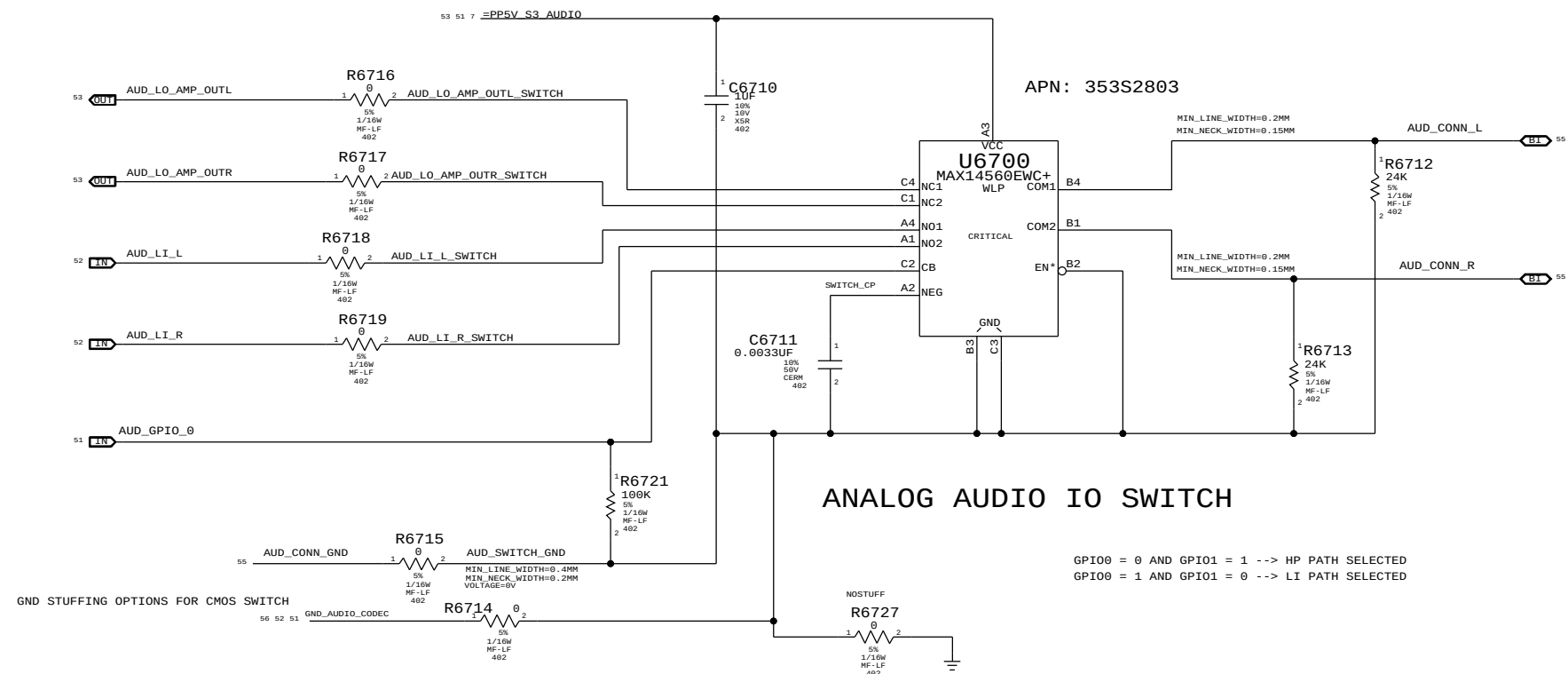
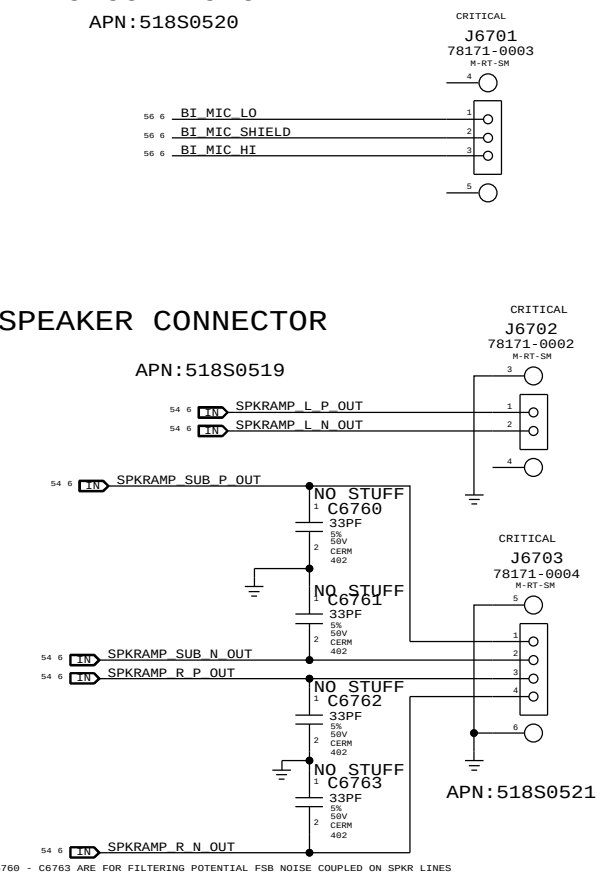
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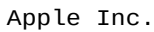
SYNC MASTER-AUDIO		SYNC DATE=07/17/2008	
PAGE TITLE		PAGE NO.	
AUDIO10: SPEAKER AMP			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-8563	D
		REVISION	
		A.13.0	
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		SHEET 54 OF 80	



SPEAKER CONNECTOR



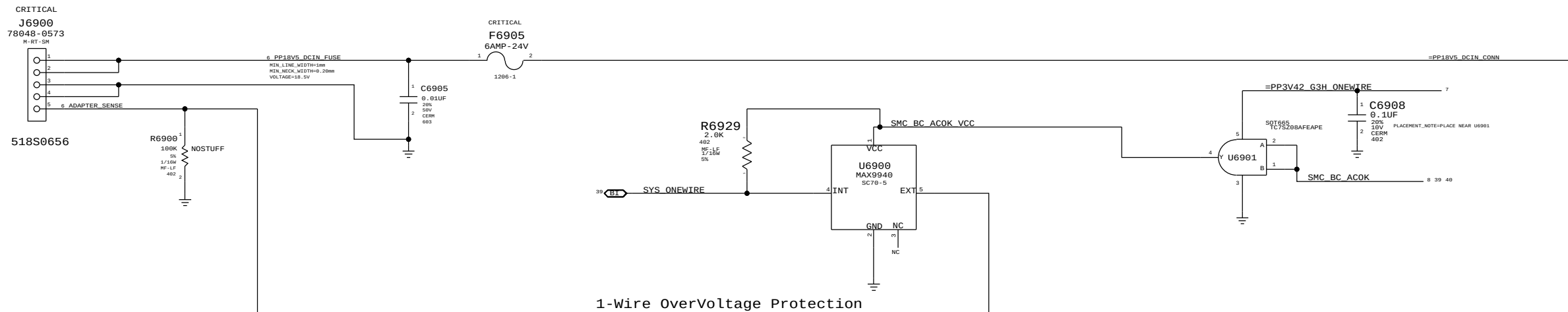
SYNC MASTER=AUDIO	SYNC DATE=08/25/2008
PAGE TITLE	
AUDIO: JACK	
 Apple Inc.	DRAWING NUMBER
	051-8563
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	BRANCH
	PAGE
	67 OF 109
	SHEET
	55 OF 80



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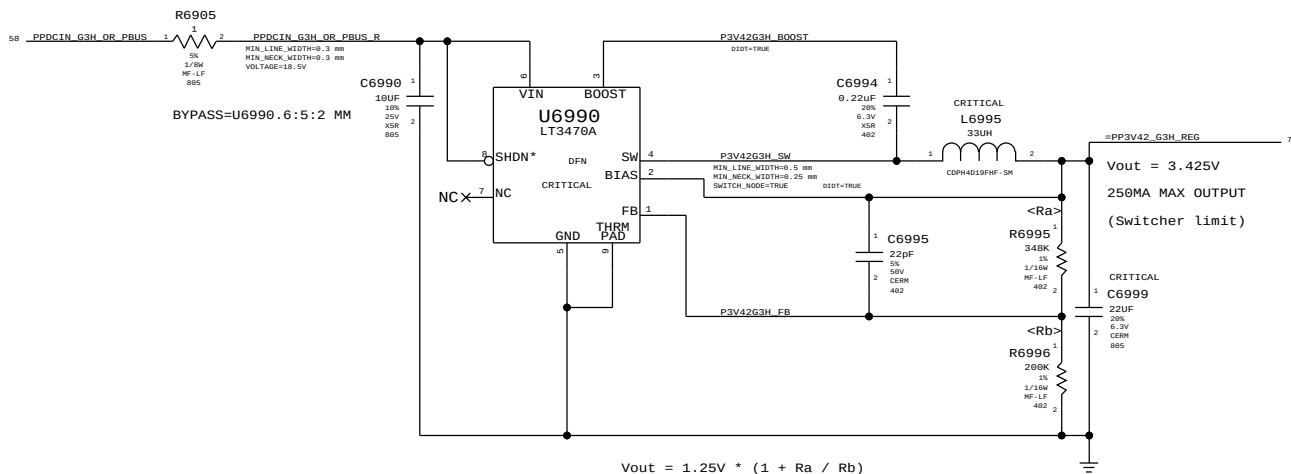


MagSafe DC Power Jack

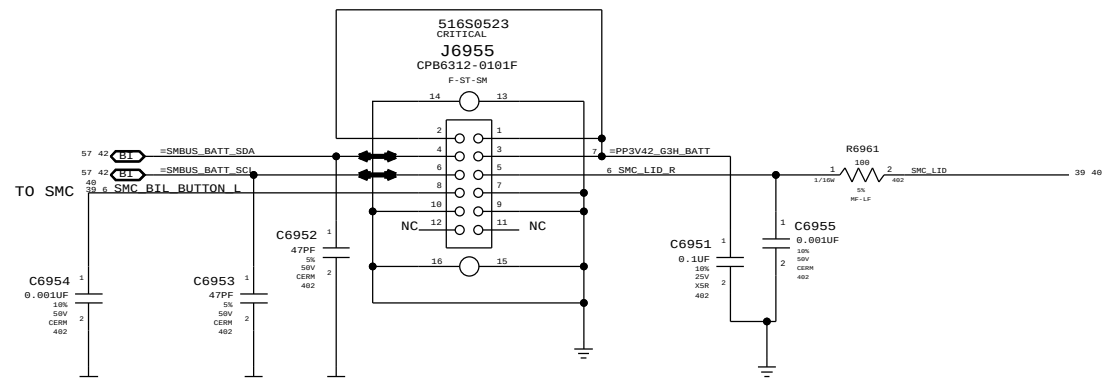


3.425V "G3Hot" Supply

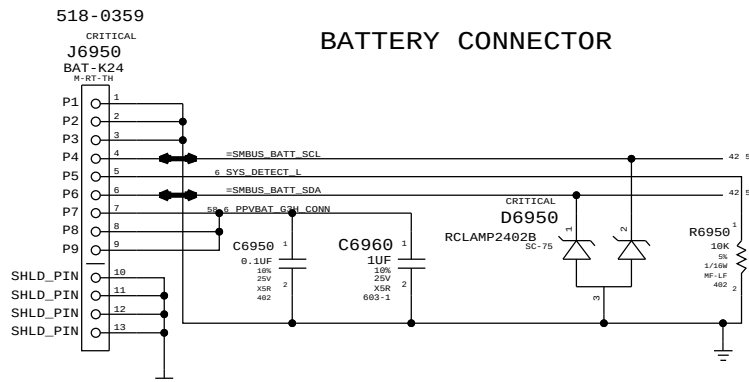
Supply needs to guarantee 3.31V delivered to SMC Vref generator



BIL CONNECTOR



BATTERY CONNECTOR

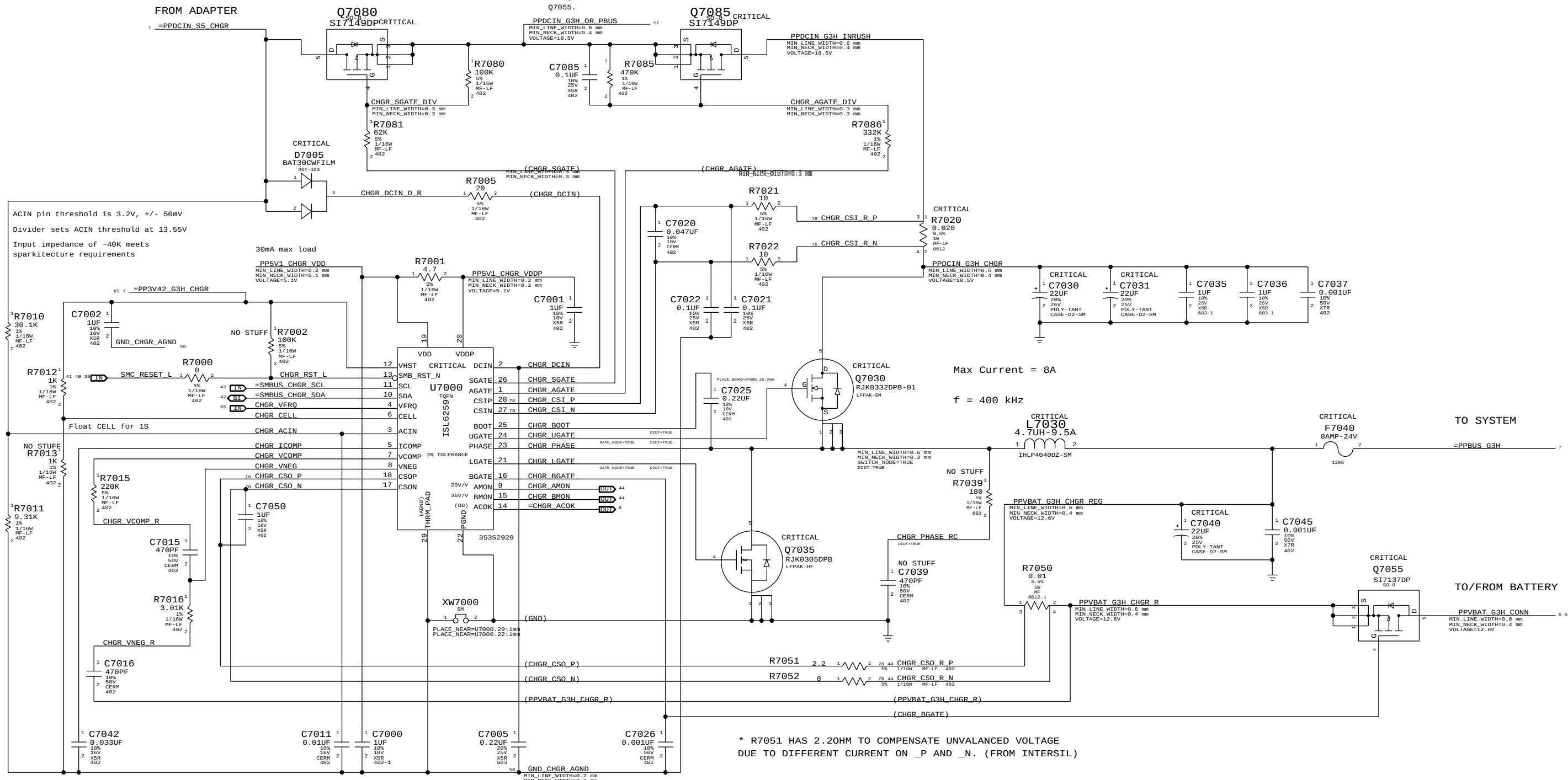


SYNC MASTER=K24 MLB		SYNC DATE=07/20/2005	
PAGE TITLE		DC-In & Battery Connectors	
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		REVISION	A.13.0
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Reverse-Current Protection

Inrush Limiter

This node is powered through body diodes:
 * DCIN through Q7080.
 * PBUS through Q7085, Charger TOP FETs and Q7055.

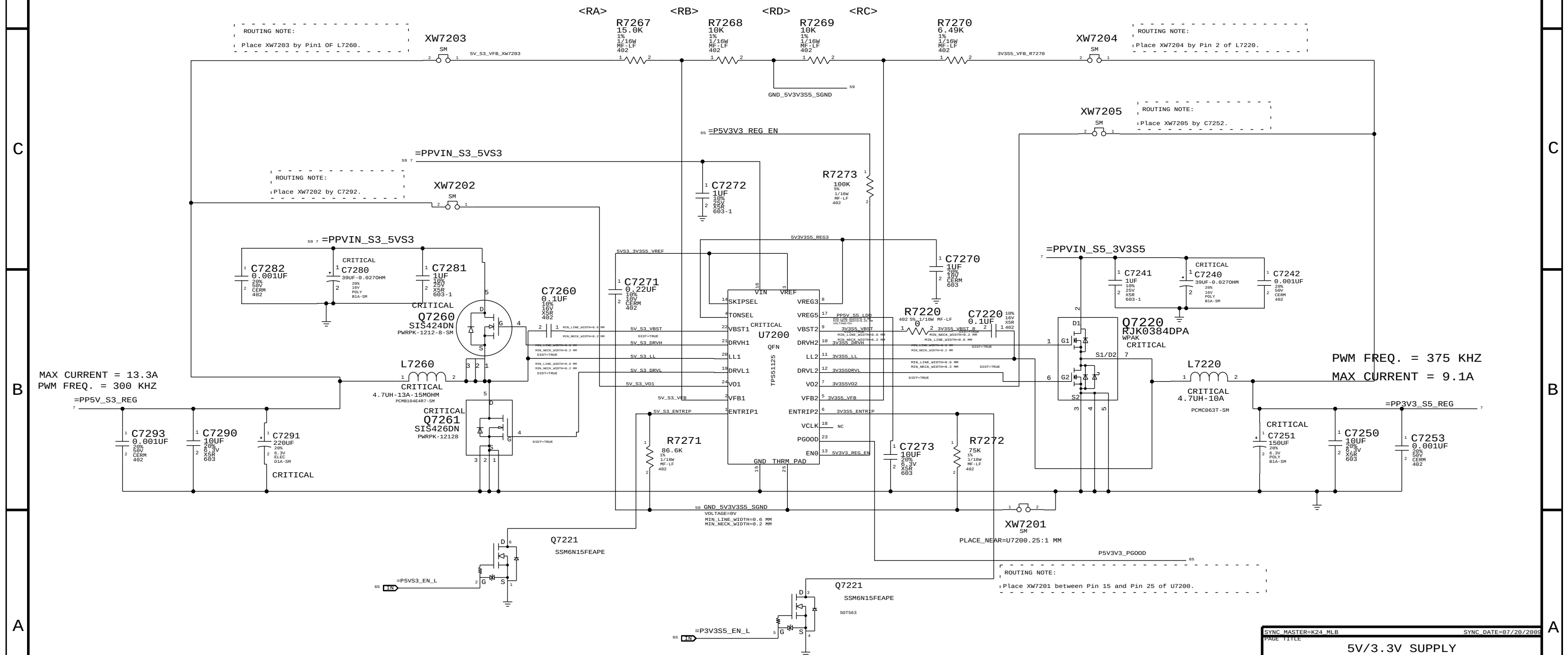


K6 NOTES : L7030 CHANGED BACK TO K24 IND DUE TO LAYOUT

SYNC MASTER=T27_MLB		SYNC DATE=07/29/2005	
PAGE TITLE			
PBus Supply & Battery Charger			
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		REVISION	A.13.0
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$$V_{OUT} = (2 * R_C / R_D) + 2$$



SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3.

SYNC MASTER-K24 MLB		SYNC DATE=07/20/2008	
PAGE TITLE			
5V/3.3V SUPPLY			
	Apple Inc.	DRAWING NUMBER	051-8563
		SIZE	D
		REVISION	A.13.0
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		SHEET	59 OF 80



C

B

A

D |

C

3 |

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SYNC MASTER=T27 MLB		SYNC DATE=08/06/2009	
PAGE TITLE		DRAWING NUMBER	
1.5V/1.35V LVDDR3 Supply		051-8563	
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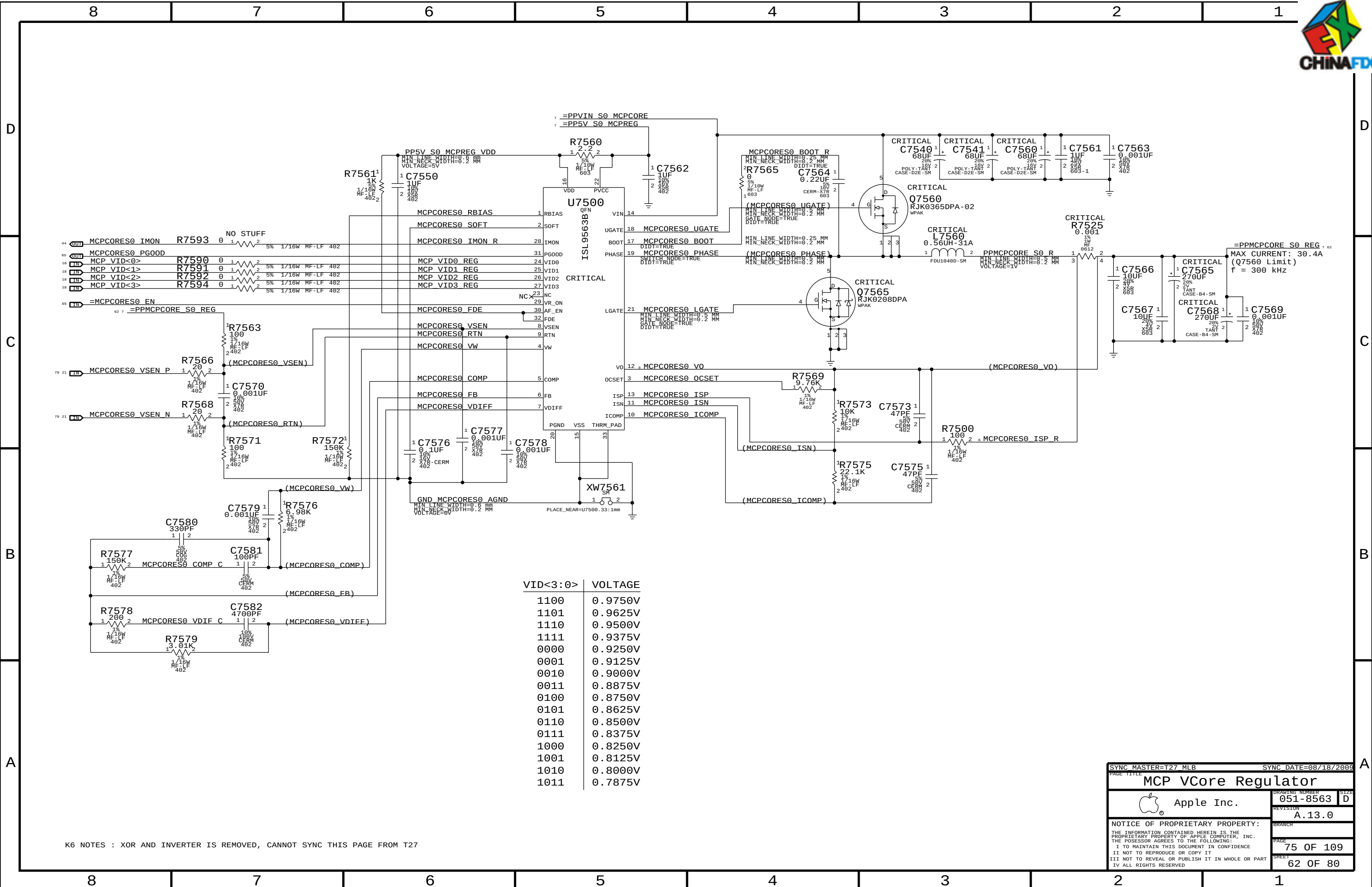
A

K6 NOTES : Q7400-Q7403 CHANGED BACK TO K24 FETS DUE TO LAYOUT
K6 NOTES : BOM OPTION ADDED TO NTC

K6 NOTES : Q7400-Q7403 CHANGED BACK TO K24 FETS DUE TO LAYOUT
K6 NOTES : BOM OPTION ADDED TO NTC

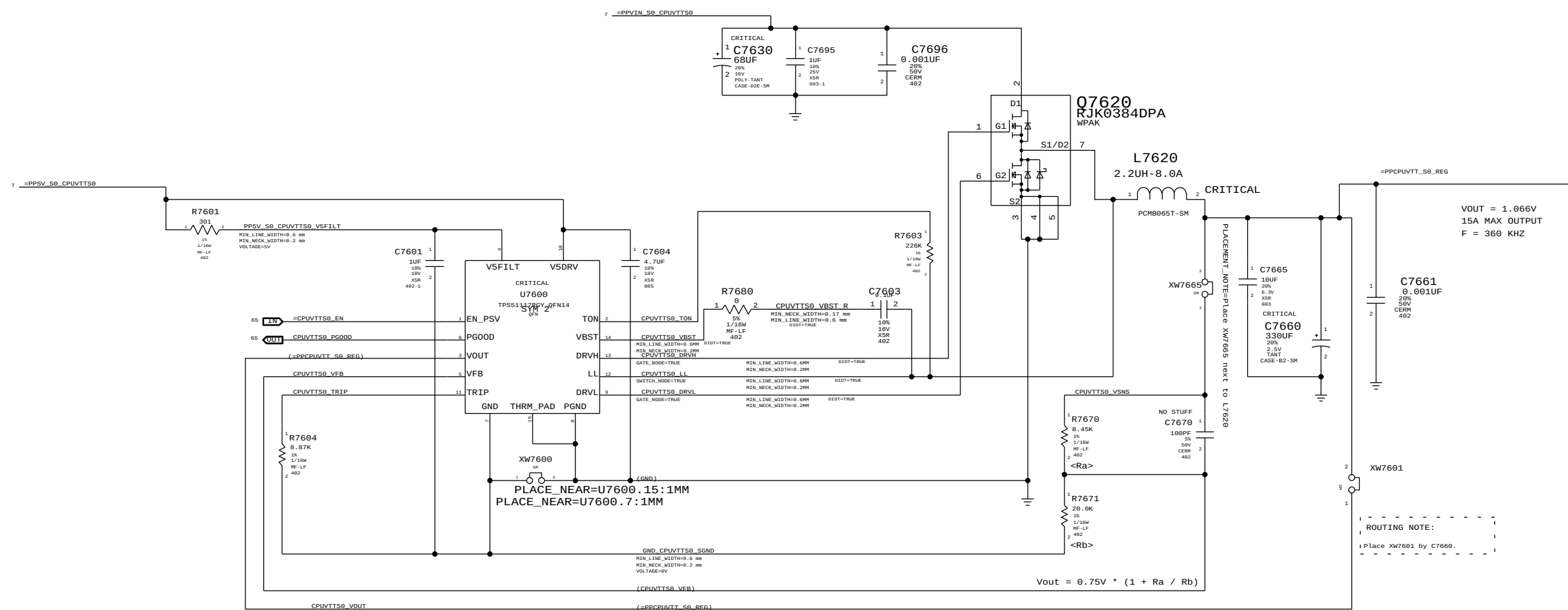
K6 NOTES : Q7400-Q7403 CHANGED BACK TO K24 FETS DUE TO LAYOUT
K6 NOTES : BOM OPTION ADDED TO NTC

K6 NOTES : Q7400-Q7403 CHANGED BACK TO K24 FETS DUE TO LAYOUT
K6 NOTES : BOM OPTION ADDED TO NTC



SYNC MASTER=T27 MLB		SYNC DATE=08/18/2009	
PAGE TITLE			
MCP VCore Regulator			
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	SIZE	D	
	REVISION	A.13.0	
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		SHEET	
		62 OF 80	

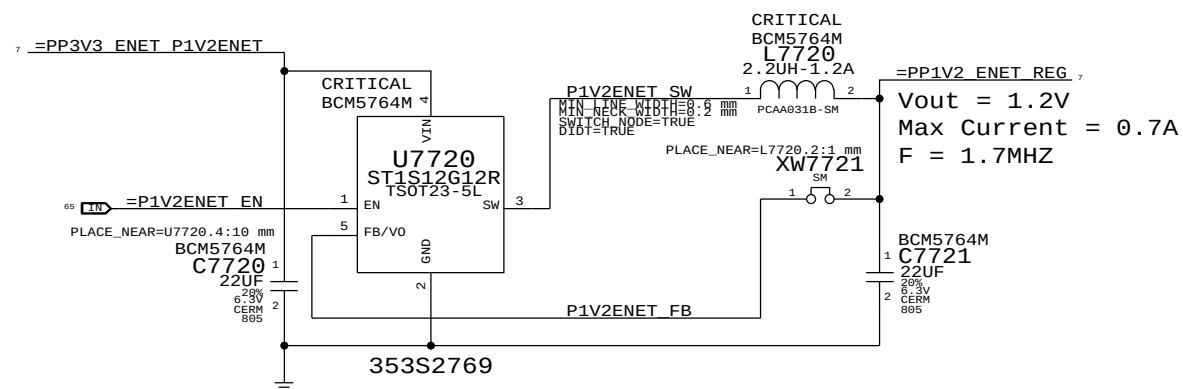
CPUVTT POWER SUPPLY



K6 NOTES : Q7620 CHANGED BACK TO K24 FETS DUE TO LAYOUT

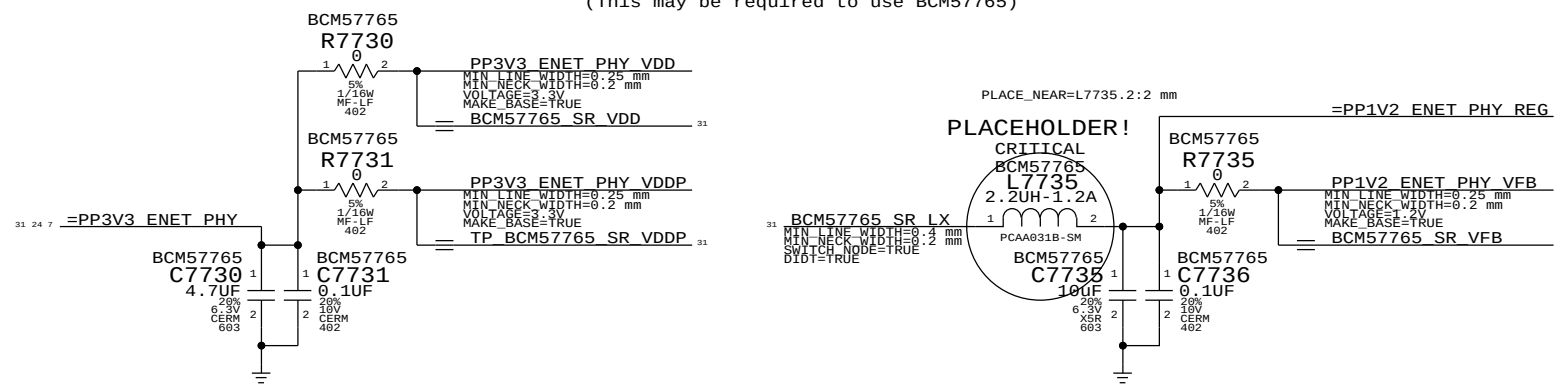
SYNC MASTER=K24 MLB		SYNC DATE=07/20/2005	
PAGE TITLE			
CPU VTT(1.05V) SUPPLY			
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		SHEET	63 OF 80

1.2V ENET Switcher

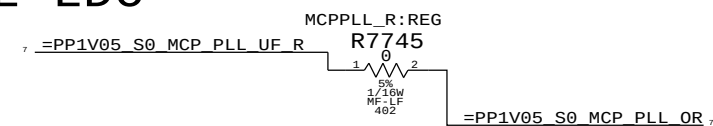


BCM57765 Internal Switcher Support

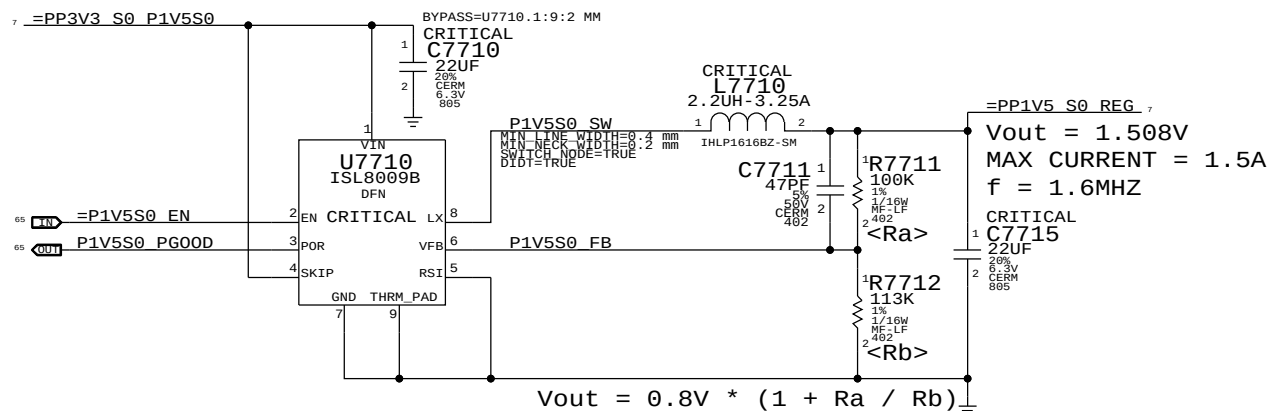
(This may be required to use BCM57765)



1.05V S0 MCP PLL LDO



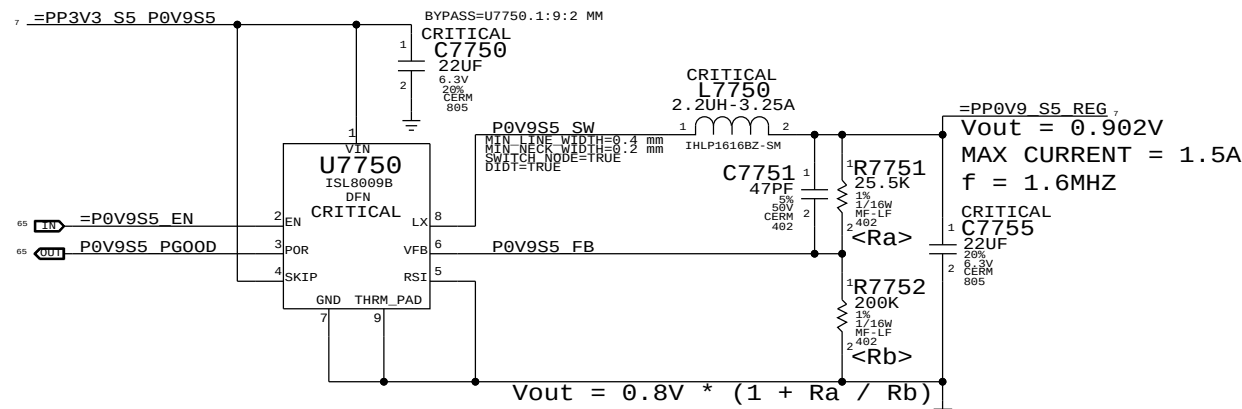
1.5V S0 Regulator



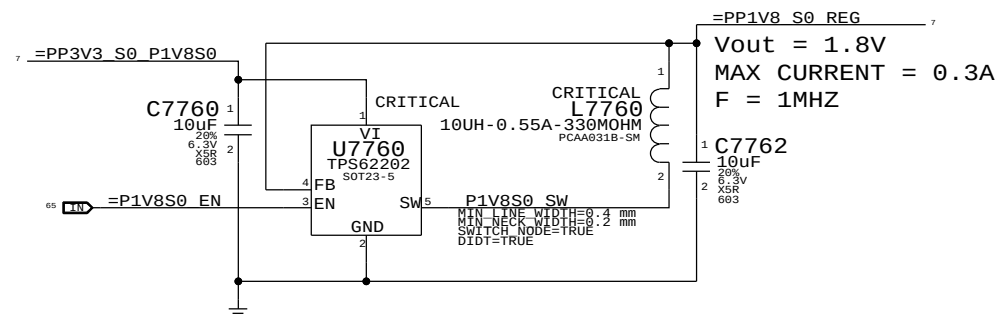
BOMOPTIONS:

MCPPLL_R:LDO - 1.05V S0 USED FOR MCP PLL LDO POWER.
MCPPLL_LDO - STUFFS U7740 AND RELATED CIRCUITRY.
TO USE U7740, MCPPLL_R:LDO AND MCPPLL_LDO MUST BE ACTIVE.
TO USE 1.05V S0, MCPPLL_R:REG MUST BE ACTIVE, MCPPLL_LDO CAN BE ACTIVE, MCPPLL_R:LDO MUST BE INACTIVE.

MCP 0.9V S5 (AUXC) Switcher



1.8V S0 Switcher



K6 NOTES : C7710 AND C7750 HAS BYPASS PROPERTY, SHOULD BE ADDED INCASE THIS PAGE IS SYNC'ED FROM T27

SYNC MASTER=T27_MLB		SYNC DATE=09/30/2006	
PAGE TITLE			
Misc Power Supplies			
		DRAWING NUMBER	051-8563
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D



B



VTT rail must ramp up in about the same time as MEMVDD rail (Q2300).

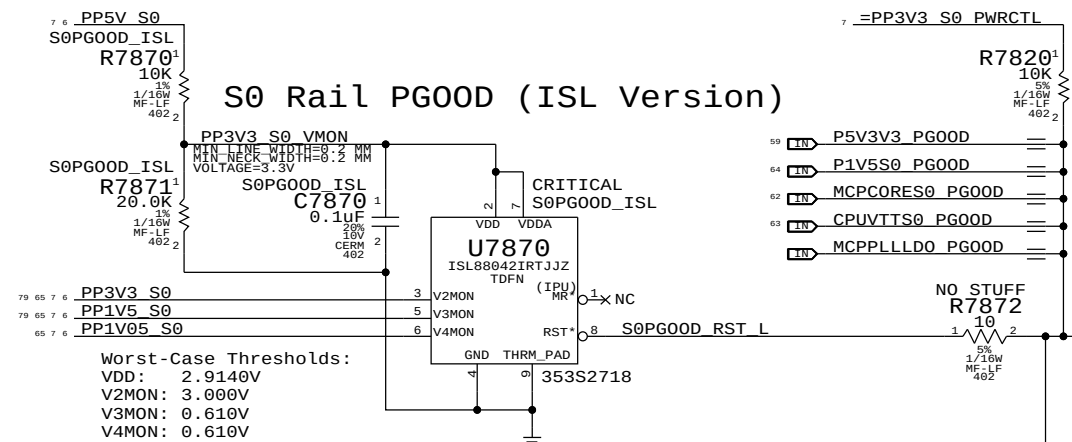
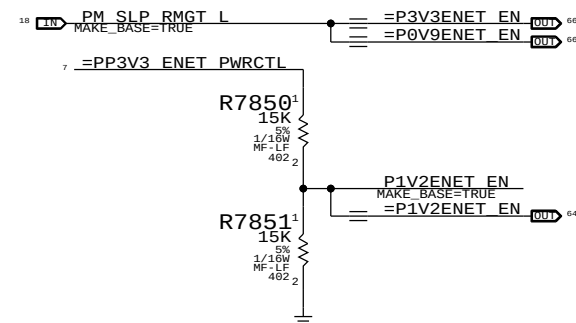
5110

WLAN Enable Generation

```
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
```

NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.

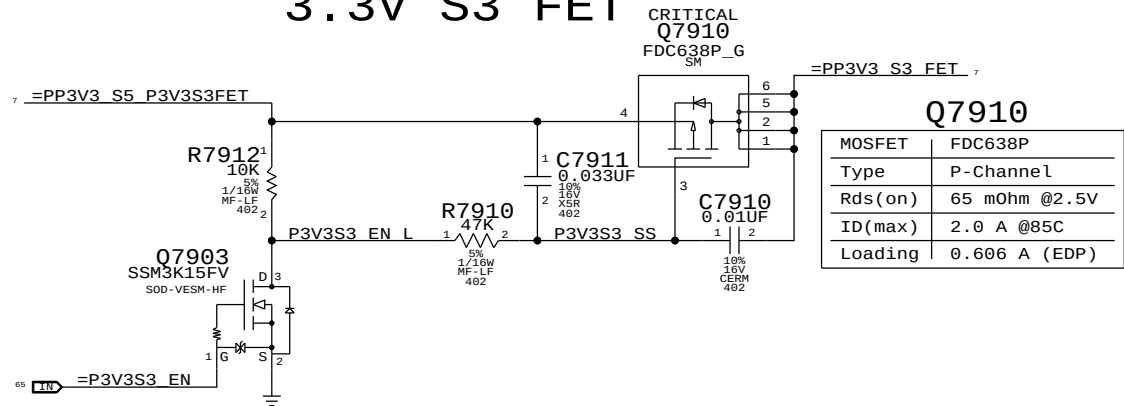
PM WLAN EN L  29
Pull-up is with power FET.

[illegible]

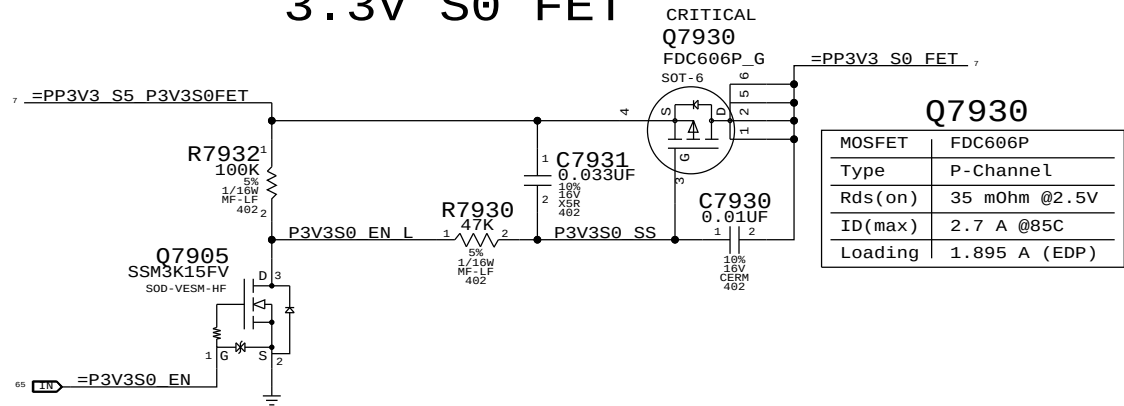
Δ



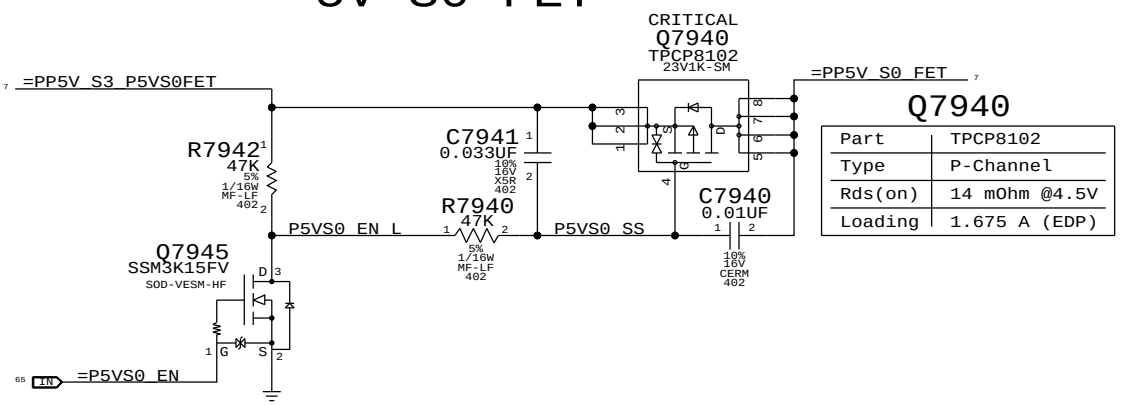
3.3V S3 FET



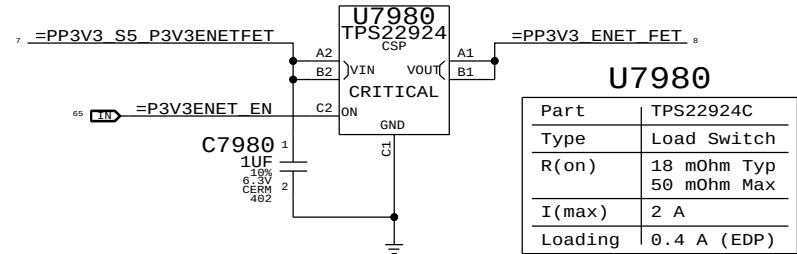
3.3V S0 FET



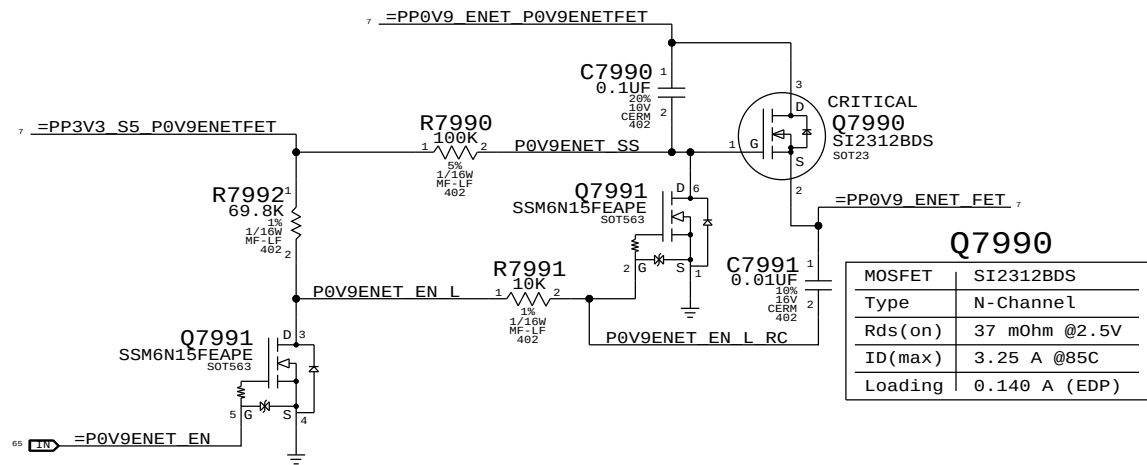
5V S0 FET



3.3V ENET Switch



0.9V ENET FET



SYNC_MASTER=T27_MLB		SYNC_DATE=08/27/2009	
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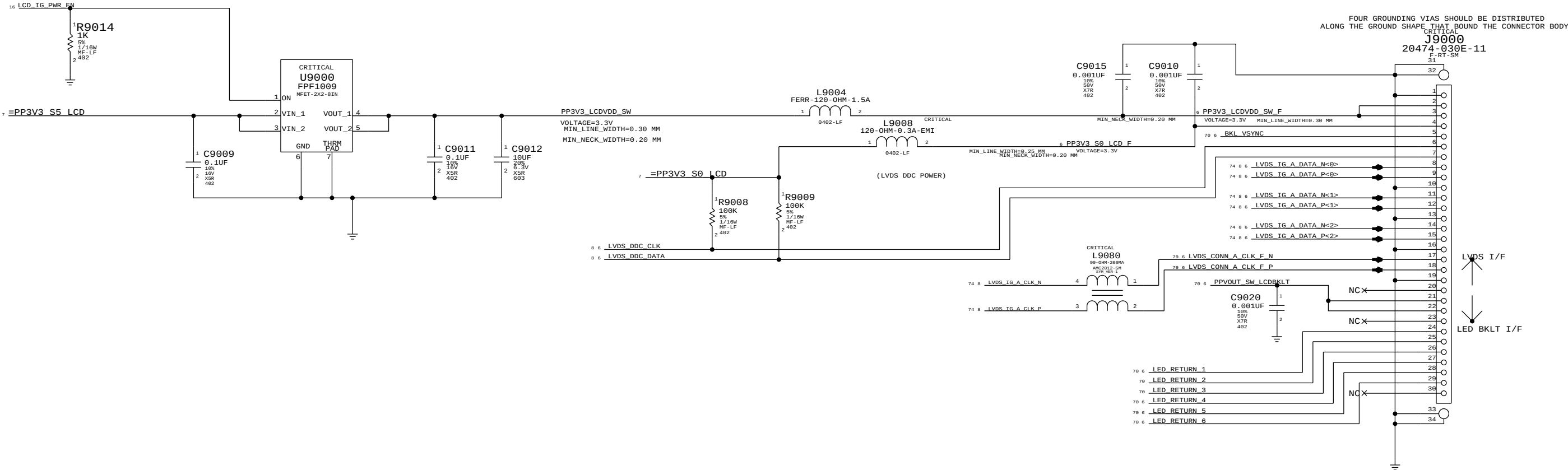
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SYNC MASTER=K24 MLB		SYNC DATE=07/20/2009	
PAGE TITLE		LVDS CONNECTOR	
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		PAGE	90 OF 109
		SHEET	67 OF 80

8

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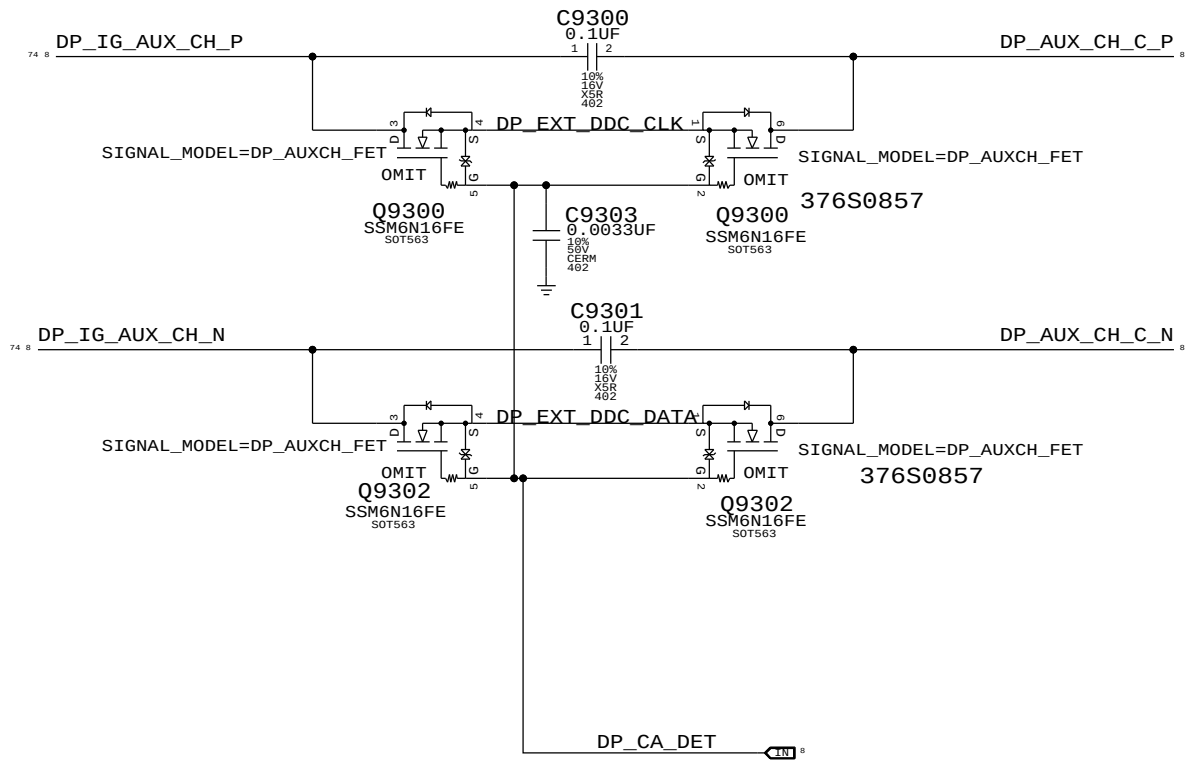
5

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2

1



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0859	2	XSTR, FT, N-CH, DUAL, SOT-563	Q9300, Q9302	CRITICAL	

SYNC MASTER=K69 MLB

SYNC DATE=08/12/2009

PAGE TITLE

DISPLAYPORT SUPPORT

Apple Inc.

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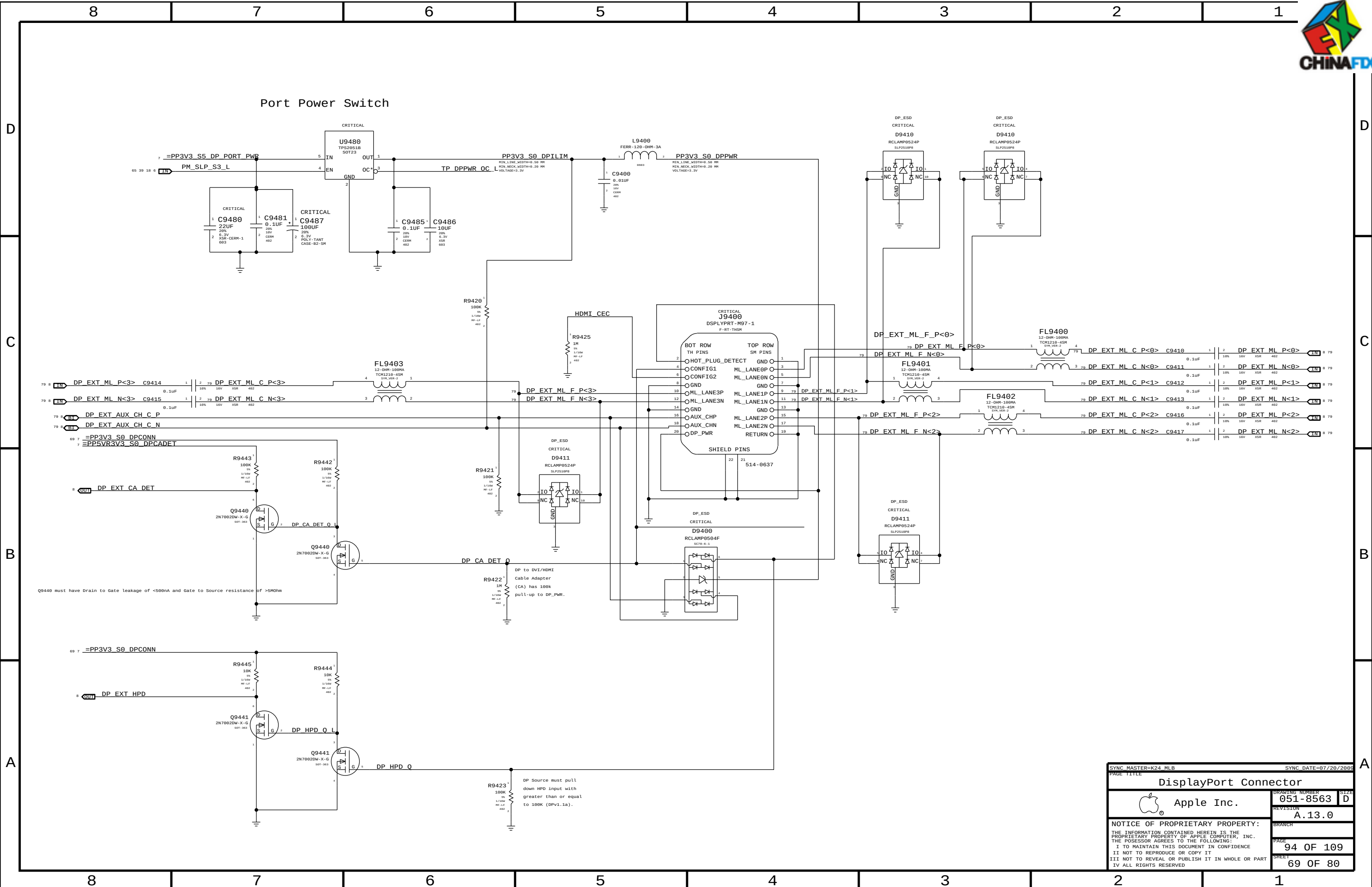
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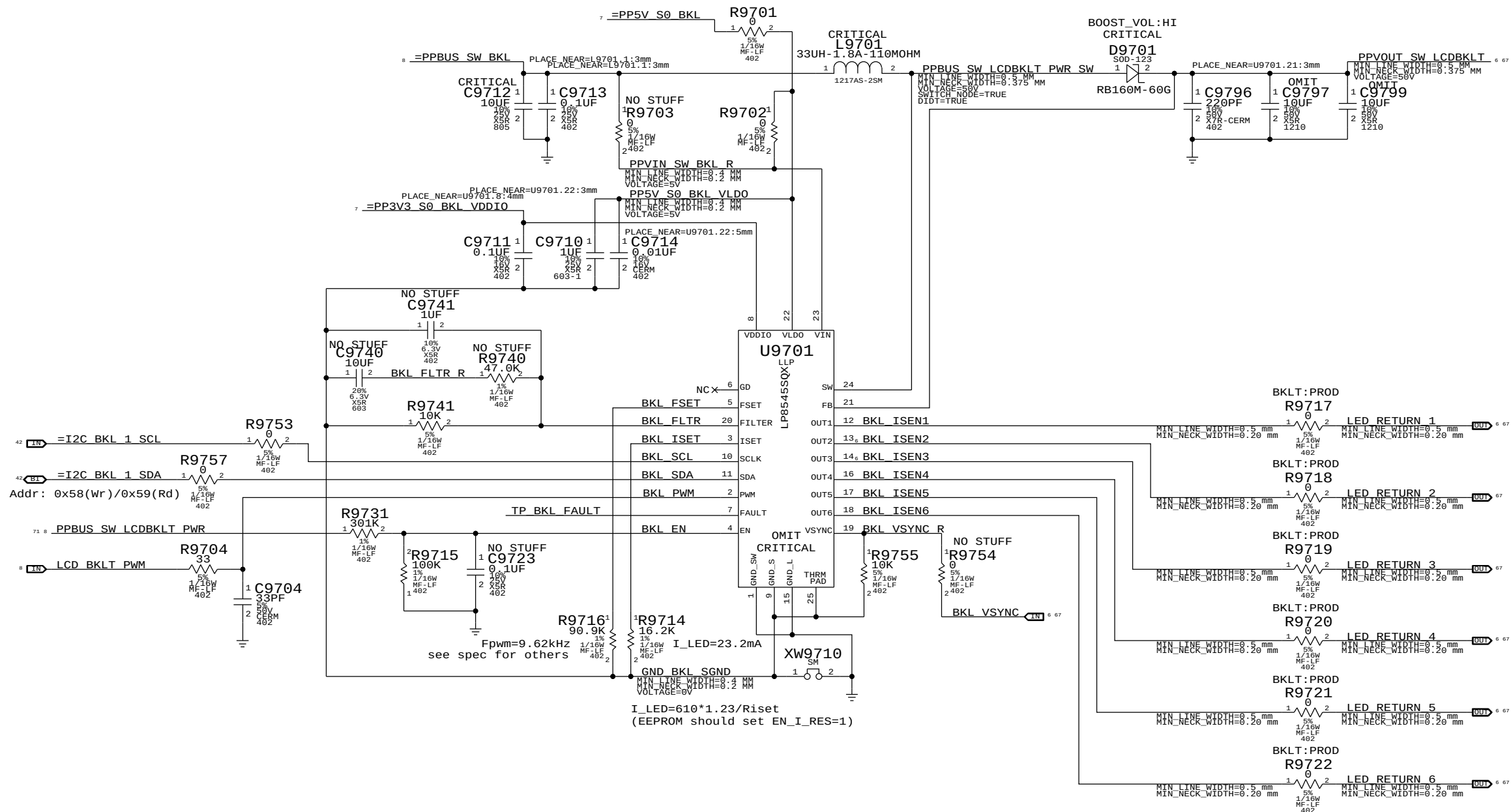
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PAGE
93 OF 109

SHEET
68 OF 80



*L9701, D9701, C9796, C9797, C9799, C9712 AND C9713 SHOULD ALL BE PLACED NEAR EACHOTHER.
 *PPBUS_SW_LCDBKLT_PWR_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
 *LCD_BKLT_PWM SHOULD BE AWAY FROM BOOST CIRCUIT

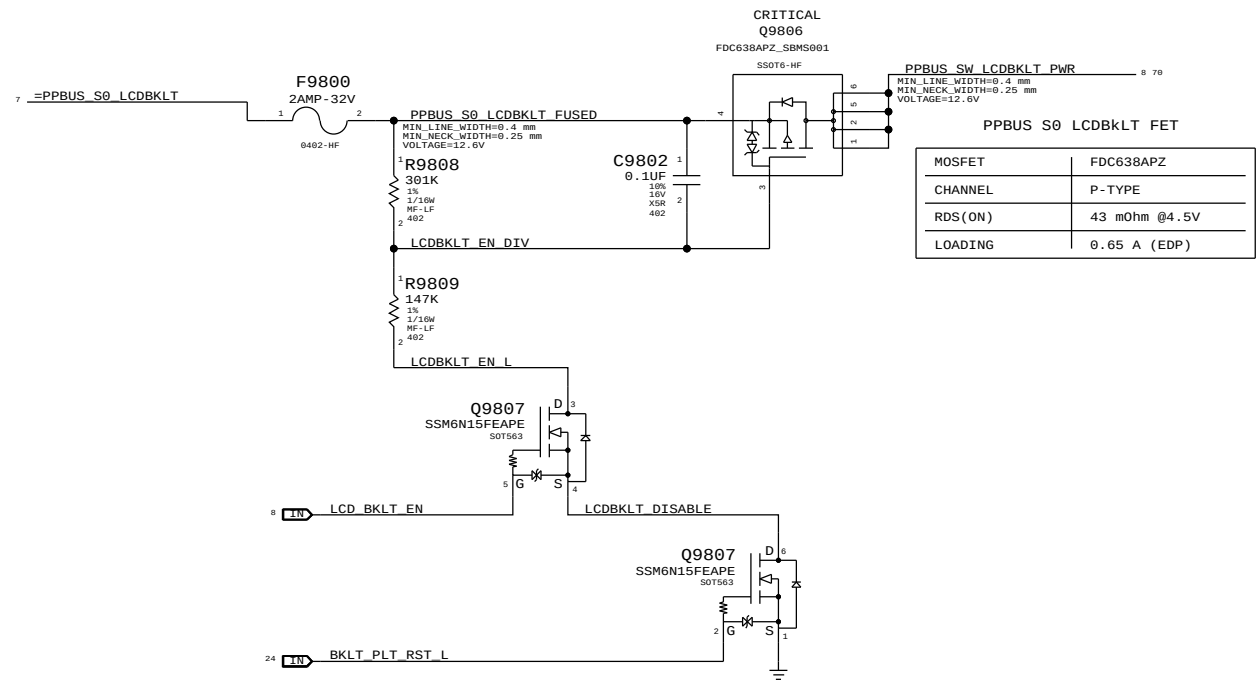


FOR LP8543:
 STUFF R9741
 NO STUFF R9740, C9740, C9741, R9754

PART NUMBER	QTY		REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG
371S0580	1	SCHOTTKY BARRIER DIODE RB160M-40	D9701		BOOST_VOL:LOW
138S0673	2	CAP, 56V, 1210, XSR, 10UF +/-10%	C9797, C9799	CRITICAL	

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=K69 MLB		SYNC DATE=08/27/2009	
PAGE TITLE		LCD Backlight Driver	
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FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2X_DIELECTRIC	?	FSB_DATA	TOP,BOTTOM	=4X_DIELECTRIC	?
FSB_DSTB	*	=3X_DIELECTRIC	?	FSB_DSTB	TOP,BOTTOM	=5X_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?	FSB_ADDR	TOP,BOTTOM	=3X_DIELECTRIC	?
FSB_ADSTB	*	=2X_DIELECTRIC	?	FSB_ADSTB	TOP,BOTTOM	=4X_DIELECTRIC	?
FSB_1X	*	=STANDARD	?	FSB_1X	TOP,BOTTOM	=3X_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.

Signals within each 4x group should be matched within 5 ps of strobe.

DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 135 ps.

Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.

FSB 2X signals / groups shown in signal table on right.

Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 270 ps.

Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.

Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1

SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP,BOTTOM	=2X_DIELECTRIC	?
CPU_8MIL	*	8 MIL	?				
CPU_COMP	*	25 MIL	?				
CPU_GTLREF	*	25 MIL	?				
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

SR DG recommends at least 25 mils, >50 mils preferred

Most CPU signals with impedance requirements are 55-ohm single-ended.

Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1

SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3X_DIELECTRIC	?	CLK_FSB	TOP,BOTTOM	=4X_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
		PHYSICAL	SPACING	
FSB 4X Signal Groups	FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB D L<15..0>
	FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB DINV L<0>
	FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<0>
	FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<0>
	FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB D L<31..16>
	FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB DINV L<1>
	FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<1>
	FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<1>
	FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB D L<47..32>
	FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB DINV L<2>
	FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<2>
	FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<2>
FSB 2X Signals	FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB D L<63..48>
	FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB DINV L<3>
	FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L P<3>
	FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB DSTB L N<3>
	FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB A L<16..3>
	FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB REQ L<4..0>
	FSB_ADSTB0	FSB_50S	FSB_ADSTB	FSB ADSTB L<0>
	FSB_ADDR_GROUP1	FSB_50S	FSB_ADDR	FSB A L<35..17>
	FSB_ADSTB1	FSB_50S	FSB_ADSTB	FSB ADSTB L<1>
	FSB_1X	FSB_50S	FSB_1X	FSB ADS L
	FSB_BREQ0_L	FSB_50S	FSB_1X	FSB_BREQ0_L
	FSB_1X	FSB_50S	FSB_1X	FSB BNR L
FSB 1X Signals	FSB_1X	FSB_50S	FSB_1X	FSB BPRT L
	FSB_1X	FSB_50S	FSB_1X	FSB DBSY L
	FSB_1X	FSB_50S	FSB_1X	FSB DEFER L
	FSB_1X	FSB_50S	FSB_1X	FSB DRDY L
	FSB_1X	FSB_50S	FSB_1X	FSB HIT L
	FSB_1X	FSB_50S	FSB_1X	FSB HITM L
	FSB_1X	FSB_50S	FSB_1X	FSB LOCK L
	FSB_CPURST_L	FSB_50S	FSB_1X	FSB_CPURST_L
	FSB_1X	FSB_50S	FSB_1X	FSB RS L<2..0>
	FSB_1X	FSB_50S	FSB_1X	FSB TRDY L
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU A20M L
	CPU_BSEL	CPU_50S	CPU_AGTL	CPU BSEL<2..0>
	CPU_FERR_L	CPU_50S	CPU_8MIT	CPU FERR L
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU IGNNE L
	CPU_TWIT_L	CPU_50S	CPU_AGTL	CPU INIT L
	CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU INTR
	CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU NMI
	CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU PROCHOT L
	CPU_PWRGD	CPU_50S	CPU_AGTL	CPU PWRGD
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU SMI L
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU STPCLK L
	PM_THRMTRIP_L	CPU_50S	CPU_8MIT	PM THRMTRIP L
	FSB_CPUSLP_L	CPU_50S	CPU_AGTL	FSB CPUSLP L
	CPU_FROM_SB	CPU_50S	CPU_AGTL	CPU DPSLP L
	CPU_DPRSTP_L	CPU_50S	CPU_AGTL	CPU DPRSTP L
	CPU_ASYNC	CPU_50S	CPU_AGTL	FSB DPWR L
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB CLK CPU P
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB CLK CPU N
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB CLK ITP P
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB CLK ITP N
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB CLK MCP P
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB CLK MCP N
	CPU_IERR_L	CPU_50S		CPU IERR L
	PM DPRSLPVR	CPU_50S	CPU_AGTL	PM DPRSLPVR
	(See above)	CPU_50S	CPU_AGTL	IMVP DPRSLPVR
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP VDD
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP GND
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP CPU COMP VCC
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP CPU COMP GND
	CPU_GTLREF	CPU_50S	CPU_GTLREF	CPU GTLREF
	CPU_COMP	CPU_50S	CPU_COMP	CPU COMP<3>
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU COMP<2>
	CPU_COMP	CPU_50S	CPU_COMP	CPU COMP<1>
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU COMP<0>
	XDP_TDI	CPU_50S	CPU_ITP	XDP TDI
	XDP_TDO	CPU_50S	CPU_ITP	XDP TDO
	XDP_TMS	CPU_50S	CPU_ITP	XDP TMS
	XDP_TCK	CPU_50S	CPU_ITP	XDP TCK
	XDP_TRST_L	CPU_50S	CPU_ITP	XDP TRST L
	XDP_BPM_L	CPU_50S	CPU_ITP	XDP BPM L<4..0>
	XDP_BPM_L5	CPU_50S	CPU_ITP	XDP BPM L<5>
	(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDP CPURST L
		CPU_50S	CPU_8MIT	CPU VID<6..0>
		CPU_50S	CPU_8MIT	IMVP6 VID<6..0>
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU VCCSENSE P
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU VCCSENSE N
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6 VSEN P
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6 VSEN N

SYNC MASTER=T27_MLB		SYNC DATE=08/03/2009	
PAGE TITLE			
CPU/FSB Constraints			
 Apple Inc.		DRAWING NUMBER	051-8563
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		PAGE	100 OF 109
		SHEET	72 OF 80



Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MIL	?

NV DG says 3x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 4x inner, 5x outer

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

DDR3:
DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 360 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
CMD/CTRL signals should be matched within 150 ps.
All memory signals maximum length is 1.030 ps.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.3
SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	=2X_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK P<5..0>	14 25
MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK N<5..0>	14 25
MEM_A_CKE	MEM_40S	MEM_CTRL	MEM A CKE<3..0>	14 26 25
MEM_A_CNT1	MEM_40S	MEM_CTRL	MEM A CS L<3..0>	14 25
MEM_A_CNT1	MEM_40S	MEM_CTRL	MEM A ODT<3..0>	14 25
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	14 25
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	14 25
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	14 25
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	14 25
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	14 25
MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM A DQ<7..0>	14 27
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DQ<15..8>	14 27
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DQ<23..16>	14 27
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DQ<31..24>	14 27
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DQ<39..32>	14 27
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DQ<47..40>	14 27
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DQ<55..48>	14 27
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DQ<63..56>	14 27
MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM A DM<0>	14 27
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DM<1>	14 27
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DM<2>	14 27
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DM<3>	14 27
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DM<4>	14 27
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DM<5>	14 27
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DM<6>	14 27
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DM<7>	14 27
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS P<0>	14 27
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS N<0>	14 27
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS P<1>	14 27
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS N<1>	14 27
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS P<2>	14 27
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS N<2>	14 27
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS P<3>	14 27
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS N<3>	14 27
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS P<4>	14 27
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS N<4>	14 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS P<5>	14 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS N<5>	14 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS P<6>	14 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS N<6>	14 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS P<7>	14 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS N<7>	14 27
MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK P<5..0>	14 26
MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK N<5..0>	14 26
MEM_B_CKE	MEM_40S	MEM_CTRL	MEM B CKE<3..0>	14 26 25
MEM_B_CNT1	MEM_40S	MEM_CTRL	MEM B CS L<3..0>	14 26
MEM_B_CNT1	MEM_40S	MEM_CTRL	MEM B ODT<3..0>	14 26
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	14 26
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	14 26
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	14 26
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	14 26
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	14 26
MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM B DQ<7..0>	14 27
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DQ<15..8>	14 27
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DQ<23..16>	14 27
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DQ<31..24>	14 27
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DQ<39..32>	14 27
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DQ<47..40>	14 27
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DQ<55..48>	14 27
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DQ<63..56>	14 27
MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM B DM<0>	14 27
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DM<1>	14 27
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DM<2>	14 27
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DM<3>	14 27
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DM<4>	14 27
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DM<5>	14 27
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DM<6>	14 27
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DM<7>	14 27
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS P<0>	14 27
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS N<0>	14 27
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS P<1>	14 27
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS N<1>	14 27
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS P<2>	14 27
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS N<2>	14 27
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS P<3>	14 27
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS N<3>	14 27
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS P<4>	14 27
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS N<4>	14 27
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS P<5>	14 27
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS N<5>	14 27
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS P<6>	14 27
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS N<6>	14 27
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS P<7>	14 27
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS N<7>	14 27
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP VDD	14
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP GND	14

MEM_A/B_CKE EC SET NAME IS CHANGED ON K6, CANNOT SYNC THIS PAGE FROM T27

SYNC MASTER=T27_MLB		SYNC DATE=08/03/2009	
PAGE TITLE			
Memory Constraints			
Apple Inc.		DRAWING NUMBER	051-8563
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		PAGE	101 OF 109
		SHEET	73 OF 80



PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?	PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?
CLK_PCIE	*	20_MIL	?				
MCP_PEX_COMP	*	8_MIL	?				

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.3

NEED PCIe Gen1/Gen2 notes!

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CRT	*	20_MIL	?	CRT	CRT	*	CRT_2CRT
CRT_2CRT	*	15_MIL	?				
CRT_2CLK	*	50_MIL	?				
CRT_2SWITCHER	*	250_MIL	?				
CRT_SYNC	*	=4X_DIELECTRIC	?				
MCP_DAC_COMP	*	=2X_DIELECTRIC	?				

CRT signal single-ended impedance varies by location:

- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from first to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).

R/G/B signals should be matched as close as possible and < 10 inches.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.1.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20_MIL	20_MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3X_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4X_DIELECTRIC	?
LVDS	*	=3X_DIELECTRIC	?	LVDS	TOP,BOTTOM	=4X_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be matched within 100 mils.
NOTE: NV DG recommends 90 ohm differential for LVDS, but cable/display assume 100 ohm.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 100 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max trace length: LVDS 10 inches, DP 8.5 inches.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.2

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=3X_DIELECTRIC	?	SATA	TOP,BOTTOM	=4X_DIELECTRIC	?
SATA_TERM	*	8_MIL	?				

SATA intra-pair matching should be 1 ps.

Max trace length: 12 inches for SATA Gen1/Gen2, TBD for SATA Gen3.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.6

MCP89 Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_P<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_N<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_P<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_N<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_P<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_N<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_C_P<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_C_N<15..0>
PCIE_AP_R2D_P	PCIE_90D	PCIE	PCIE_AP_R2D_P
PCIE_AP_R2D_N	PCIE_90D	PCIE	PCIE_AP_R2D_N
PCIE_AP_R2D_C_P	PCIE_90D	PCIE	PCIE_AP_R2D_C_P
PCIE_AP_R2D_C_N	PCIE_90D	PCIE	PCIE_AP_R2D_C_N
PCIE_AP_D2R_P	PCIE_90D	PCIE	PCIE_AP_D2R_P
PCIE_AP_D2R_N	PCIE_90D	PCIE	PCIE_AP_D2R_N
PCIE_ENET_R2D_P	PCIE_90D	PCIE	PCIE_ENET_R2D_P
PCIE_ENET_R2D_N	PCIE_90D	PCIE	PCIE_ENET_R2D_N
PCIE_ENET_R2D_C_P	PCIE_90D	PCIE	PCIE_ENET_R2D_C_P
PCIE_ENET_R2D_C_N	PCIE_90D	PCIE	PCIE_ENET_R2D_C_N
PCIE_ENET_D2R_P	PCIE_90D	PCIE	PCIE_ENET_D2R_P
PCIE_ENET_D2R_N	PCIE_90D	PCIE	PCIE_ENET_D2R_N
PCIE_ENET_D2R_C_P	PCIE_90D	PCIE	PCIE_ENET_D2R_C_P
PCIE_ENET_D2R_C_N	PCIE_90D	PCIE	PCIE_ENET_D2R_C_N
PCIE_FW_R2D_P	PCIE_90D	PCIE	PCIE_FW_R2D_P
PCIE_FW_R2D_N	PCIE_90D	PCIE	PCIE_FW_R2D_N
PCIE_FW_R2D_C_P	PCIE_90D	PCIE	PCIE_FW_R2D_C_P
PCIE_FW_R2D_C_N	PCIE_90D	PCIE	PCIE_FW_R2D_C_N
PCIE_FW_D2R_P	PCIE_90D	PCIE	PCIE_FW_D2R_P
PCIE_FW_D2R_N	PCIE_90D	PCIE	PCIE_FW_D2R_N
PCIE_FW_D2R_C_P	PCIE_90D	PCIE	PCIE_FW_D2R_C_P
PCIE_FW_D2R_C_N	PCIE_90D	PCIE	PCIE_FW_D2R_C_N
MCP_PEG_REECLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_P
MCP_PEG_REECLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_N
MCP_PEG1_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_P
MCP_PEG1_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_N
MCP_PEG2_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_P
MCP_PEG2_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_N
MCP_PEG3_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P
MCP_PEG3_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N
MCP_PEX_CLK_COMP		MCP_PEX_COMP	MCP_PEX0_TERM
CRT_RED	CRT_50S	CRT	CRT_IG_R_C_PR
CRT_GREEN	CRT_50S	CRT	CRT_IG_G_Y_Y
CRT_BLUE	CRT_50S	CRT	CRT_IG_B_COMP_PB
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_HSYNC
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_VSYNC
MCP_DAC_RSET		MCP_DAC_COMP	MCP_TV_DAC_RSET
MCP_DAC_VREF		MCP_DAC_COMP	MCP_TV_DAC_VREF
TMDS_IG_TXC	DP_90D	DISPLAYPORT	TMDS_IG_TXC_P
TMDS_IG_TXC	DP_90D	DISPLAYPORT	TMDS_IG_TXC_N
TMDS_IG_TXD	DP_90D	DISPLAYPORT	TMDS_IG_TXD_P<5..0>
TMDS_IG_TXD	DP_90D	DISPLAYPORT	TMDS_IG_TXD_N<5..0>
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML_P<3..0>
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML_N<3..0>
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH_P
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH_N
MCP_TMDS0_RSET	MCP_DV_COMP		MCP_TMDS0_RSET
MCP_TMDS0_VPROBE			MCP_TMDS0_VPROBE
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_P
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_N
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<2..0>
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<2..0>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<3>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<3>
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_P
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_N
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<2..0>
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<2..0>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<3>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<3>
MCP_IFPAB_RSET	MCP_DV_COMP		MCP_IFPAB_RSET
MCP_IFPAB_VPROBE			MCP_IFPAB_VPROBE
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_P
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_N
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_P
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_N
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_P
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_N
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_P
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_N
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_P
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_N
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_P
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_N
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_P
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_N
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_C_P
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_C_N
MCP_SATA_TERM		SATA_TERM	MCP_SATA_TERM

SYNC_MASTER=T27_MLB		SYNC_DATE=08/03/2009	
PAGE TITLE			
MCP Constraints 1			
 Apple Inc.		DRAWING NUMBER	051-8563
		SIZE	D
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		BRANCH	
		PAGE	102 OF 109
		SHEET	74 OF 80



LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
LPC	*	=1.5x_DIELECTRIC	?
CLK_LPC	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.7

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.8

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.9

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.10

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_SLOW	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.11

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SPI	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.12

MCP89 Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
LPC_AD	LPC_55S	LPC	LPC AD<3..0> 18 39 41
LPC_FRAME_L	LPC_55S	LPC	LPC FRAME L 18 39 41
LPC_RESET_L	LPC_55S	LPC	LPC RESET L 18 24
MCP_LPC_CLK0	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC R 18 24
	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC 24 39
	CLK_LPC_55S	CLK_LPC	LPC CLK33M LPCPLUS 24 41
USB_EXT_A	USB_90D	USB	USB EXT_A P 17 37
	USB_90D	USB	USB EXT_A N 17 37
	USB_90D	USB	USB EXT_A MUXED P 37 79
	USB_90D	USB	USB EXT_A MUXED N 37 79
USB_MINI	USB_90D	USB	USB MINI P 8 17
	USB_90D	USB	USB MINI N 8 17
USB_EXTD	USB_90D	USB	USB EXT_D P 8 17
	USB_90D	USB	USB EXT_D N 8 17
USB_CAMERA	USB_90D	USB	USB CAMERA P 17 29
	USB_90D	USB	USB CAMERA N 17 29
USB_BT	USB_90D	USB	USB BT P 17 29
	USB_90D	USB	USB BT N 17 29
USB_TPAD	USB_90D	USB	USB TPAD P 17 47
	USB_90D	USB	USB TPAD N 17 47
USB_IR	USB_90D	USB	USB IR P 17 38
	USB_90D	USB	USB IR N 17 38
USB_EXTB	USB_90D	USB	USB EXT_B P 17 37
	USB_90D	USB	USB EXT_B N 17 37
USB_T57	USB_90D	USB	USB T57 P 6 38
	USB_90D	USB	USB T57 N 6 38
USB_EXTC	USB_90D	USB	USB EXT_C P 8 17
	USB_90D	USB	USB EXT_C N 8 17
USB_SDCARD	USB_90D	USB	USB SDCARD P 17 38
	USB_90D	USB	USB SDCARD N 17 38
USB_WM	USB_90D	USB	USB WM P 8 17
	USB_90D	USB	USB WM N 8 17
MCP_USB_RBIAS	MCP_USB_RBIAS		MCP USB RBIAS GND 17
SMBUS_MCP_0_CLK	SMB_55S	SMB	SMBUS MCP_0_CLK 12 18 42
SMBUS_MCP_0_DATA	SMB_55S	SMB	SMBUS MCP_0_DATA 12 18 42
(SMBUS_SMC_MGMT_SCL)	SMB_55S	SMB	SMBUS MCP_1_CLK 18 42
(SMBUS_SMC_MGMT_SDA)	SMB_55S	SMB	SMBUS MCP_1_DATA 18 42
HDA_BIT_CLK	HDA_55S	HDA	HDA BIT_CLK 18 51
	HDA_55S	HDA	HDA BIT_CLK R 18
HDA_SYNC	HDA_55S	HDA	HDA SYNC 18 51
	HDA_55S	HDA	HDA SYNC R 18
HDA_RST_L	HDA_55S	HDA	HDA RST_R_L 18
	HDA_55S	HDA	HDA RST_L 18 51
HDA_SDIN0	HDA_55S	HDA	HDA SDIN0 18 51
	HDA_55S	HDA	HDA SDIN CODEC 18
HDA_SDOUT	HDA_55S	HDA	HDA SDOUT 18 51
	HDA_55S	HDA	HDA SDOUT_R 18
MCP_HDA_PULLDN_COMP	MCP_HDA_COMP		MCP HDA PULLDN_COMP 18
MCP_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM CLK32K SUSCLK_R 18 24
	CLK_SLOW_55S	CLK_SLOW	PM CLK32K SUSCLK 24 39
SPT_CLK	SPT_55S	SPT	SPI_CLK_R 18 41
	SPT_55S	SPT	SPI_CLK 6 41
SPT_MOST	SPT_55S	SPT	SPI_MOST_R 18 41
	SPT_55S	SPT	SPI_MOST 6 41
SPT_MISO	SPT_55S	SPT	SPI_MISO 6 18 41
SPT_CS0	SPT_55S	SPT	SPI_CS0_R_L 18 41
	SPT_55S	SPT	SPI_CS0_L 6 41
	SPT_55S	SPT	SPI_MLB_CLK 41 98
	SPT_55S	SPT	SPI_MLB_MOSI 41 98
	SPT_55S	SPT	SPI_MLB_MISO 41 98
	SPT_55S	SPT	SPI_MLB_CS_L 41 98
	SPT_55S	SPT	SPI_ALT_CLK 41
	SPT_55S	SPT	SPI_ALT_MOSI 41
	SPT_55S	SPT	SPI_ALT_MISO 41
	SPT_55S	SPT	SPI_ALT_CS_L 41

SYNC_MASTER=T27_MLB		SYNC_DATE=08/27/2009	
PAGE TITLE			
MCP Constraints 2			
	DRAWING NUMBER		SHEET
	051-8563		D
	REVISION		A.13.0
BRANCH			
PAGE			
103 OF 109			
SHEET			
75 OF 80			

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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_MII_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_MDI_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

SD Card Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

RGMII Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
MCP_MII_COMP	MCP_MII_COMP		MCP MII COMP VDD 17
MCP_MII_COMP	MCP_MII_COMP		MCP MII COMP GND 17
MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	MCP CLK25M BUF0 R
	ENET_MII_55S	MCP_BUF0_CLK	RTL8211 CLK25M CKXTAL1
ENET_INTR_L	ENET_MII_55S	ENET_MII	ENET INTR_L
ENET_MDIO	ENET_MII_55S	ENET_MII	ENET MDIO 8 17
ENET_MDC	ENET_MII_55S	ENET_MII	ENET MDC
ENET_PWRDWN_L	ENET_MII_55S	ENET_MII	ENET PWRDWN_L
ENET_RXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK R
	ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK 8 17
	ENET_MII_55S	ENET_MII	ENET RXD R<3..0>
ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET RXD<0> 8 17
ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET RXD<3..1> 8 17
ENET_RXD	ENET_MII_55S	ENET_MII	ENET RX_CTRL 8 17
ENET_TXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M TXCLK
ENET_TXD	ENET_MII_55S	ENET_MII	ENET TXD<0>
ENET_TXD	ENET_MII_55S	ENET_MII	ENET TXD<3..1>
ENET_TXD	ENET_MII_55S	ENET_MII	ENET TX_CTRL
	ENET_MII_55S	ENET_MII	ENET RESET_L 24 31

Ethernet Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
ENET_MDI	ENET_MDI_100D	ENET_MDI	ENET MDI P<3..0> 31 32
	ENET_MDI_100D	ENET_MDI	ENET MDI N<3..0> 31 32

SD Card Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
SD_DATA	SD_55S	SD_INTERFACE	SD D<4..0> 30
	SD_55S	SD_INTERFACE	SDCONN_DATA<4..0> 30 31
	SD_55S	SD_INTERFACE	BCM57765_CR_DATA<4> 31
SD_DATA_R	SD_55S	SD_INTERFACE	SD D<7..5> 30
	SD_55S	SD_INTERFACE	SDCONN_DATA<7..5> 30 31
	SD_55S	SD_INTERFACE	BCM57765_CR_DATA<7..5> 31
SD_CLK	SD_55S	SD_INTERFACE	SD_CLK 30
	SD_55S	SD_INTERFACE	SD_CLK_R 30
	SD_55S	SD_INTERFACE	SDCONN_CLK 30 31
SD_CMD	SD_55S	SD_INTERFACE	SD_CMD 30
	SD_55S	SD_INTERFACE	SDCONN_CMD 30 31
	SD_55S	SD_INTERFACE	BCM57765_CR_CMD 31

NOTE: SD_D<7..5> are different to support BCM5764M/BCM57765 co-layout.

SYNC_MASTER=T27_MLB

SYNC_DATE=11/23/2009

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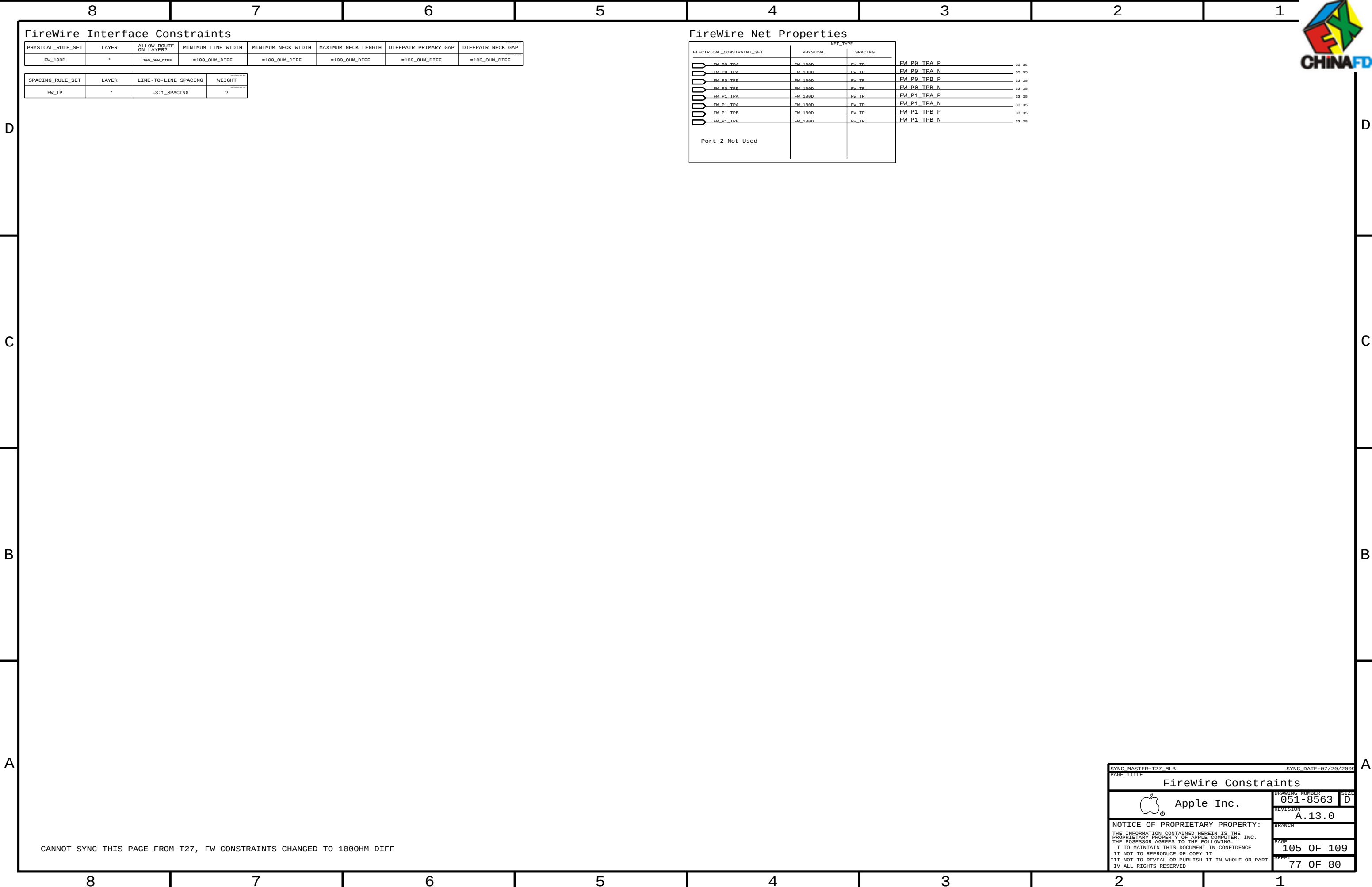
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DRAWING NUMBER
051-8563

REVISION
A.13.0

PAGE
104 OF 109

SHEET
76 OF 80



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SYNC_MASTER=T27_MLB

SYNC_DATE=07/20/2009

PAGE TITLE

FireWire Constraints

Apple Inc.

DRAWING NUMBER

051-8563

SIZE

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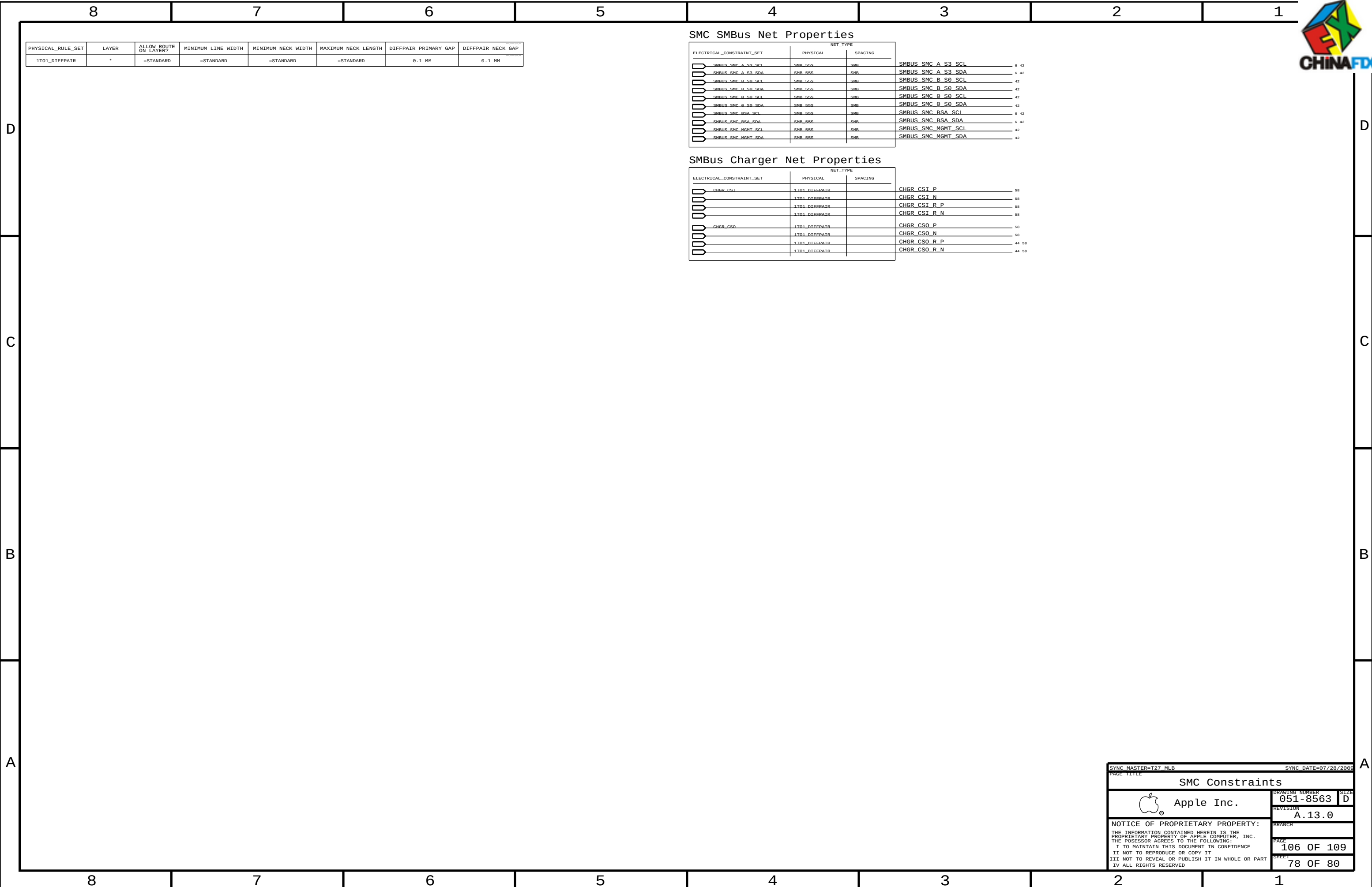
PAGE

105 OF 109

SHEET

77 OF 80

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SMBUS_SMC_A_S3_SCL	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	6 42
	SMBUS_SMC_A_S3_SDA	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	6 42
	SMBUS_SMC_B_S0_SCL	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	42
	SMBUS_SMC_B_S0_SDA	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	42
	SMBUS_SMC_0_S0_SCL	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	42
	SMBUS_SMC_0_S0_SDA	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	42
	SMBUS_SMC_BSA_SCL	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	6 42
	SMBUS_SMC_BSA_SDA	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	6 42
	SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	42
	SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	42

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
    	CHGR_CS1	1701_DIEFEATR		CHGR_CS1_P	58
		1701_DIEFEATR		CHGR_CS1_N	98
		1701_DIEFEATR		CHGR_CS1_R_P	98
		1701_DIEFEATR		CHGR_CS1_R_N	98
    	CHGR_CS0	1701_DIEFEATR		CHGR_CS0_P	58
		1701_DIEFEATR		CHGR_CS0_N	
		1701_DIEFEATR		CHGR_CS0_R_P	44 58
		1701_DIEFEATR		CHGR_CS0_R_N	44 58
		1701_DIEFEATR			

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		106 OF 109	
		SHEET	
		78 OF 80	



PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
MEM_POWER	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_POWER	*	PWR_P2MM
MEM_CMD	MEM_POWER	*	PWR_P2MM
MEM_CTRL	MEM_POWER	*	PWR_P2MM
MEM_DATA	MEM_POWER	*	PWR_P2MM
MEM_DQS	MEM_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

MCP Fanout Constraint Relaxations

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.09 MM_OVERRIDE	5.8 MM_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE
MCP_DV_COMP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE
MCP_MEM_COMP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE
MCP_MII_COMP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE
MCP_USB_RBIAIS_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE
MCP_DV_COMP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE	0.25 MM_OVERRIDE	250 MIL_OVERRIDE	TOP_OVERRIDE	TOP_OVERRIDE

Misc Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
(PCIE_AP)	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M AP CONN P 6 29
	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M AP CONN N 6 29
(USB_EXT_A)	USB_90D	USB	USB EXT_A MUXED P 37 75
(USB_EXT_A)	USB_90D	USB	USB EXT_A MUXED N 37 75
(USB_EXT_A)	USB_90D	USB	USB LT1 P 37
(USB_EXT_A)	USB_90D	USB	USB LT1 N 37
(USB_TPAD)	USB_90D	USB	USB TPAD R P 47
(USB_TPAD)	USB_90D	USB	USB TPAD R N 47
(USB_CAMERA)	USB_90D	USB	USB CAMERA CONN P 6 29
(USB_CAMERA)	USB_90D	USB	USB CAMERA CONN N 6 29
	USB_90D	USB	USB BT CONN P 6 29
	USB_90D	USB	USB BT CONN N 6 29
	USB_90D	USB	USB LT2 P 37
	USB_90D	USB	USB LT2 N 37
	ENET_MDI_100D	ENETCONN	ENETCONN P<3..0> 32
	ENET_MDI_100D	ENETCONN	ENETCONN N<3..0> 32
	SATA_90D	SATA	SATA ODD R2D UF P 36
	SATA_90D	SATA	SATA ODD R2D UF N 36
	SATA_90D	SATA	SATA ODD D2R UF P 6 36
	SATA_90D	SATA	SATA ODD D2R UF N 6 36
	SATA_90D	SATA	SATA HDD D2R UF P 36
	SATA_90D	SATA	SATA HDD D2R UF N 36
	SATA_90D	SATA	SATA HDD R2D UF P 36
	SATA_90D	SATA	SATA HDD R2D UF N 36
	SATA_90D	SATA	SATA HDD D2R RDRV IN P 36
	SATA_90D	SATA	SATA HDD D2R RDRV IN N 36
	SATA_90D	SATA	SATA HDD R2D RDRV IN P 36
	SATA_90D	SATA	SATA HDD R2D RDRV IN N 36
	SATA_90D	SATA	SATA HDD D2R RDRV OUT P 36
	SATA_90D	SATA	SATA HDD D2R RDRV OUT N 36
	SATA_90D	SATA	SATA HDD R2D RDRV OUT P 36
	SATA_90D	SATA	SATA HDD R2D RDRV OUT N 36
	SATA_90D	SATA	SATA HDD D2R NORDRV P 36
	SATA_90D	SATA	SATA HDD D2R NORDRV N 36
	SATA_90D	SATA	SATA HDD R2D NORDRV P 36
	SATA_90D	SATA	SATA HDD R2D NORDRV N 36

Graphics Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	LVDS_100D	LVDS	LVDS CONN A CLK P
	LVDS_100D	LVDS	LVDS CONN A CLK N
	LVDS_100D	LVDS	LVDS CONN A CLK F P 6 67
	LVDS_100D	LVDS	LVDS CONN A CLK F N 6 67
	LVDS_100D	LVDS	LVDS CONN A DATA P<2..0>
	LVDS_100D	LVDS	LVDS CONN A DATA N<2..0>
	LVDS_100D	LVDS	LVDS CONN B CLK P
	LVDS_100D	LVDS	LVDS CONN B CLK N
	LVDS_100D	LVDS	LVDS CONN B CLK F P
	LVDS_100D	LVDS	LVDS CONN B CLK F N
	LVDS_100D	LVDS	LVDS CONN B DATA P<2..0>
	LVDS_100D	LVDS	LVDS CONN B DATA N<2..0>
(DP_EXT_ML)	DP_90D	DISPLAYPORT	DP EXT_ML P<3..0> 8 69
	DP_90D	DISPLAYPORT	DP EXT_ML N<3..0> 8 69
	DP_90D	DISPLAYPORT	DP EXT_ML C P<3..0> 69
	DP_90D	DISPLAYPORT	DP EXT_ML C N<3..0> 69
	DP_90D	DISPLAYPORT	DP EXT_ML F P<3..0> 69
	DP_90D	DISPLAYPORT	DP EXT_ML F N<3..0> 69
(DP_EXT_AUX_CH)	DP_90D	DISPLAYPORT	DP EXT_AUX_CH C P 8 69
	DP_90D	DISPLAYPORT	DP EXT_AUX_CH C N 8 69
	DP_90D	DISPLAYPORT	DP_AUX_CH_SW P
	DP_90D	DISPLAYPORT	DP_AUX_CH_SW N

Power Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	CPU_THMSNS_D2	THERM_1T01_55S	CPU_THMSNS_D2 P 45
		THERM_1T01_55S	CPU_THMSNS_D2 N 45
	CPU_THERMD	THERM_1T01_55S	CPU_THERMD P 9 45
		THERM_1T01_55S	CPU_THERMD N 9 45
	MCPTHMSNS_D2	THERM_1T01_55S	MCPTHMSNS_D2 P
		THERM_1T01_55S	MCPTHMSNS_D2 N
	MCP_THMDIODE	THERM_1T01_55S	MCP_THMDIODE P 18 45
		THERM_1T01_55S	MCP_THMDIODE N 18 45
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_1V5_S3 P
		SENSE_1T01_55S	ISNS_1V5_S3 N
		SENSE_1T01_55S	ISNS_1V5_S3_R P
		SENSE_1T01_55S	ISNS_1V5_S3_R N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_AIRPORT P
		SENSE_1T01_55S	ISNS_AIRPORT N
		SENSE_1T01_55S	ISNS_AIRPORT_R P
		SENSE_1T01_55S	ISNS_AIRPORT_R N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_HDD P
		SENSE_1T01_55S	ISNS_HDD N
		SENSE_1T01_55S	ISNS_HDD_R P
		SENSE_1T01_55S	ISNS_HDD_R N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_LCDBKLT P
		SENSE_1T01_55S	ISNS_LCDBKLT N
		SENSE_1T01_55S	ISNS_LCDBKLT_R P
		SENSE_1T01_55S	ISNS_LCDBKLT_R N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_ODD P
		SENSE_1T01_55S	ISNS_ODD N
		SENSE_1T01_55S	ISNS_ODD_R P
		SENSE_1T01_55S	ISNS_ODD_R N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_CPUVTT P 44
		SENSE_1T01_55S	ISNS_CPUVTT N 44
	SENSE_DIEFPATR	SENSE_1T01_55S	MCPCORES0_VSEN P 21 62
		SENSE_1T01_55S	MCPCORES0_VSEN N 21 62
		MEM_POWER	PP1V5R1V35_S3 6 7
		SB_POWER	PP3V3_S5 6 7 65
		SB_POWER	PP3V3_S0 6 7 65
		SB_POWER	PP1V5_S0 6 7 65
		GND	GND

Audio Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	DIEFPATR	AUDIO	AUD_SPKRAMP_LIN P
	DIEFPATR	AUDIO	AUD_SPKRAMP_LIN N
	DIEFPATR	AUDIO	AUD_SPKRAMP_SUBIN P
	DIEFPATR	AUDIO	AUD_SPKRAMP_SUBIN N
	DIEFPATR	AUDIO	AUD_SPKRAMP_RIN P
	DIEFPATR	AUDIO	AUD_SPKRAMP_RIN N
	DIEFPATR	AUDIO	SSM2315L_P
	DIEFPATR	AUDIO	SSM2315L_N
	DIEFPATR	AUDIO	SSM2315S_P
	DIEFPATR	AUDIO	SSM2315S_N
	DIEFPATR	AUDIO	SSM2315R_P
	DIEFPATR	AUDIO	SSM2315R_N
	DIEFPATR	AUDIO	SPKRCONN_L_OUT P
SPK_OUT	DIEFPATR	AUDIO	SPKRCONN_L_OUT N
	DIEFPATR	AUDIO	SPKRCONN_S_OUT P
SPK_OUT	DIEFPATR	AUDIO	SPKRCONN_S_OUT N
	DIEFPATR	AUDIO	SPKRCONN_R_OUT P
SPK_OUT	DIEFPATR	AUDIO	SPKRCONN_R_OUT N
	DIEFPATR	AUDIO	BI_MIC P
	DIEFPATR	AUDIO	BI_MIC N
	DIEFPATR	AUDIO	HS_MIC P
	DIEFPATR	AUDIO	HS_MIC N

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		PAGE	108 OF 109
		SHEET	79 OF 80

